

Chapter 6 Wall Construction: 2023 FBC - Residential, 8th edition | UpCodes

Chapter 6 Wall Construction

Section R601 General

R601.1 Application

The provisions of this chapter shall control the design and construction of walls and partitions for buildings.

Exception: Buildings and structures located within the High-Velocity Hurricane Zone shall comply with the provisions of Chapter 44.

R601.2 Requirements

Wall construction shall be capable of accommodating all loads imposed in accordance with Section R301 and of transmitting the resulting loads to the supporting structural elements.

R601.2.1 Compressible Floor-Covering Materials

Compressible floor-covering materials that compress more than $\frac{1}{32}$ inch (0.8 mm) when subjected to 50 pounds (23 kg) applied over 1 inch square (645 mm) of material and are greater than $\frac{1}{8}$ inch (3.2 mm) in thickness in the uncompressed state shall not extend beneath walls, partitions or columns, which are fastened to the floor.

Section R602 Wood Wall Framing

R602.1 General

Wood and wood-based products used for load-supporting purposes shall conform to the applicable provisions of this section.

R602.1.1 Sawn Lumber

Sawn lumber shall be identified by a grade mark of an accredited lumber grading or inspection agency and have design values certified by an accreditation body that complies with DOC PS 20. In lieu of a grade mark, a certification of inspection issued by a lumber grading or inspection agency meeting the requirements of this section shall be accepted.

R602.1.2 End-Jointed Lumber

Approved end-jointed lumber identified by a grade mark conforming to Section R602.1 shall be permitted to be used interchangeably with solid-sawn members of the same species and grade. End-jointed lumber used in an assembly required elsewhere in this code to have a fire-resistance rating shall have the designation "Heat Resistant Adhesive" or "HRA" included in its grade mark.

R602.1.3 Structural Glued-Laminated Timbers

Glued-laminated timbers shall be manufactured and identified as required in ANSI A190.1, ANSI 117 and ASTM D3737.

R602.1.4 Structural Log Members

Structural log members shall comply with the provisions of ICC 400.

R602.1.5 Structural Composite Lumber

Structural capacities for structural composite lumber shall be established and monitored in accordance with ASTM D5456.

R602.1.6 Cross-Laminated Timber

Cross-laminated timber shall be manufactured and identified as required by ANSI/APA PRG 320.

R602.1.7 Engineered Wood Rim Board

Engineered wood rim boards shall conform to ANSI/APA PRR 410 or shall be evaluated in accordance with ASTM D7672. Structural capacities shall be in accordance with either ANSI/APA PRR 410 or established in accordance with ASTM D7672. Rim boards conforming to ANSI/APA PRR 410 shall be marked in accordance with that standard.

R602.1.8 Wood Structural Panels

Wood structural panel sheathing shall conform to DOC PS 1, DOC PS 2 or, when manufactured in Canada, CSA O437 or CSA O325.

Panels shall be identified for grade, bond classification, and performance category by a grade mark or certificate of inspection issued by an approved agency.

R602.1.9 Particleboard

Particleboard shall conform to ANSI A208.1. Particleboard shall be identified by the grade mark or certificate of inspection issued by an approved agency.

R602.1.10 Fiberboard

Fiberboard shall conform to ASTM C208. Fiberboard sheathing, where used structurally, shall be identified by an approved agency as conforming to ASTM C208.

R602.1.11 Structural Insulated Panels

Structural insulated panels shall be manufactured and identified in accordance with ANSI/APA PRS 610.1.

R602.2 Grade

Studs shall be a minimum No. 3, standard or stud grade lumber.

Exception: Bearing studs not supporting floors and nonbearing studs shall be permitted to be utility grade lumber, provided the studs are spaced in accordance with Section R602.3.

R602.3 Design and Construction

Exterior walls of woodframe construction shall be designed in accordance with Section R301.2.1.1 or ANSI AWC NDS.

Table R602.3(1) Fastening Schedule. Reserved

Figure R602.3(1) Typical Wall, Floor and Roof Framing. Reserved.

Table R602.3(2) Alternate Attachments to Table R602.3(1). Reserved.

Figure R602.3(2) Framing Details. Reserved.

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Table R602.3(4) Allowable Spans for Particleboard Wall Sheathing. Reserved.

Table R602.3(5) Size, Height and Spacing of Wood Studs. Reserved.

R602.3.1 Stud Size, Height and Spacing

Reserved.

R602.3.2 Top Plate

Reserved.

Table R602.3.2 Single Top-Plate Splice Connection Details. Reserved.

R602.3.3 Bearing Studs

Reserved.

R602.3.4 Bottom (Sole) Plate

Reserved.

R602.3.5 Braced Wall Panel Uplift Load Path

Reserved.

R602.4 Interior Load-Bearing Walls

Reserved.

R602.5 Interior Nonbearing Walls

Reserved.

R602.6 Drilling and Notching of Studs

Reserved.

Figure R602.6(1) Notching and Bored Hole Limitations for Exterior Walls and Bearing Walls. Reserved.

Figure R602.6(2) Notching and Bored Hole Limitations for Interior Nonbearing Walls. Reserved.

R602.6.1 Drilling and Notching of Top Plate

Reserved.

Figure R602.6.1 Top Plate Framing to Accommodate Piping. Reserved.

R602.7 Headers

Reserved.

Table R602.7(1) Girder Spans and Header Spans for Exterior Bearing Walls. Reserved.

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R602.7.1 Single Member Headers

Reserved.

Figure R602.7.1(1) Single Member Header in Exterior Bearing Wall. Reserved.

Figure R602.7.1(2) Alternative Single Member Header Without Cripple. Reserved.

R602.7.2 Rim Board Headers

Reserved.

Figure R602.7.2 Rim Board Header Construction. Reserved.

R602.7.3 Wood Structural Panel Box Headers

Reserved.

Table R602.7.3 Maximum Spans for Wood Structural Panel Box Headers. Reserved.

Figure R602.7.3 Typical Wood Structural Panel Box Header Construction. Reserved.

R602.7.4 Nonbearing Walls

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R602.7.5 Supports for Headers

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R602.8 Fireblocking Required

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R602.9 Cripple Walls

Reserved.

R602.10 Wall Bracing

Reserved.

R602.10.1 Braced Wall Lines

Reserved.

R602.10.1.1 Length of a Braced Wall Line

Reserved.

R602.10.1.2 Offsets Along a Braced Wall Line
Reserved.

R602.10.1.3 Spacing of Braced Wall Lines
Reserved.

Table R602.10.1.3 Braced Wall Line Spacing. Reserved.

R602.10.1.4 Angled Walls
Reserved.

Figure R602.10.1.4 Angled Walls. Reserved.

R602.10.2 Braced Wall Panels
Reserved.

R602.10.2.1 Braced Wall Panel Uplift Load Path
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R602.10.2.2 Locations of Braced Wall Panels
Reserved.

Figure R602.10.2.2 Location of Braced Wall Panels. Reserved.

R602.10.2.2.1 Location of Braced Wall Panels in Seismic Design Categories D₀, D₁ and D₂
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R602.10.2.3 Minimum Number of Braced Wall Panels
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Table R602.10.3(1) Bracing Requirements Based on Wind Speed. Reserved.

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Table R602.10.4 Bracing Methods. Reserved.

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R602.10.4.3 Braced Wall Panel Interior Finish Material
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R602.10.5 Minimum Length of a Braced Wall Panel
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Table R602.10.5 Minimum Length of Braced Wall Panels. Reserved.

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R602.10.5.1 Contributing Length

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Reserved.

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Reserved.

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Reserved.

Table R602.10.6.1 Minimum Hold-Down Forces for Method ABW Braced Wall Panels. Reserved.

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R602.10.6.2 Method PFH: Portal Frame With Hold-Downs

Reserved.

Figure R602.10.6.2 Method PFH—Portal Frame with Hold-Downs. Reserved.

R602.10.6.3 Method PFG: Portal Frame at Garage Door Openings in Seismic Design Categories A, B and C

Reserved.

Figure R602.10.6.3 Method PFG—Portal Frame at Garage Door Openings in Seismic Design Categories A, B and C. Reserved.

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Reserved.

Figure R602.10.6.4 Method CS-PF—Continuously Sheathed Portal Frame Panel Construction. Reserved.

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R602.10.6.5 Wall Bracing for Dwellings With Stone and Masonry Veneer in Seismic Design Categories D₀, D₁ and D₂

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Table R602.10.6.5 Method BV-WSP Wall Bracing Requirements. Reserved.

Figure R602.10.6.5 Method BV-WSP—Wall Bracing for Dwellings with Stone and Masonry Veneer in Seismic Design Categories D₀, D₁ and D₂. Reserved.

R602.10.6.5.1 Length of Bracing

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R602.10.7 Ends of Braced Wall Lines With Continuous Sheathing

Reserved.

Figure R602.10.7 End Conditions for Braced Wall Lines with Continuous Sheathing. Reserved.

R602.10.8 Braced Wall Panel Connections

Reserved.

Figure R602.10.8(1) Braced Wall Panel Connection When Perpendicular to Floor/Ceiling Framing. Reserved.

Figure R602.10.8(2) Braced Wall Panel Connection When Parallel to Floor/Ceiling Framing. Reserved.

R602.10.8.1 Braced Wall Panel Connections for Seismic Design Categories D₀, D₁ and D₂

Reserved.

R602.10.8.2 Connections to Roof Framing

Reserved.

Figure R602.10.8.2(1) Braced Wall Panel Connection to Perpendicular Rafters. Reserved.

Figure R602.10.8.2(2) Braced Wall Panel Connection Option to Perpendicular Rafters or Roof Trusses. Reserved.

Figure R602.10.8.2(3) Braced Wall Panel Connection Option to Perpendicular Rafters or Roof Trusses. Reserved.

R602.10.9 Braced Wall Panel Support

Reserved.

Figure R602.10.9 Masonry Stem Walls Supporting Braced Wall Panels. Reserved.

R602.10.9.1 Braced Wall Panel Support for Seismic Design Categories D₀, D₁ and D₂

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R602.10.10 Panel Joints

Reserved.

R602.10.11 Cripple Wall Bracing

Reserved.

R602.10.11.1 Cripple Wall Bracing for Seismic Design Categories D₀ and D₁ and Townhouses in Seismic Design Category C

Reserved.

R602.10.11.2 Cripple Wall Bracing for Seismic Design Category D₂

Reserved.

R602.10.11.3 Redesignation of Cripple Walls

Reserved.

R602.11 Wall Anchorage

Reserved.

R602.11.1 Wall Anchorage for All Buildings in Seismic Design Categories D₀, D₁ and D₂ and Townhouses in Seismic Design Category C

Reserved.

R602.11.2 Stepped Foundations in Seismic Design Categories D₀, D₁ and D₂

Reserved.

Figure R602.11.2 Stepped Foundation Construction. Reserved.

R602.12 Simplified Wall Bracing

Reserved.

R602.12.1 Circumscribed Rectangle

Reserved.

Figure R602.12.1 Rectangle Circumscribing an Enclosed Building. Reserved.

R602.12.2 Sheathing Materials

Reserved.

R602.12.3 Bracing Unit

Reserved.

R602.12.3.1 Multiple Bracing Units

Reserved.

R602.12.4 Number of Bracing Units

Reserved.

Table R602.12.4 Minimum Number of Bracing Units on Each Side of the Circumscribed Rectangle. Reserved.

R602.12.5 Distribution of Bracing Units

Reserved.

Figure R602.12.5 Bracing Unit Distribution. Reserved.

R602.12.6 Narrow Panels

Reserved.

R602.12.6.1 Method CS-G

Reserved.

R602.12.6.2 Method CS-PF

Reserved.

R602.12.6.3 Methods ABW, PFH and PFG

Reserved.

R602.12.7 Lateral Support

Reserved.

R602.12.8 Stem Walls

Reserved.

Section R603 Cold-Formed Steel Wall Framing

R603.1 General

In accordance with Section R301.2.1.1, the design of cold-formed steel wall framing shall be in accordance with AISI S230, *Standard for Cold-Formed Steel Framing—Prescriptive Method For One- and Two-Family Dwellings*.

Section R604 Wood Structural Panels

R604.1 Identification and Grade

Wood structural panels shall conform to DOC PS 1, DOC PS 2 or ANSI/APA PRP 210, CSA O437 or CSA O325. Panels shall be identified by a grade mark or certificate of inspection issued by an *approved* agency.

R604.2 Allowable Spans

The maximum allowable spans for wood structural panel wall sheathing shall not exceed the values set forth in Section R602.3.

R604.3 Installation

Wood structural panel wall sheathing shall be attached to framing in accordance with Section R602.3.

Section R605 Particleboard

R605.1 Identification and Grade

Particleboard shall conform to ANSI A208.1 and shall be so identified by a grade mark or certificate of inspection issued by an *approved* agency.

Section R606 General Masonry Construction

R606.1 General

Masonry construction shall be designed and constructed in accordance with the provisions of this section and Section R301.2.1.1, TMS 402, TMS 403 or TMS 404.

R606.1.1 Professional Registration Not Required

When the empirical design provisions of Appendix A of TMS 402, the provisions of TMS 403, or the provisions of this section are used to design masonry, project drawings, typical details and specifications are not required to bear the seal of the architect or engineer responsible for design, unless otherwise required by the state law of the *jurisdiction* having authority.

R606.2 Masonry Construction Materials

R606.2.1 Concrete Masonry Units

Concrete masonry units shall conform to the following standards: ASTM C55 for concrete brick; ASTM C73 for calcium silicate face brick; ASTM C90 for load-bearing concrete masonry units; ASTM C744 for prefaced concrete and calcium silicate masonry units; or ASTM C1634 for concrete facing brick.

R606.2.2 Clay or Shale Masonry Units

Clay or shale masonry units shall conform to the following standards: ASTM C34 for structural clay *load-bearing wall* tile; ASTM C56 for structural clay nonload-bearing wall tile; ASTM C62 for building brick (solid masonry units made from clay or shale); ASTM C1088 for solid units of thin veneer brick; ASTM C126 for ceramic-glazed structural clay facing tile, facing brick and solid masonry units; ASTM C212 for structural clay facing tile; ASTM C216 for facing brick (solid masonry units made from clay or shale); ASTM C652 for hollow brick (hollow masonry units made from clay or shale); or ASTM C1405 for glazed brick (single-fired solid brick units).

Exception: Structural clay tile for nonstructural use in fireproofing of structural members and in wall furring shall not be required to meet the compressive strength specifications. The fire-resistance rating shall be determined in accordance with ASTM E119 or UL 263 and shall comply with the requirements of Section R302.

R606.2.3 AAC Masonry

AAC masonry units shall conform to ASTM C1691 and ASTM C1693 for the strength class specified.

R606.2.4 Stone Masonry Units

Stone masonry units shall conform to the following standards: ASTM C503 for marble building stone (exterior); ASTM C568 for limestone building stone; ASTM C615 for granite building stone; ASTM C616 for sandstone building stone; or ASTM C629 for slate building stone.

R606.2.5 Architectural Cast Stone

Architectural cast stone shall conform to ASTM C1364.

R606.2.6 Adhered Manufactured Stone Masonry Veneer Units

Adhered manufactured stone masonry veneer units shall conform to ASTM C1670.

R606.2.7 Second Hand Units

Second hand masonry units shall not be reused unless they conform to the requirements of new units. The units shall be of whole, sound materials and free from cracks and other defects that will interfere with proper laying or use. Old mortar shall be cleaned from the unit before reuse.

R606.2.8 Mortar

Except for mortars listed in Sections R606.2.9, R606.2.10 and R606.2.11, mortar for use in masonry construction shall meet the proportion specifications of Table R606.2.8 or the property specifications of ASTM C270. The type of mortar shall be in accordance with Sections R606.2.8.1 and R606.2.8.2.

TABLE R606.2.8
MORTAR PROPORTIONS^{a, b}

		PROPORTIONS BY VOLUME (cementitious materials)	
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MORTAR	TYPE	Portland cement or blended cement	Mortar cement			Masonry cement			Hydrated lime ^c or lime putty	Aggregate ratio (measured in damp, loose conditions)
			M	S	N	M	S	N		
Cement-lime	M	1	—	—	—	—	—	—	1/4	Not less than 2 1/4 and not more than 3 times the sum of separate volumes of lime, if used, and cement
	S	1	—	—	—	—	—	—	over 1/4 to 1/2	
	N	1	—	—	—	—	—	—	over 1/2 to 1 1/4	
	O	1	—	—	—	—	—	—	over 1 1/4 to 2 1/2	
Mortar cement	M	1	—	—	1	—	—	—	—	
	M	—	1	—	—	—	—	—		
	S	1/2	—	—	1	—	—	—		
	S	—	—	1	—	—	—	—		
	N	—	—	—	1	—	—	—		
	O	—	—	—	1	—	—	—		
Masonry cement	M	1				—	—	1	—	
	M	—				1	—	—		
	S	1/2				—	—	1		
	S	—				—	1	—		
	N	—				—	—	1		
	O	—				—	—	1		

For SI: 1 cubic foot = 0.0283 m³, 1 pound = 0.454 kg.

- a. For the purpose of these specifications, the weight of 1 cubic foot of the respective materials shall be considered to be as follows:

Portland Cement	94 pounds	Masonry Cement	Weight printed on bag
Mortar Cement	Weight printed on bag	Hydrated Lime	40 pounds
Lime Putty (Quicklime)	80 pounds	Sand, damp and loose	80 pounds of dry sand

- b. Two air-entraining materials shall not be combined in mortar.

- c. Hydrated lime conforming to the requirements of ASTM C207.

R606.2.8.1 Foundation Walls

Mortar for masonry foundation walls constructed as set forth in Tables R404.1.1(1) through R404.1.1(4) shall be Type M or S mortar.

R606.2.8.2 Masonry Serving as the Lateral-Force-Resisting System

Mortar for masonry serving as the lateral-force-resisting system shall be Type M, S or N mortar.

R606.2.9 Surface-Bonding Mortar

Surface-bonding mortar shall comply with ASTM C887. Surface bonding of concrete masonry units shall comply with ASTM C946.

R606.2.10 Mortar for AAC Masonry

Thin-bed mortar for AAC masonry shall comply with Article 2.1 C.1 of TMS 602. Mortar used for the leveling courses of AAC masonry shall comply with Article 2.1 C.2 of TMS 602.

R606.2.11 Mortar for Adhered Masonry Veneer

Mortar for use with adhered masonry veneer shall conform to ASTM C270 Type S or Type N or shall comply with ANSI A118.4 for latex-modified portland cement mortar.

R606.2.12 Grout

Grout shall consist of cementitious material and aggregate in accordance with ASTM C476 or the proportion specifications of Table R606.2.12. Type M or Type S mortar to which sufficient water has been added to produce pouring consistency shall be permitted to be used as grout.

TABLE R606.2.12

GROUT PROPORTIONS BY VOLUME FOR MASONRY CONSTRUCTION

TYPE	PORTLAND CEMENT OR BLENDED CEMENT SLAG CEMENT	HYDRATED LIME OR LIME PUTTY	AGGREGATE MEASURED IN A DAMP, LOOSE CONDITION	
			Fine	Coarse
Fine	1	0 to 1/10	2 ¹ / ₄ to 3 times the sum of the volume of the cementitious materials	—
Coarse	1	0 to 1/10	2 ¹ / ₄ to 3 times the sum of the volume of the cementitious materials	1 to 2 times the sum of the volumes of the cementitious materials

R606.2.13 Metal Reinforcement and Accessories

Metal reinforcement and accessories shall conform to Article 2.4 of TMS 602. Where provided in exterior walls, joint reinforcement shall be a minimum No. 9-gauge ladder-type stainless steel, hot dipped galvanized, or epoxy coated in accordance with TMS 602 Section 2.4E1, 2.4F1b or 2.4F2a as appropriate.

R606.3 Construction Requirements

R606.3.1 Bed and Head Joints

Unless otherwise required or indicated on the project drawings, head and bed joints shall be ³/₈ inch (9.5 mm) thick, except that the thickness of the bed joint of the starting course placed over foundations shall be not less than ¹/₄ inch (6.4 mm) and not more than ³/₄ inch (19.1 mm). Mortar joint thickness for load-bearing masonry shall be within the following tolerances from the specified dimensions:

1. Bed joint: + ¹/₈ inch (3.2 mm).
2. Head joint: - ¹/₄ inch (6.4 mm), + ³/₈ inch (9.5 mm).
3. Collar joints: - ¹/₄ inch (6.4 mm), + ³/₈ inch (9.5 mm).

R606.3.1.1 Masonry Opening Tolerances

Masonry rough openings may vary in the cross section dimension or elevation dimension specified on the approved plans from - ¹/₄ inch (6.4 mm) to + ¹/₂ inch (12.7 mm). For exterior window and door installation provisions, see Section R609.3.

R606.3.2 Masonry Unit Placement

The mortar shall be sufficiently plastic and units shall be placed with sufficient pressure to extrude mortar from the joint and produce a tight joint. Deep furrowing of bed joints that produces voids shall not be permitted. Any units disturbed to the extent that initial bond is

broken after initial placement shall be removed and relaid in fresh mortar. Surfaces to be in contact with mortar shall be clean and free of deleterious materials.

R606.3.2.1 Solid Masonry

Solid masonry units shall be laid with full head and bed joints and all interior vertical joints that are designed to receive mortar shall be filled.

R606.3.2.2 Hollow Masonry

For hollow masonry units, head and bed joints shall be filled solidly with mortar for a distance in from the face of the unit not less than the thickness of the face shell.

R606.3.3 Installation of Wall Ties

The installation of wall ties shall be as follows:

1. The ends of wall ties shall be embedded in mortar joints. Wall ties shall have not less than $\frac{5}{8}$ -inch (15.9 mm) mortar coverage from the exposed face.
2. Wall ties shall not be bent after being embedded in grout or mortar.
3. For solid masonry units, solid grouted hollow units, or hollow units in anchored masonry veneer, wall ties shall be embedded in mortar bed not less than $1\frac{1}{2}$ inches (38 mm).
4. For hollow masonry units in other than anchored masonry veneer, wall ties shall engage outer face shells by not less than $\frac{1}{2}$ inch (13 mm).

R606.3.4 Protection for Reinforcement

Bars shall be completely embedded in mortar or grout. Joint reinforcement embedded in horizontal mortar joints shall not have less than $\frac{5}{8}$ -inch (15.9 mm) mortar coverage from the exposed face. Other reinforcement shall have a minimum coverage of one bar diameter over all bars, but not less than $\frac{3}{4}$ inch (19 mm), except where exposed to weather or soil, in which case the minimum coverage shall be 2 inches (51 mm).

R606.3.4.1 Corrosion Protection

Minimum corrosion protection of joint reinforcement, anchor ties and wire fabric for use in masonry wall construction shall conform to Table R606.3.4.1.

TABLE R606.3.4.1

MINIMUM CORROSION PROTECTION

MASONRY METAL ACCESSORY	STANDARD
Joint reinforcement, interior walls	ASTM A641, Class 1
Wire ties or anchors in exterior walls completely embedded in mortar or grout	ASTM A641, Class 3
Wire ties or anchors in exterior walls not completely embedded in mortar or grout	ASTM A153, Class B-2
Joint reinforcement in exterior walls or interior walls exposed to moist environment	ASTM A153, Class B-2
Sheet metal ties or anchors exposed to weather	ASTM A153, Class B-2
Sheet metal ties or anchors completely embedded in mortar or grout	ASTM A653, Coating Designation G60

Stainless steel hardware for any exposure

ASTM A167, Type 304

R606.3.5 Grouting Requirements**R606.3.5.1 Grout Placement**

Grout shall be a plastic mix suitable for pumping without segregation of the constituents and shall be mixed thoroughly. Grout shall be placed by pumping or by an approved alternate method and shall be placed before any initial set occurs and not more than $1\frac{1}{2}$ hours after water has been added. Grout shall be consolidated by puddling or mechanical vibrating during placing and reconsolidated after excess moisture has been absorbed but before plasticity is lost. Grout shall not be pumped through aluminum pipes.

Maximum pour heights and the minimum dimensions of spaces provided for grout placement shall conform to Table R606.3.5.1. Grout shall be poured in lifts of 8-foot (2438 mm) maximum height. Where a total grout pour exceeds 8 feet (2438 mm) in height, the grout shall be placed in lifts not exceeding 64 inches (1626 mm) and special inspection during grouting shall be required. If the work is stopped for 1 hour or longer, the horizontal construction joints shall be formed by stopping all tiers at the same elevation and with the grout 1 inch (25 mm) below the top.

TABLE R606.3.5.1**GROUT SPACE DIMENSIONS AND POUR HEIGHTS**

GROUT TYPE	GROUT POUR MAXIMUM HEIGHT (feet)	MINIMUM WIDTH OF GROUT SPACES ^{a, b} (inches)	MINIMUM GROUT ^{b, c} SPACE DIMENSIONS FOR GROUTING CELLS OF HOLLOW UNITS (inches × inches)
Fine	1	0.75	1.5 × 2
	5	2	2 × 3
	12	2.5	2.5 × 3
	24	3	3 × 3
Coarse	1	1.5	1.5 × 3
	5	2	2.5 × 3
	12	2.5	3 × 3
	24	3	3 × 4

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

a. For grouting between masonry wythes.

b. Grout space dimension is the clear dimension between any masonry protrusion and shall be increased by the horizontal projection of the diameters of the horizontal bars within the cross section of the grout space.

c. Area of vertical reinforcement shall not exceed 6 percent of the area of the grout space.

R606.3.5.2 Cleanouts

Provisions shall be made for cleaning the space to be grouted. Mortar that projects more than $\frac{1}{2}$ inch (12.7 mm) into the grout space and any other foreign matter shall be removed from the grout space prior to inspection and grouting. Where required by the building official, cleanouts shall be provided in the bottom course of masonry for each grout pour where the grout pour height exceeds 64 inches (1626 mm). In solid grouted masonry, cleanouts shall be spaced horizontally not more than 32 inches (813 mm) on center. The cleanouts shall be sealed before grouting and after inspection.

R606.3.5.3 Construction

Requirements for grouted masonry construction shall be as follows:

1. Masonry shall be built to preserve the unobstructed vertical continuity of the cells or spaces to be filled. In partially grouted construction, cross webs forming cells to be filled shall be full-bedded in mortar to prevent leakage of grout. Head and end joints shall be solidly filled with mortar for a distance in from the face of the wall or unit not less than the thickness of the longitudinal face shells.
2. Vertical reinforcement shall be held in position at top and bottom and at intervals not exceeding 200 diameters of the reinforcement.
3. Cells containing reinforcement shall be filled solidly with grout.
4. The thickness of grout or mortar between masonry units and reinforcement shall be not less than $\frac{1}{4}$ inch (6.4 mm), except that $\frac{1}{4}$ -inch (6.4 mm) bars shall be permitted to be laid in horizontal mortar joints not less than $\frac{1}{2}$ inch (12.7 mm) thick, and steel wire reinforcement shall be permitted to be laid in horizontal mortar joints not less than twice the thickness of the wire diameter.

R606.3.6 Grouted Multiple-Wythe Masonry

Grouted multiple-wythe masonry shall conform to all the requirements specified in Section R606.3.5 and the requirements of this section.

R606.3.6.1 Bonding of Backup Wythe

Where all interior vertical spaces are filled with grout in multiple-wythe construction, masonry headers shall not be permitted. Metal wall ties shall be used in accordance with Section R606.13.2 to prevent spreading of the wythes and to maintain the vertical alignment of the wall. Wall ties shall be installed in accordance with Section R606.13.2 where the backup wythe in multiple-wythe construction is fully grouted.

R606.3.6.2 Grout Barriers

Vertical grout barriers or dams shall be built of *solid masonry* across the grout space the entire height of the wall to control the flow of the grout horizontally. Grout barriers shall be not more than 25 feet (7620 mm) apart. The grouting of any section of a wall between control barriers shall be completed in one day without interruptions greater than 1 hour.

R606.3.7 Masonry Bonding Pattern

Masonry laid in running and stack bond shall conform to Sections R606.3.7.1 and R606.3.7.2.

R606.3.7.1 Masonry Laid in Running Bond

In each wythe of masonry laid in running bond, head joints in successive courses shall be offset by not less than one-fourth the unit length, or the masonry walls shall be reinforced longitudinally as required in Section R606.3.7.2.

R606.3.7.2 Masonry Laid in Stack Bond

Where unit masonry is laid with less head joint offset than in Section R606.3.7.1, the minimum area of horizontal reinforcement placed in mortar bed joints or in bond beams spaced not more than 48 inches (1219 mm) apart shall be 0.0007 times the vertical cross-sectional area of the wall.

R606.4 Thickness of Masonry

The nominal thickness of masonry walls shall conform to the requirements of Sections R606.4.1 through R606.4.4.

R606.4.1 Minimum Thickness

The minimum thickness of masonry bearing walls more than one story high shall be 8 inches (203 mm). *Solid masonry* walls of one-story dwellings and garages shall be not less than 6 inches (152 mm) in thickness where not greater than 9 feet (2743 mm) in height, provided that where gable construction is used, an additional 6 feet (1829 mm) is permitted to the peak of the gable. Masonry walls shall be laterally supported in either the horizontal or vertical direction at intervals as required by Section R606.6.4.

R606.4.2 Rubble Stone Masonry Wall

The minimum thickness of rough, random or coursed rubble stone masonry walls shall be 16 inches (406 mm).

R606.4.3 Change in Thickness

Where walls of masonry of hollow units or masonry-bonded hollow walls are decreased in thickness, a course of solid masonry or masonry units filled with mortar or grout shall be constructed between the wall below and the thinner wall above, or special units or

construction shall be used to transmit the loads from face shells or wythes above to those below.

R606.4.4 Parapet Walls

Unreinforced solid masonry parapet walls shall be not less than 8 inches (203 mm) thick and their height shall not exceed four times their thickness. Unreinforced hollow unit masonry parapet walls shall be not less than 8 inches (203 mm) thick, and their height shall not exceed three times their thickness.

R606.5 Corbeled Masonry

Corbeled masonry shall be in accordance with Sections R606.5.1 through R606.5.3.

R606.5.1 Units

Solid masonry units or masonry units filled with mortar or grout shall be used for corbeling.

R606.5.2 Corbel Projection

The maximum projection of one unit shall not exceed one-half the height of the unit or one-third the thickness at right angles to the wall. The maximum corbeled projection beyond the face of the wall shall not exceed:

1. One-half of the wall thickness for multiwythe walls bonded by mortar or grout and wall ties or masonry headers.
2. One-half the wythe thickness for single wythe walls, masonry-bonded hollow walls, multiwythe walls with open collar joints and veneer walls.

R606.5.3 Corbeled Masonry Supporting Floor or Roof-Framing Members

Where corbeled masonry is used to support floor or roof-framing members, the top course of the corbel shall be a header course or the top course bed joint shall have ties to the vertical wall.

R606.6 Support Conditions

Bearing and support conditions shall be in accordance with Sections R606.6.1 through R606.6.4.

R606.6.1 Bearing on Support

Each masonry wythe shall be supported by at least two-thirds of the wythe thickness.

R606.6.2 Support at Foundation

Cavity wall or masonry veneer construction shall be permitted to be supported on an 8-inch (203 mm) foundation wall, provided the 8-inch (203 mm) wall is corbeled to the width of the wall system above with masonry constructed of solid masonry units or masonry units filled with mortar or grout. The total horizontal projection of the corbel shall not exceed 2 inches (51 mm) with individual corbels projecting not more than one-third the thickness of the unit or one-half the height of the unit. The hollow space behind the corbeled masonry shall be filled with mortar or grout.

R606.6.3 Beam Supports

Beams, girders or other concentrated loads supported by a wall or column shall have a bearing of not less than 3 inches (76 mm) in length measured parallel to the beam upon *solid masonry* not less than 4 inches (102 mm) in thickness, or upon a metal bearing plate of adequate design and dimensions to distribute the load safely, or upon a continuous reinforced masonry member projecting not less than 4 inches (102 mm) from the face of the wall.

R606.6.3.1 Joist Bearing

Joists shall have a bearing of not less than 1½ inches (38 mm), except as provided in Section R606.6.3, and shall be supported in accordance with Section R606.1.

R606.6.4 Lateral Support

Masonry walls shall be laterally supported in either the horizontal or the vertical direction in accordance with Section R301.2.1.1, TMS 402, TMS 403 or TMS 404.

Table R606.6.4 Spacing of Lateral Support for Masonry Walls. Reserved.

R606.6.4.1 Horizontal Lateral Support

Reserved.

R606.6.4.1.1 Bonding Pattern

Reserved.

R606.6.4.1.2 Metal Reinforcement

Reserved.

R606.6.4.2 Vertical Lateral Support

Reserved.

R606.6.4.2.1 Roof Structures

Reserved.

R606.6.4.2.2 Floor Diaphragms

Reserved.

R606.7 Piers

The unsupported height of masonry piers shall not exceed 10 times their least dimension. Where structural clay tile or hollow concrete masonry units are used for isolated piers to support beams and girders, the cellular spaces shall be filled solidly with grout or Type M or S mortar, except that unfilled hollow piers shall be permitted to be used if their unsupported height is not more than four times their least dimension. Where hollow masonry units are solidly filled with grout or Type M, S or N mortar, the allowable compressive stress shall be permitted to be increased as provided in Table R606.9.

R606.7.1 Pier Cap

Hollow piers shall be capped with 4 inches (102 mm) of *solid masonry* or concrete, a masonry cap block, or shall have cavities of the top course filled with concrete or grout.

R606.8 Chases

Chases and recesses in masonry walls shall not be deeper than one-third the wall thickness, and the maximum length of a horizontal chase or horizontal projection shall not exceed 4 feet (1219 mm), and shall have not less than 8 inches (203 mm) of masonry in back of the chases and recesses and between adjacent chases or recesses and the jambs of openings. Chases and recesses in masonry walls shall be designed and constructed so as not to reduce the required strength or required fire resistance of the wall and in no case shall a chase or recess be permitted within the required area of a pier. Masonry directly above chases or recesses wider than 12 inches (305 mm) shall be supported on noncombustible lintels.

R606.9 Allowable Stresses

Allowable compressive stresses in masonry shall not exceed the values prescribed in Table R606.9. In determining the stresses in masonry, the effects of all loads and conditions of loading and the influence of all forces affecting the design and strength of the several parts shall be taken into account.

TABLE R606.9

ALLOWABLE COMPRESSIVE STRESSES FOR EMPIRICAL DESIGN OF MASONRY

CONSTRUCTION; COMPRESSIVE STRENGTH OF UNIT, GROSS AREA	ALLOWABLE COMPRESSIVE STRESSES ^a GROSS CROSS-SECTIONAL AREA ^b	
	Type M or S mortar	Type N mortar
Solid masonry of brick and other solid units of clay or shale; sand-lime or concrete brick:		
8,000 + psi	350	300
4,500 psi	225	200

2,500 psi	160	140
1,500 psi	115	100
Grouted ^c masonry, of clay or shale; sand-lime or concrete:		
4,500 + psi	225	200
2,500 psi	160	140
1,500 psi	115	100
Solid masonry of solid concrete masonry units:		
3,000 + psi	225	200
2,000 psi	160	140
1,200 psi	115	100
Masonry of hollow load-bearing units:		
2,000 + psi	140	120
1,500 psi	115	100
1,000 psi	75	70
700 psi	60	55
Hollow walls (cavity or masonry bonded ^d) solid units:		
2,500 + psi	160	140
1,500 psi	115	100
Hollow units	75	70
Stone ashlar masonry:		
Granite	720	640
Limestone or marble	450	400
Sandstone or cast stone	360	320
Rubble stone masonry:		
Coarse, rough or random	120	100

For SI: 1 pound per square inch = 6.895 kPa.

- a. Linear interpolation shall be used for determining allowable stresses for masonry units having compressive strengths that are intermediate between those given in the table.
- b. Gross cross-sectional area shall be calculated on the actual rather than nominal dimensions.
- c. See Section R606.13.

- d. Where floor and roof loads are carried upon one wythe, the gross cross-sectional area is that of the wythe under load; if both wythes are loaded, the gross cross-sectional area is that of the wall minus the area of the cavity between the wythes. Walls bonded with metal ties shall be considered as cavity walls unless the collar joints are filled with mortar or grout.

R606.9.1 Combined Units

In walls or other structural members composed of different kinds or grades of units, materials or mortars, the maximum stress shall not exceed the allowable stress for the weakest of the combination of units, materials and mortars of which the member is composed. The net thickness of any facing unit that is used to resist stress shall be not less than $1\frac{1}{2}$ inches (38 mm).

R606.10 Lintels

Masonry over openings shall be supported by steel lintels, reinforced concrete or masonry lintels or masonry arches, designed to support load imposed.

R606.11 Anchorage

Reserved.

Figure R606.11(1) Anchorage Requirements for Masonry Walls Located in Seismic Design Category A, B or C and Where Wind Loads Are Less Than 30 PSF. Reserved.

Figure R606.11(2) Requirements for Reinforced Grouted Masonry Construction in Seismic Design Category C. Reserved.

Figure R606.11(3) Requirements for Reinforced Masonry Construction in Seismic Design Category D₀, D₁ or D₂. Reserved.

R606.12 Seismic Requirements

Reserved.

R606.12.1 General

Reserved.

R606.12.1.1 Floor and Roof Diaphragm Construction

Reserved.

R606.12.2 Seismic Design Category C

Reserved.

R606.12.2.1 Minimum Length of Wall Without Openings

Reserved.

Table R606.12.2.1 Minimum Solid Wall Length Along Exterior Wall Lines. Reserved.

R606.12.2.2 Design of Elements Not Part of the Lateral Force-Resisting System

Reserved.

R606.12.2.2.1 Load-Bearing Frames or Columns

Reserved.

R606.12.2.2.2 Masonry Partition Walls

Reserved.

R606.12.2.2.3 Reinforcement Requirements for Masonry Elements

Reserved.

R606.12.2.3 Design of Elements Part of the Lateral Force-Resisting System

Reserved.

R606.12.2.3.1 Connections to Masonry Shear Walls

Reserved.

R606.12.2.3.2 Connections to Masonry Columns

Reserved.

R606.12.2.3.3 Minimum Reinforcement Requirements for Masonry Shear Walls

Reserved.

R606.12.3 Seismic Design Category D₀ or D₁

Reserved.

R606.12.3.1 Design Requirements

Reserved.

R606.12.3.2 Minimum Reinforcement Requirements for Masonry Walls

Reserved.

Table R606.12.3.2 Minimum Distributed Wall Reinforcement for Buildings Assigned to Seismic Design Category D₀ or D₁.

Reserved.

R606.12.3.2.1 Shear Wall Reinforcement Requirements

Reserved.

R606.12.3.3 Minimum Reinforcement for Masonry Columns

Reserved.

R606.12.3.4 Material Restrictions

Reserved.

R606.12.3.5 Lateral Tie Anchorage

Reserved.

R606.12.4 Seismic Design Category D₂

Reserved.

R606.12.4.1 Design of Elements Not Part of the Lateral Force-Resisting System

Reserved.

Table R606.12.4.1 Minimum Reinforcing for Stacked Bonded Masonry Walls in Seismic Design Category D₂. Reserved.

R606.12.4.2 Design of Elements Part of the Lateral Force-Resisting System

Reserved.

Table R606.12.4.2 Minimum Reinforcing for Stacked Bonded Masonry Walls in Seismic Design Category D₂. Reserved.

R606.13 Multiple-Wythe Masonry

The facing and backing of multiple-wythe masonry walls shall be bonded in accordance with Section R606.13.1, R606.13.2 or R606.13.3. In cavity walls, neither the facing nor the backing shall be less than 3 inches (76 mm) nominal in thickness and the cavity shall be not more than 4 inches (102 mm) nominal in width. The backing shall be not less than as thick as the facing.

Exception: Cavities shall be permitted to exceed the 4-inch (102 mm) nominal dimension provided tie size and tie spacing have been established by calculation.

R606.13.1 Bonding With Masonry Headers

Bonding with solid or hollow masonry headers shall comply with Sections R606.13.1.1 and R606.13.1.2.

R606.13.1.1 Solid Units

Where the facing and backing (adjacent wythes) of *solid masonry* construction are bonded by means of masonry headers, not less than 4 percent of the wall surface of each face shall be composed of headers extending not less than 3 inches (76 mm) into the backing. The distance between adjacent full-length headers shall not exceed 24 inches (610 mm) either vertically or horizontally. In walls in which a single header does not extend through the wall, headers from the opposite sides shall overlap not less than 3 inches (76 mm), or headers from opposite sides shall be covered with another header course overlapping the header below not less than 3 inches (76 mm).

R606.13.1.2 Hollow Units

Where two or more hollow units are used to make up the thickness of a wall, the stretcher courses shall be bonded at vertical intervals not exceeding 34 inches (864 mm) by lapping not less than 3 inches (76 mm) over the unit below, or by lapping at vertical intervals not exceeding 17 inches (432 mm) with units that are not less than 50 percent thicker than the units below.

R606.13.2 Bonding With Wall Ties or Joint Reinforcement

Bonding with wall ties or joint reinforcement shall comply with Section R606.13.2.3.

R606.13.2.1 Bonding With Wall Ties

Bonding with wall ties, except as required by Section R607, where the facing and backing (adjacent wythes) of masonry walls are bonded with $\frac{3}{16}$ -inch-diameter (5 mm) wall ties embedded in the horizontal mortar joints, there shall be not less than one metal tie for each $4\frac{1}{2}$ square feet (0.418 m²) of wall area. Ties in alternate courses shall be staggered. The maximum vertical distance between ties shall not exceed 24 inches (610 mm), and the maximum horizontal distance shall not exceed 36 inches (914 mm). Rods or ties bent to rectangular shape shall be used with hollow masonry units laid with the cells vertical. In other walls, the ends of ties shall be bent to 90-degree (0.79 rad) angles to provide hooks not less than 2 inches (51 mm) long. Additional bonding ties shall be provided at all openings, spaced not more than 3 feet (914 mm) apart around the perimeter and within 12 inches (305 mm) of the opening.

R606.13.2.2 Bonding With Adjustable Wall Ties

Where the facing and backing (adjacent wythes) of masonry are bonded with adjustable wall ties, there shall be not less than one tie for each 2.67 square feet (0.248 m²) of wall area. Neither the vertical nor the horizontal spacing of the adjustable wall ties shall exceed 24 inches (610 mm). The maximum vertical offset of bed joints from one wythe to the other shall be 1.25 inches (32 mm). The maximum clearance between connecting parts of the ties shall be $\frac{1}{16}$ inch (2 mm). Where pintle legs are used, ties shall have not less than two $\frac{3}{16}$ -inch-diameter (5 mm) legs.

R606.13.2.3 Bonding With Prefabricated Joint Reinforcement

Where the facing and backing (adjacent wythes) of masonry are bonded with prefabricated joint reinforcement, there shall be not less than one cross wire serving as a tie for each 2.67 square feet (0.248 m²) of wall area. The vertical spacing of the joint reinforcement shall not exceed 16 inches (406 mm). Cross wires on prefabricated joint reinforcement shall not be smaller than No. 9 gage. The longitudinal wires shall be embedded in the mortar.

R606.13.3 Bonding With Natural or Cast Stone

Bonding with natural and cast stone shall conform to Sections R606.13.3.1 and R606.13.3.2.

R606.13.3.1 Ashlar Masonry

In ashlar masonry, bonder units, uniformly distributed, shall be provided to the extent of not less than 10 percent of the wall area. Such bonder units shall extend not less than 4 inches (102 mm) into the backing wall.

R606.13.3.2 Rubble Stone Masonry

Rubble stone masonry 24 inches (610 mm) or less in thickness shall have bonder units with a maximum spacing of 3 feet (914 mm) vertically and 3 feet (914 mm) horizontally, and if the masonry is of greater thickness than 24 inches (610 mm), shall have one bonder unit for each 6 square feet (0.557 m²) of wall surface on both sides.

R606.14 Anchored and Adhered Masonry Veneer

R606.14.1 Anchored Veneer

Anchored masonry veneer installed over a backing of wood or cold-formed steel shall meet the requirements of Section R703.8.

R606.14.2 Adhered Veneer

Adhered masonry veneer shall be installed in accordance with the requirements of Section R703.12.

Section R607 Glass Unit Masonry

R607.1 General

Panels of glass unit masonry located in load-bearing and nonload-bearing exterior and interior walls shall be constructed in accordance with this section.

R607.2 Materials

Hollow glass units shall be partially evacuated and have a minimum average glass face thickness of 3/16 inch (5 mm). The surface of units in contact with mortar shall be treated with a polyvinyl butyral coating or latex-based paint. The use of reclaimed units is prohibited.

R607.3 Units

Hollow or solid glass block units shall be standard or thin units.

R607.3.1 Standard Units

The specified thickness of standard units shall be not less than 3 7/8 inches (98 mm).

R607.3.2 Thin Units

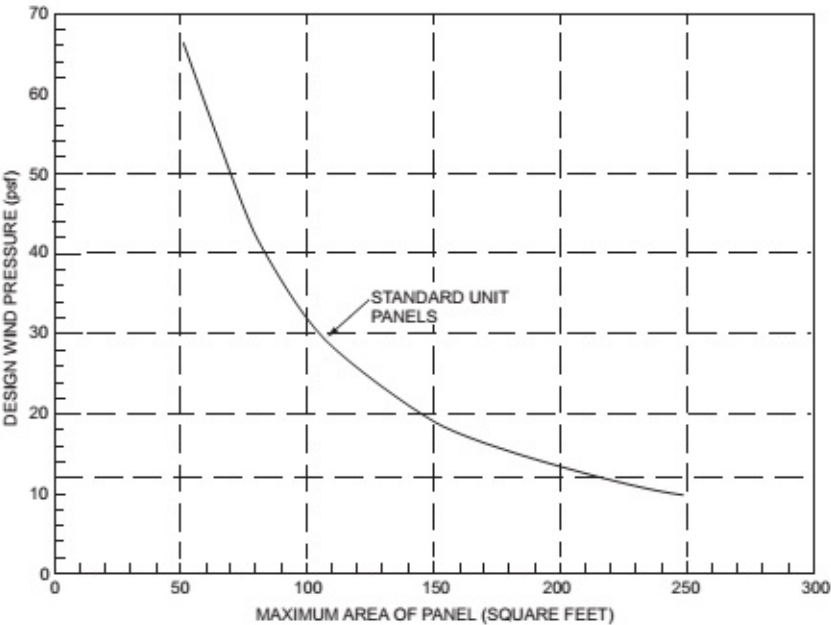
The specified thickness of thin units shall be not less than 3 1/8 inches (79 mm) for hollow units and not less than 3 inches (76 mm) for solid units.

R607.4 Isolated Panels

Isolated panels of glass unit masonry shall conform to the requirements of this section.

R607.4.1 Exterior Standard-Unit Panels

The maximum area of each individual standard-unit panel shall be 144 square feet (13.4 m²) where the design wind pressure is 20 pounds per square foot (958 Pa). The maximum area of such panels subjected to design wind pressures other than 20 pounds per square foot (958 Pa) shall be in accordance with Figure R607.4.1. The maximum panel dimension between structural supports shall be 25 feet (7620 mm) in width or 20 feet (6096 mm) in height.



For SI: 1 square foot = 0.0929 m², 1 pound per square foot = 0.0479 kPa.

FIGURE R607.4.1
GLASS UNIT MASONRY DESIGN WIND LOAD RESISTANCE

R607.4.2 Exterior Thin-Unit Panels

The maximum area of each individual thin-unit panel shall be 85 square feet (7.9 m²). The maximum dimension between structural supports shall be 15 feet (4572 mm) in width or 10 feet (3048 mm) in height. Thin units shall not be used in applications where the

design wind pressure as stated in Table R301.2(1) exceeds 20 pounds per square foot (958 Pa).

R607.4.3 Interior Panels

The maximum area of each individual standard-unit panel shall be 250 square feet (23.2 m²). The maximum area of each thin-unit panel shall be 150 square feet (13.9 m²). The maximum dimension between structural supports shall be 25 feet (7620 mm) in width or 20 feet (6096 mm) in height.

R607.4.4 Curved Panels

The width of curved panels shall conform to the requirements of Sections R607.4.1, R607.4.2 and R607.4.3, except additional structural supports shall be provided at locations where a curved section joins a straight section, and at inflection points in multicurved walls.

R607.5 Panel Support

Glass unit masonry panels shall conform to the support requirements of this section.

R607.5.1 Deflection

The maximum total deflection of structural members that support glass unit masonry shall not exceed $\frac{1}{600}$.

R607.5.2 Lateral Support

Glass unit masonry panels shall be laterally supported along the top and sides of the panel. Lateral supports for glass unit masonry panels shall be designed to resist not less than 200 pounds per lineal feet (2918 N/m) of panel, or the actual applied loads, whichever is greater. Except for single unit panels, lateral support shall be provided by panel anchors along the top and sides spaced not greater than 16 inches (406 mm) on center or by channel-type restraints. Single unit panels shall be supported by channel-type restraints.

Exceptions:

1. Lateral support is not required at the top of panels that are one unit wide.
2. Lateral support is not required at the sides of panels that are one unit high.

R607.5.2.1 Panel Anchor Restraints

Panel anchors shall be spaced not greater than 16 inches (406 mm) on center in both jambs and across the head. Panel anchors shall be embedded not less than 12 inches (305 mm) and shall be provided with two fasteners so as to resist the loads specified in Section R607.5.2.

R607.5.2.2 Channel-Type Restraints

Glass unit masonry panels shall be recessed not less than 1 inch (25 mm) within channels and chases. Channel-type restraints shall be oversized to accommodate expansion material in the opening, packing and sealant between the framing restraints, and the glass unit masonry perimeter units.

R607.6 Sills

Before bedding of glass units, the sill area shall be covered with a water base asphaltic emulsion coating. The coating shall be not less than $\frac{1}{8}$ inch (3 mm) thick.

R607.7 Expansion Joints

Glass unit masonry panels shall be provided with expansion joints along the top and sides at all structural supports. Expansion joints shall be not less than $\frac{3}{8}$ inch (10 mm) in thickness and shall have sufficient thickness to accommodate displacements of the supporting structure. Expansion joints shall be entirely free of mortar and other debris and shall be filled with resilient material.

R607.8 Mortar

Glass unit masonry shall be laid with Type S or N mortar. Mortar shall not be retempered after initial set. Mortar unused within $1\frac{1}{2}$ hours after initial mixing shall be discarded.

R607.9 Reinforcement

Glass unit masonry panels shall have horizontal joint reinforcement spaced not greater than 16 inches (406 mm) on center located in the mortar bed joint. Horizontal joint reinforcement shall extend the entire length of the panel but shall not extend across expansion joints. Longitudinal wires shall be lapped not less than 6 inches (152 mm) at splices. Joint reinforcement shall be placed in the bed joint

immediately below and above openings in the panel. The reinforcement shall have not less than two parallel longitudinal wires of size W1.7 or greater, and have welded cross wires of size W1.7 or greater.

R607.10 Placement

Glass units shall be placed so head and bed joints are filled solidly. Mortar shall not be furrowed. Head and bed joints of glass unit masonry shall be 1/4 inch (6.4 mm) thick, except that vertical joint thickness of radial panels shall be not less than 1/8 inch (3 mm) or greater than 5/8 inch (16 mm). The bed joint thickness tolerance shall be minus 1/16 inch (1.6 mm) and plus 1/8 inch (3 mm). The head joint thickness tolerance shall be plus or minus 1/8 inch (3 mm).

Section R608 Exterior Concrete Wall Construction

R608.1 General

Exterior concrete walls shall be designed and constructed in accordance with the provisions of this section or in accordance with the provisions of PCA 100, ACI 318 or ACI 332. Where PCA 100, ACI 318, ACI 332 or the provisions of this section are used to design concrete walls, project drawings, typical details and specifications are not required to bear the seal of the architect or engineer responsible for design, unless otherwise required by *Florida Statutes*.

R608.1.1 Interior Construction

These provisions are based on the assumption that interior walls and partitions, both load-bearing and nonload-bearing, floors and roof/ceiling assemblies are constructed of *light-framed construction* complying with the limitations of this code and the additional limitations of Section R608.2. Design and construction of light-framed assemblies shall be in accordance with the applicable provisions of this code. Where second-story exterior walls are of *light-framed construction*, they shall be designed and constructed as required by this code.

Aspects of concrete construction not specifically addressed by this code, including interior concrete walls, shall comply with ACI 318.

R608.1.2 Other Concrete Walls

Exterior concrete walls constructed in accordance with this code shall comply with the shapes and minimum concrete cross-sectional dimensions of Table R608.3. Other types of forming systems resulting in concrete walls not in compliance with this section shall be designed in accordance with ACI 318.

R608.2 Applicability Limits

The provisions of this section shall apply to the construction of exterior concrete walls for buildings not greater than 60 feet (18 288 mm) in plan dimensions, floors with clear spans not greater than 32 feet (9754 mm) and roofs with clear spans not greater than 40 feet (12 192 mm). Buildings shall not exceed 35 feet (10 668 mm) in mean roof height or two stories in height above grade. Floor/ceiling dead loads shall not exceed 10 pounds per square foot (479 Pa), roof/ceiling dead loads shall not exceed 15 pounds per square foot (718 Pa) and *attic* live loads shall not exceed 20 pounds per square foot (958 Pa). Roof overhangs shall not exceed 2 feet (610 mm) of horizontal projection beyond the exterior wall and the dead load of the overhangs shall not exceed 8 pounds per square foot (383 Pa).

Walls constructed in accordance with the provisions of this section shall be limited to buildings subjected to a maximum design wind speed of 160 mph (72 m/s) Exposure B, 136 mph (61 m/s) Exposure C and 125 mph (56 m/s) Exposure D.

Buildings that are not within the scope of this section shall be designed in accordance with PCA 100 or ACI 318.

R608.3 Concrete Wall Systems

Concrete walls constructed in accordance with these provisions shall comply with the shapes and minimum concrete cross-sectional dimensions of Table R608.3.

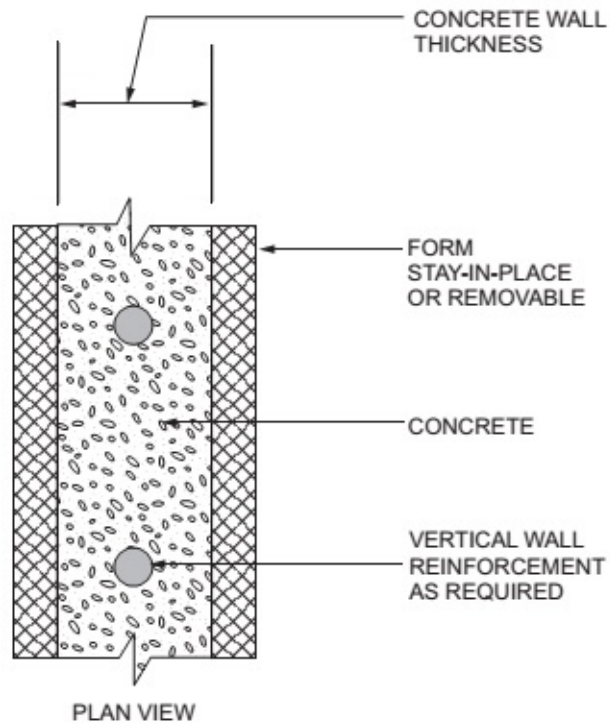
TABLE R608.3
DIMENSIONAL REQUIREMENTS FOR WALLS^{a, b}

WALL TYPE AND NOMINAL THICKNESS	MAXIMUM WALL WEIGHT ^c (psf)	MINIMUM WIDTH, W, OF VERTICAL CORES (inches)	MINIMUM THICKNESS, T, OF VERTICAL CORES (inches)	MAXIMUM SPACING OF VERTICAL CORES (inches)	MAXIMUM SPACING OF HORIZONTAL CORES (inches)	MINIMUM WEB THICKNESS (inches)
4" Flat ^d	50	N/A	N/A	N/A	N/A	N/A

6" Flat ^d	75	N/A	N/A	N/A	N/A	N/A
8" Flat ^d	100	N/A	N/A	N/A	N/A	N/A
10" Flat ^d	125	N/A	N/A	N/A	N/A	N/A
6" Waffle-grid	56	8 ^e	5.5 ^e	12	16	2
8" Waffle-grid	76	8 ^f	8 ^f	12	16	2
6" Screen-grid	53	6.25 ^g	6.25 ^g	12	12	N/A

For SI: 1 inch = 25.4 mm; 1 pound per square foot = 0.0479 kPa, 1 pound per cubic foot = 2402.77 kg/m³, 1 square inch = 645.16 mm², 1 inch⁴ = 42 cm⁴.

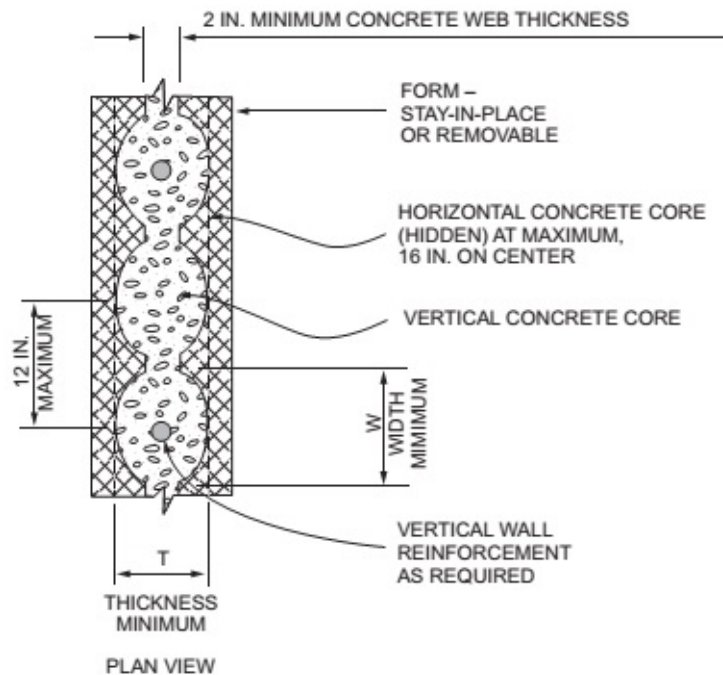
- a. Width "W," thickness "T," spacing and web thickness, refer to Figures R608.3(2) and R608.3(3).
- b. N/A indicates not applicable.
- c. Wall weight is based on a unit weight of concrete of 150 pcf. For flat walls the weight is based on the nominal thickness. The tabulated values do not include any allowance for interior and exterior finishes.
- d. Nominal wall thickness. The actual as-built thickness of a flat wall shall not be more than 1/2 inch less or more than 1/4 inch more than the nominal dimension indicated.
- e. Vertical core is assumed to be elliptical-shaped. Another shape core is permitted provided the minimum thickness is 5 inches, the moment of inertia, *I*, about the centerline of the wall (ignoring the web) is not less than 65 inch⁴, and the area, *A*, is not less than 31.25 square inches. The width used to calculate *A* and *I* shall not exceed 8 inches.
- f. Vertical core is assumed to be circular. Another shape core is permitted provided the minimum thickness is 7 inches, the moment of inertia, *I*, about the centerline of the wall (ignoring the web) is not less than 200 inch⁴, and the area, *A*, is not less than 49 square inches. The width used to calculate *A* and *I* shall not exceed 8 inches.
- g. Vertical core is assumed to be circular. Another shape core is permitted provided the minimum thickness is 5.5 inches, the moment of inertia, *I*, about the centerline of the wall is not less than 76 inch⁴, and the area, *A*, is not less than 30.25 square inches. The width used to calculate *A* and *I* shall not exceed 6.25 inches.



SEE TABLE 608.3 FOR MINIMUM DIMENSIONS

FIGURE R608.3(1)

FLAT WALL SYSTEM

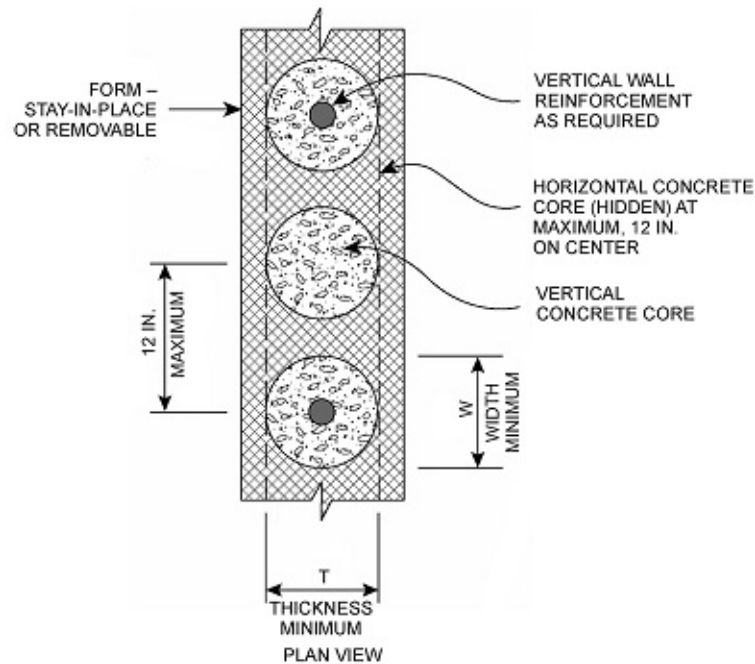


SEE TABLE R608.3 FOR MINIMUM DIMENSIONS

For SI: 1 inch = 25.4 mm.

FIGURE R608.3(2)

WAFFLE-GRID WALL SYSTEM



SEE TABLE R608.3 FOR MINIMUM DIMENSIONS

For SI: 1 inch = 25.4 mm.

FIGURE R608.3(3)

SCREEN-GRID WALL SYSTEM

R608.3.1 Flat Wall Systems

Flat concrete wall systems shall comply with Table R608.3 and Figure R608.3(1) and have a minimum nominal thickness of 4 inches (102 mm).

R608.3.2 Waffle-Grid Wall Systems

Waffle-grid wall systems shall comply with Table R608.3 and Figure R608.3(2) and shall have a minimum nominal thickness of 6 inches (152 mm) for the horizontal and vertical concrete members (cores). The core and web dimensions shall comply with Table R608.3. The maximum weight of waffle-grid walls shall comply with Table R608.3.

R608.3.3 Screen-Grid Wall Systems

Screen-grid wall systems shall comply with Table R608.3 and Figure R608.3(3) and shall have a minimum nominal thickness of 6 inches (152 mm) for the horizontal and vertical concrete members (cores). The core dimensions shall comply with Table R608.3. The maximum weight of screen-grid walls shall comply with Table R608.3.

R608.4 Stay-in-Place Forms

Stay-in-place concrete forms shall comply with this section.

R608.4.1 Surface Burning Characteristics

The flame spread index and smoke-developed index of forming material, other than foam plastic, left exposed on the interior shall comply with Section R302.9. The surface burning characteristics of foam plastic used in insulating concrete forms shall comply with Section R316.3.

R608.4.2 Interior Covering

Stay-in-place forms constructed of rigid foam plastic shall be protected on the interior of the building as required by Sections R316.4 and R702.3.4. Where gypsum board is used to protect the foam plastic, it shall be installed with a mechanical fastening system. Use of adhesives is permitted in addition to mechanical fasteners.

R608.4.3 Exterior Wall Covering

Stay-in-place forms constructed of rigid foam plastics shall be protected from sunlight and physical damage by the application of an *approved* exterior wall covering complying with this code. Exterior surfaces of other stay-in-place forming systems shall be protected

in accordance with this code.

Requirements for installation of masonry veneer, stucco and other finishes on the exterior of concrete walls and other construction details not covered in this section shall comply with the requirements of this code.

R608.4.4 Flat ICF Wall Systems

Flat ICF wall system forms shall conform to ASTM E2634.

R608.5 Materials

Materials used in the construction of concrete walls shall comply with this section.

R608.5.1 Concrete and Materials for Concrete

Materials used in concrete, and the concrete itself, shall conform to requirements of this section, PCA 100, ACI 318 or ACI 332.

R608.5.1.1 Cements

The following standards as referenced in Chapter 46 shall be permitted to be used.

1. ASTM C150
2. ASTM C595
3. ASTM C1157

R608.5.1.2 Concrete Mixing and Delivery

Mixing and delivery of concrete shall comply with ASTM C94 or ASTM C685.

R608.5.1.3 Maximum Aggregate Size

The nominal maximum size of coarse aggregate shall not exceed one-fifth the narrowest distance between sides of forms, or three-fourths the clear spacing between reinforcing bars or between a bar and the side of the form.

Exception: When *approved*, these limitations shall not apply where removable forms are used and workability and methods of consolidation permit concrete to be placed without honeycombs or voids.

R608.5.1.4 Proportioning and Slump of Concrete

Proportions of materials for concrete shall be established to provide workability and consistency to permit concrete to be worked readily into forms and around reinforcement under conditions of placement to be employed, without segregation or excessive bleeding. Slump of concrete placed in removable forms shall not exceed 6 inches (152 mm).

Exception: When *approved*, the slump is permitted to exceed 6 inches (152 mm) for concrete mixtures that are resistant to segregation, and are in accordance with the form manufacturer's recommendations.

Slump of concrete placed in stay-in-place forms shall exceed 6 inches (152 mm). Slump of concrete shall be determined in accordance with ASTM C143.

R608.5.1.5 Compressive Strength

The minimum specified compressive strength of concrete, f'_c , shall comply with Section R402.2 and shall be not less than 2,500 pounds per square inch (17.2 MPa) at 28 days.

R608.5.1.6 Consolidation of Concrete

Concrete shall be consolidated by suitable means during placement and shall be worked around embedded items and reinforcement and into corners of forms. Where stay-in-place forms are used, concrete shall be consolidated by internal vibration.

Exception: When *approved*, self-consolidating concrete mixtures with slumps equal to or greater than 8 inches (203 mm) that are specifically designed for placement without internal vibration need not be internally vibrated.

R608.5.2 Steel Reinforcement and Anchor Bolts

R608.5.2.1 Steel Reinforcement

Steel reinforcement shall comply with ASTM A615, ASTM A706, or ASTM A996. ASTM A996 bars produced from rail steel shall be Type R.

R608.5.2.2 Anchor Bolts

Anchor bolts for use with connection details in accordance with Figures R608.9(1) through R608.9(12) shall be bolts with heads complying with ASTM A307 or ASTM F1554. ASTM A307 bolts shall be Grade A with heads. ASTM F1554 bolts shall be Grade 36 minimum. Instead of bolts with heads, it is permissible to use rods with threads on both ends fabricated from steel complying with ASTM A36. The threaded end of the rod to be embedded in the concrete shall be provided with a hex or square nut.

R608.5.2.3 Sheet Steel Angles and Tension Tie Straps

Angles and tension tie straps for use with connection details in accordance with Figures R608.9(1) through R608.9(12) shall be fabricated from sheet steel complying with ASTM A653 SS, ASTM A792 SS, or ASTM A875 SS. The steel shall be minimum Grade 33 unless a higher grade is required by the applicable figure.

R608.5.3 Form Materials and Form Ties

Forms shall be made of wood, steel, aluminum, plastic, a composite of cement and foam insulation, a composite of cement and wood chips, or other *approved* material suitable for supporting and containing concrete. Forms shall provide sufficient strength to contain concrete during the concrete placement operation.

Form ties shall be steel, solid plastic, foam plastic, a composite of cement and wood chips, a composite of cement and foam plastic, or other suitable material capable of resisting the forces created by fluid pressure of fresh concrete.

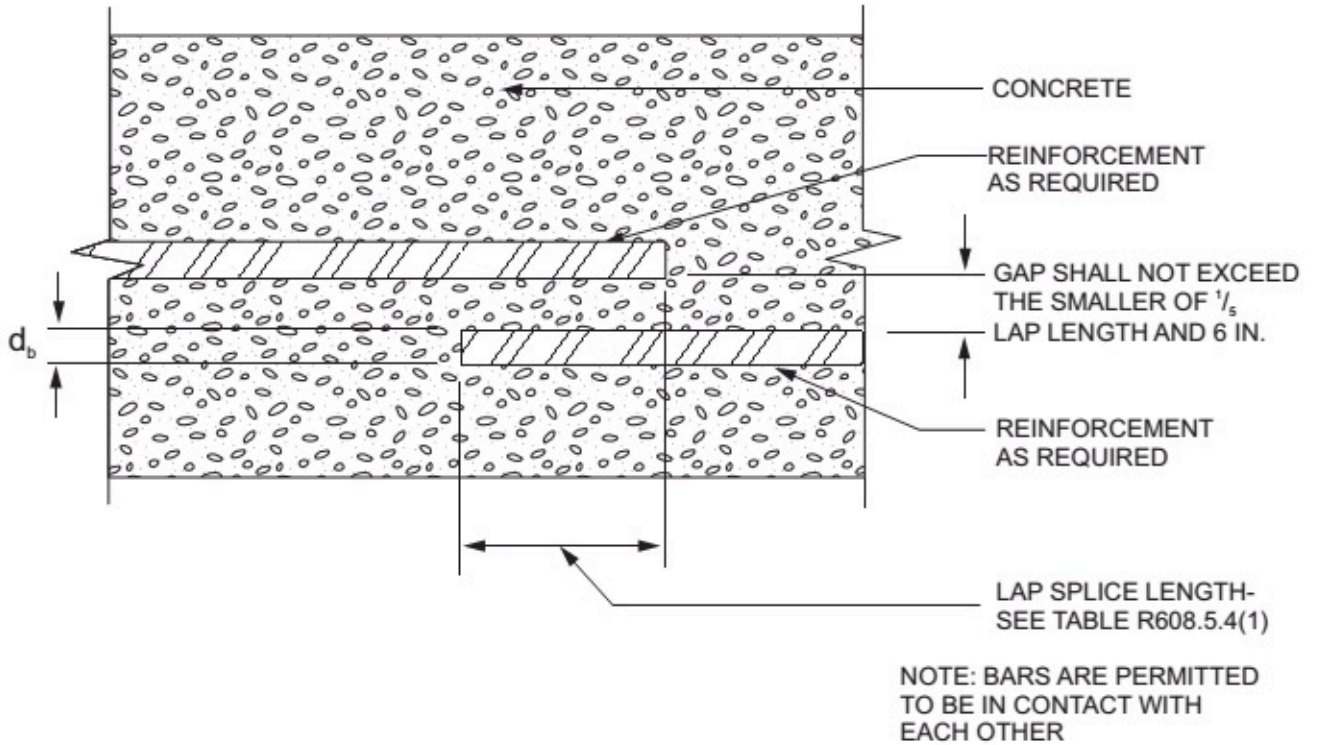
R608.5.4 Reinforcement Installation Details

TABLE R608.5.4(1)
LAP SPLICE AND TENSION DEVELOPMENT LENGTHS

	BAR SIZE NO.	YIELD STRENGTH OF STEEL, f_y - psi (MPa)	
		40,000 (280)	60,000 (420)
		Splice length or tension development length (inches)	
Lap splice length-tension	4	20	30
	5	25	38
	6	30	45
Tension development length for straight bar	4	15	23
	5	19	28
	6	23	34
Tension development length for: a. 90-degree and 180-degree standard hooks with not less than 2 ¹ / ₂ inches of side cover perpendicular to plane of hook, and b. 90-degree standard hooks with not less than 2 inches of cover on the bar extension beyond the hook.	4	6	9
	5	7	11
	6	8	13
Tension development length for bar with 90-degree or 180-degree standard hook having less cover than required above.	4	8	12
	5	10	15

	6	12	18
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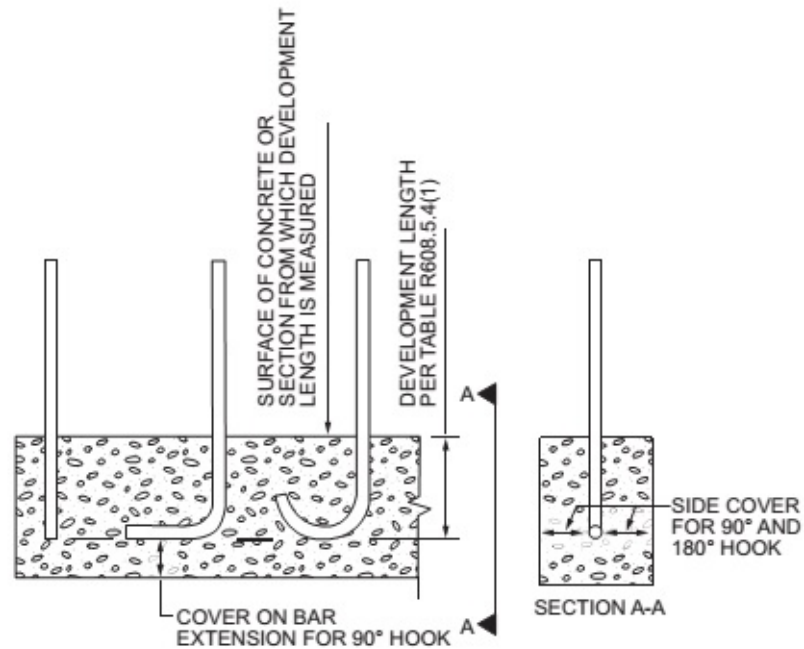
For SI: 1 inch = 25.4 mm.



For SI: 1 inch = 25.4 mm.

FIGURE R608.5.4(1)

LAP SPLICES



For SI: 1 degree = 0.0175 rad.

FIGURE R608.5.4(2)

DEVELOPMENT LENGTH AND COVER FOR HOOKS AND BAR EXTENSION

TABLE R608.5.4(2)

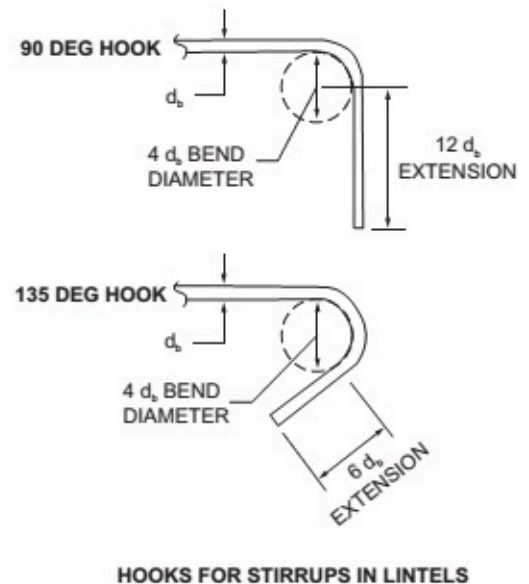
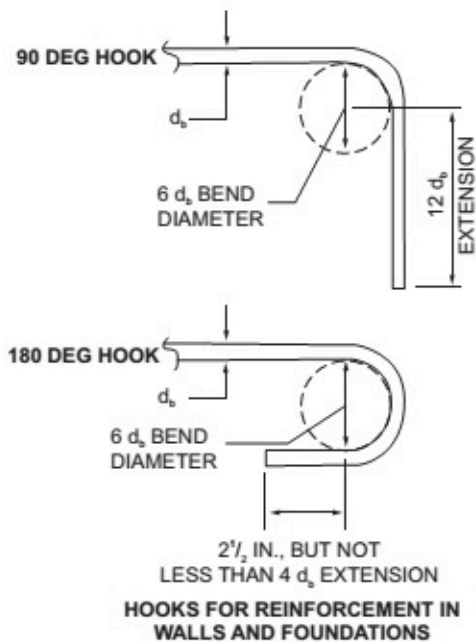
MAXIMUM SPACING FOR ALTERNATIVE BAR SIZE AND/OR ALTERNATIVE GRADE OF STEEL^{a, b, c}

BAR SPACING FROM APPLICABLE TABLE IN SECTION R608.6 (inches)	BAR SIZE FROM APPLICABLE TABLE IN SECTION R608.6														
	#4					#5					#6				
	Alternate bar size and/or alternate grade of steel desired														
	Grade 60		Grade 40			Grade 60		Grade 40			Grade 60		Grade 40		
	#5	#6	#4	#5	#6	#4	#6	#4	#5	#6	#4	#5	#4	#5	#6
	Maximum spacing for alternate bar size and/or alternate grade of steel (inches)														
8	12	18	5	8	12	5	11	3	5	8	4	6	2	4	5
9	14	20	6	9	13	6	13	4	6	9	4	6	3	4	6
10	16	22	7	10	15	6	14	4	7	9	5	7	3	5	7
11	17	24	7	11	16	7	16	5	7	10	5	8	3	5	7
12	19	26	8	12	18	8	17	5	8	11	5	8	4	6	8
13	20	29	9	13	19	8	18	6	9	12	6	9	4	6	9
14	22	31	9	14	21	9	20	6	9	13	6	10	4	7	9
15	23	33	10	16	22	10	21	6	10	14	7	11	5	7	10
16	25	35	11	17	23	10	23	7	11	15	7	11	5	8	11
17	26	37	11	18	25	11	24	7	11	16	8	12	5	8	11
18	28	40	12	19	26	12	26	8	12	17	8	13	5	8	12
19	29	42	13	20	28	12	27	8	13	18	9	13	6	9	13
20	31	44	13	21	29	13	28	9	13	19	9	14	6	9	13
21	33	46	14	22	31	14	30	9	14	20	10	15	6	10	14
22	34	48	15	23	32	14	31	9	15	21	10	16	7	10	15
23	36	48	15	24	34	15	33	10	15	22	10	16	7	11	15
24	37	48	16	25	35	15	34	10	16	23	11	17	7	11	16
25	39	48	17	26	37	16	35	11	17	24	11	18	8	12	17
26	40	48	17	27	38	17	37	11	17	25	12	18	8	12	17
27	42	48	18	28	40	17	38	12	18	26	12	19	8	13	18
28	43	48	19	29	41	18	40	12	19	26	13	20	8	13	19
29	45	48	19	30	43	19	41	12	19	27	13	20	9	14	19
30	47	48	20	31	44	19	43	13	20	28	14	21	9	14	20
31	48	48	21	32	45	20	44	13	21	29	14	22	9	15	21
32	48	48	21	33	47	21	45	14	21	30	15	23	10	15	21
33	48	48	22	34	48	21	47	14	22	31	15	23	10	16	22

34	48	48	23	35	48	22	48	15	23	32	15	24	10	16	23
35	48	48	23	36	48	23	48	15	23	33	16	25	11	16	23
36	48	48	24	37	48	23	48	15	24	34	16	25	11	17	24
37	48	48	25	38	48	24	48	16	25	35	17	26	11	17	25
38	48	48	25	39	48	25	48	16	25	36	17	27	12	18	25
39	48	48	26	40	48	25	48	17	26	37	18	27	12	18	26
40	48	48	27	41	48	26	48	17	27	38	18	28	12	19	27
41	48	48	27	42	48	26	48	18	27	39	19	29	12	19	27
42	48	48	28	43	48	27	48	18	28	40	19	30	13	20	28
43	48	48	29	44	48	28	48	18	29	41	20	30	13	20	29
44	48	48	29	45	48	28	48	19	29	42	20	31	13	21	29
45	48	48	30	47	48	29	48	19	30	43	20	32	14	21	30
46	48	48	31	48	48	30	48	20	31	44	21	32	14	22	31
47	48	48	31	48	48	30	48	20	31	44	21	33	14	22	31
48	48	48	32	48	48	31	48	21	32	45	22	34	15	23	32

For SI: 1 inch = 25.4 mm.

- This table is for use with tables in Section R608.6 that specify the minimum bar size and maximum spacing of vertical wall reinforcement for foundation walls and above-grade walls. Reinforcement specified in tables in Section R608.6 is based on Grade 60 (420 MPa) steel reinforcement.
- Bar spacing shall not exceed 48 inches on center and shall be not less than one-half the nominal wall thickness.
- For Grade 50 (350 MPa) steel bars (ASTM A996, Type R), use spacing for Grade 40 (280 MPa) bars or interpolate between Grade 40 (280 MPa) and Grade 60 (420 MPa).



For SI: 1 inch = 25.4 mm, 1 degree = 0.0175 rad.

FIGURE R608.5.4(3)

STANDARD HOOKS

R608.5.4.1 Support and Cover

Reinforcement shall be secured in the proper location in the forms with tie wire or other bar support system such that displacement will not occur during the concrete placement operation. Steel reinforcement in concrete cast against the earth shall have a minimum cover of 3 inches (76 mm). Minimum cover for reinforcement in concrete cast in removable forms that will be exposed to the earth or weather shall be $1\frac{1}{2}$ inches (38 mm) for No. 5 bars and smaller, and 2 inches (50 mm) for No. 6 bars and larger. For concrete cast in removable forms that will not be exposed to the earth or weather, and for concrete cast in stay-in-place forms, minimum cover shall be $\frac{3}{4}$ inch (19 mm). The minus tolerance for cover shall not exceed the smaller of one-third the required cover and $\frac{3}{8}$ inch (10 mm). See Section R608.5.4.4 for cover requirements for hooks of bars developed in tension.

R608.5.4.2 Location of Reinforcement in Walls

For location of reinforcement in foundation walls and above-grade walls, see Sections R404.1.3.3.7.2 and R608.6.5, respectively.

R608.5.4.3 Lap Splices

Vertical and horizontal wall reinforcement required by Sections R608.6 and R608.7 shall be the longest lengths practical. Where splices are necessary in reinforcement, the length of lap splices shall be in accordance with Table R608.5.4(1) and Figure R608.5.4(1). The maximum gap between noncontact parallel bars at a lap splice shall not exceed the smaller of one-fifth the required lap length and 6 inches (152 mm). See Figure R608.5.4(1).

R608.5.4.4 Development of Bars in Tension

Where bars are required to be developed in tension by other provisions of this code, development lengths and cover for hooks and bar extensions shall comply with Table R608.5.4(1) and Figure R608.5.4(2). The development lengths shown in Table R608.5.4(1) shall apply to bundled bars in lintels installed in accordance with Section R608.8.2.2.

R608.5.4.5 Standard Hooks

Where reinforcement is required by this code to terminate with a standard hook, the hook shall comply with Figure R608.5.4(3).

R608.5.4.6 Webs of Waffle-Grid Walls

Reinforcement, including stirrups, shall not be placed in webs of waffle-grid walls, including lintels. Webs are permitted to have form ties.

R608.5.4.7 Alternate Grade of Reinforcement and Spacing

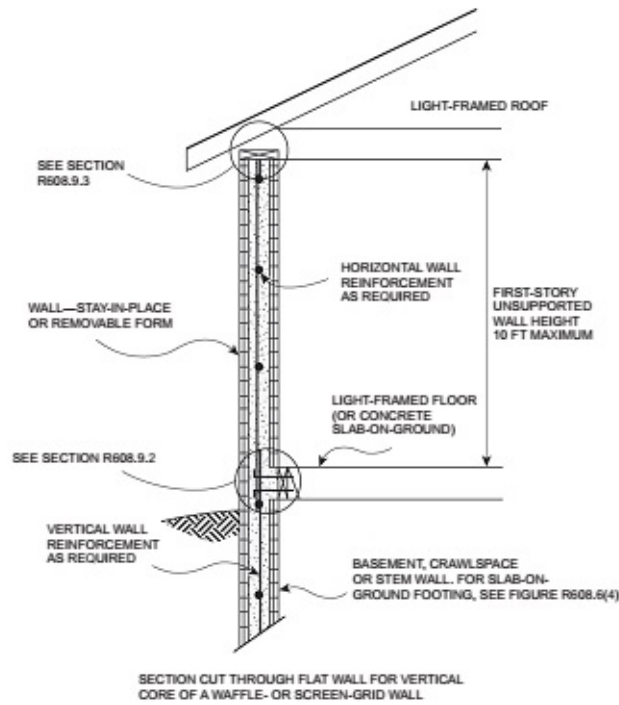
Where tables in Sections R404.1.3 and R608.6 specify vertical wall reinforcement based on minimum bar size and maximum spacing, which are based on Grade 60 (420 MPa) steel reinforcement, different size bars or bars made from a different grade of steel are permitted provided an equivalent area of steel per linear foot of wall is provided. Use of Table R608.5.4(2) is permitted to determine the maximum bar spacing for different bar sizes than specified in the tables and/or bars made from a different grade of steel. Bars shall not be spaced less than one-half the wall thickness, or more than 48 inches (1219 mm) on center.

R608.5.5 Construction Joints in Walls

Construction joints shall be made and located to not impair the strength of the wall. Construction joints in plain concrete walls, including walls required to have not less than No. 4 bars at 48 inches (1219 mm) on center by Section R608.6, shall be located at points of lateral support, and not less than one No. 4 bar shall extend across the construction joint at a spacing not to exceed 24 inches (610 mm) on center. Construction joint reinforcement shall have not less than 12 inches (305 mm) embedment on both sides of the joint. Construction joints in reinforced concrete walls shall be located in the middle third of the span between lateral supports, or located and constructed as required for joints in plain concrete walls.

Exception: Vertical wall reinforcement required by this code is permitted to be used in lieu of construction joint reinforcement, provided the spacing does not exceed 24 inches (610 mm), or the combination of wall reinforcement and No. 4 bars described in Section R608.5.5 does not exceed 24 inches (610 mm).

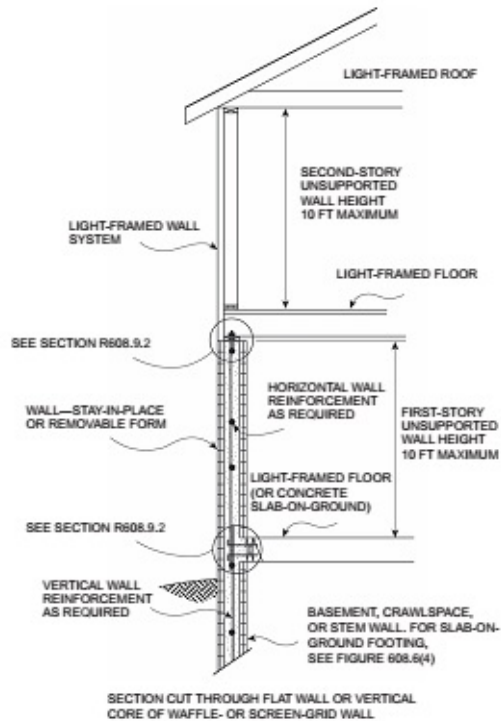
R608.6 Above-Grade Wall Requirements



For SI: 1 foot = 304.8 mm.

FIGURE R608.6(1)

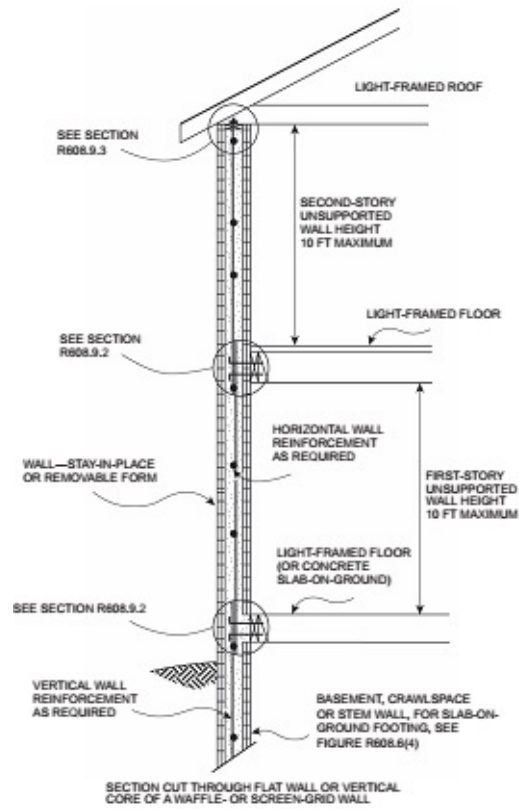
ABOVE-GRADE CONCRETE WALL CONSTRUCTION ONE STORY



For SI: 1 foot = 304.8 mm.

FIGURE R608.6(2)

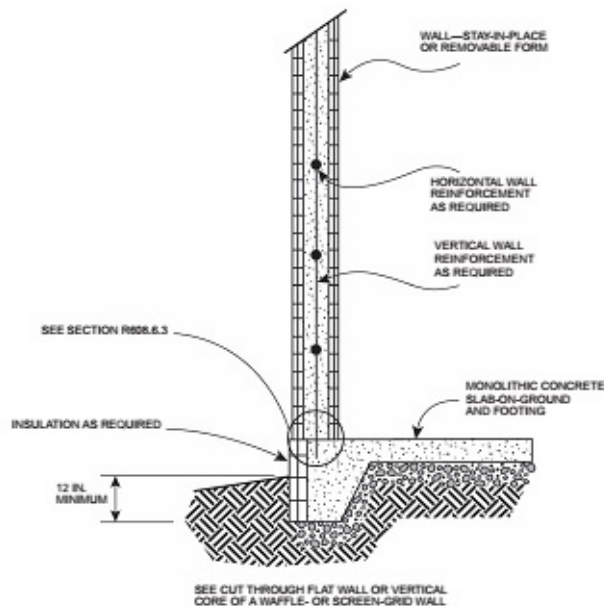
ABOVE-GRADE CONCRETE WALL CONSTRUCTION CONCRETE FIRST STORY AND LIGHT-FRAMED SECOND STORY



For SI: 1 foot = 304.8 mm.

FIGURE R608.6(3)

ABOVE-GRADE CONCRETE WALL CONSTRUCTION TWO-STORY



For SI: 1 inch = 25.4 mm.

FIGURE R608.6(4)

ABOVE-GRADE CONCRETE WALL SUPPORTED ON MONOLITHIC SLAB-ON-GROUND FOOTING

TABLE R608.6(1)

MINIMUM VERTICAL REINFORCEMENT FOR FLAT ABOVE-GRADE WALLS^{a, b, c, d, e}

MAXIMUM WIND SPEED SPEED (mph)			MAXIMUM UNSUPPORTED WALL HEIGHT PER STORY (feet)	MINIMUM VERTICAL REINFORCEMENT-BAR SIZE AND SPACING (inches) ^f g							
				Nominal ^h wall thickness (inches)							
Exposure Category				4		6		8		10	
				Top ⁱ	Side ⁱ	Top ⁱ	Side ⁱ	Top ⁱ	Side ⁱ	Top ⁱ	Side ⁱ
115			8	4@48	4@48	4@48	4@48	4@48	4@48	4@48	4@48
			9	4@48	4@39	4@48	4@48	4@48	4@48	4@48	4@48
			10	4@41	4@34	4@48	4@48	4@48	4@48	4@48	4@48
120			8	4@48	4@43	4@48	4@48	4@48	4@48	4@48	4@48
			9	4@48	4@36	4@48	4@48	4@48	4@48	4@48	4@48
			10	4@37	4@34	4@48	4@48	4@48	4@48	4@48	4@48
130	110		8	4@48	4@38	4@48	4@48	4@48	4@48	4@48	4@48
			9	4@39	4@34	4@48	4@48	4@48	4@48	4@48	4@48
			10	4@34	4@34	4@48	4@48	4@48	4@48	4@48	4@48
140	119	110	8	4@43	4@34	4@48	4@48	4@48	4@48	4@48	4@48
			9	4@34	4@34	4@48	4@48	4@48	4@48	4@48	4@48
			10	4@34	4@31	4@48	4@48	4@48	4@48	4@48	4@48
150	127	117	8	4@37	4@34	4@48	4@48	4@48	4@48	4@48	4@48
			9	4@34	4@33	4@48	4@48	4@48	4@48	4@48	4@48
			10	4@31	4@27	4@48	4@48	4@48	4@48	4@48	4@48
160	136	125	8	4@34	4@34	4@48	4@48	4@48	4@48	4@48	4@48
			9	4@34	4@29	4@48	4@48	4@48	4@48	4@48	4@48
			10	4@27	4@24	4@48	4@48	4@48	4@48	4@48	4@48

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s, 1 pound per square inch = 1.895 kPa, 1 square foot = 0.0929 m².

- Table is based on ASCE 7 components and cladding wind pressures for an enclosed building using a mean roof height of 35 feet, interior wall area 4, an effective wind area of 10 square feet, topographic factor, K_{zt} , equal to 1.0, and Risk Category II.
- Table is based on concrete with a minimum specified compressive strength of 2,500 psi.
- See Section R608.6.5 for location of reinforcement in wall.
- Deflection criterion is $L/240$, where L is the unsupported height of the wall in inches.
- Interpolation is not permitted.
- Where No. 4 reinforcing bars at a spacing of 48 inches are specified in the table as indicated by shaded cells, use of bars with a minimum yield strength of 40,000 psi or 60,000 psi is permitted.
- Other than for No. 4 bars spaced at 48 inches on center, table values are based on reinforcing bars with a minimum yield strength of 60,000 psi. Vertical reinforcement with a yield strength of less than 60,000 psi or bars of a different size than

specified in the table are permitted in accordance with Section R608.5.4.7 and Table R608.5.4(2).

h. See Table R608.3 for tolerances on nominal thicknesses.

i. "Top" means gravity load from roof or floor construction bears on top of wall. "Side" means gravity load from floor construction is transferred to wall from a wood ledger or cold-formed steel track bolted to side of wall. For nonload-bearing walls where floor framing members span parallel to the wall, use of the "Top" bearing condition is permitted.

TABLE R608.6(2)

MINIMUM VERTICAL REINFORCEMENT FOR WAFFLE-GRID ABOVE-GRADE WALLS^{a, b, c, d, e}

MAXIMUM WIND SPEED (mph)			MAXIMUM UNSUPPORTED WALL HEIGHT PER STORY (feet)	MINIMUM VERTICAL REINFORCEMENT-BAR SIZE AND SPACING (inches) ^{f, g}			
				Nominal ^h wall thickness (inches)			
Exposure Category				6		8	
				Top ⁱ	Side ⁱ	Top ⁱ	Side ⁱ
115			8	4@48	4@48	4@48	4@48
			9	4@48	5@43	4@48	4@48
			10	5@47	5@37	4@48	4@48
120			8	4@48	5@48	4@48	4@48
			9	4@48	5@40	4@48	4@48
			10	5@43	5@37	4@48	4@48
130	110		8	4@48	5@42	4@48	4@48
			9	5@45	5@37	4@48	4@48
			10	5@37	5@37	4@48	4@48
140	119	110	8	4@48	5@38	4@48	4@48
			9	5@39	5@37	4@48	4@48
			10	5@37	5@35	4@48	4@48
150	127	117	8	5@43	5@37	4@48	4@48
			9	5@37	5@37	4@48	4@48
			10	5@36	6@44	4@48	4@48
160	136	125	8	5@38	5@37	4@48	4@48
			9	5@37	6@47	4@48	4@48
			10	6@45	6@39	4@48	6@46

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s, 1 pound per square inch = 6.895 kPa, 1 square foot = 0.0929 m².

a. Table is based on ASCE 7 components and cladding wind pressures for an enclosed building using a mean roof height of 35 feet, interior wall area 4, an effective wind area of 10 square feet, topographic factor, K_{zt} , equal to 1.0, and Risk Category II.

- b. Table is based on concrete with a minimum specified compressive strength of 2,500 psi.
- c. See Section R608.6.5 for location of reinforcement in wall.
- d. Deflection criterion is $L/240$, where L is the unsupported height of the wall in inches.
- e. Interpolation is not permitted.
- f. Where No. 4 reinforcing bars at a spacing of 48 inches are specified in the table as indicated by shaded cells, use of bars with a minimum yield strength of 40,000 psi or 60,000 psi is permitted.
- g. Other than for No. 4 bars spaced at 48 inches on center, table values are based on reinforcing bars with a minimum yield strength of 60,000 psi. Maximum spacings shown are the values calculated for the specified bar size. Where the bar used is Grade 60 and the size specified in the table, the actual spacing in the wall shall not exceed a whole-number multiple of 12 inches such as, 12, 24, 36 and 48, that is less than or equal to the tabulated spacing. Vertical reinforcement with a yield strength of less than 60,000 psi or bars of a different size than specified in the table are permitted in accordance with Section R608.5.4.7 and Table R608.5.4(2).
- h. See Table R608.3 for minimum core dimensions and maximum spacing of horizontal and vertical cores.
- i. "Top" means gravity load from roof or floor construction bears on top of wall. "Side" means gravity load from floor construction is transferred to wall from a wood ledger or cold-formed steel track bolted to side of wall. For nonload-bearing walls and where floor framing members span parallel to the wall, the "top" bearing condition is permitted to be used.

TABLE R608.6(3)

MINIMUM VERTICAL REINFORCEMENT FOR 6-INCH SCREEN-GRID ABOVE-GRADE WALLS^{a, b, c, d, e}

MAXIMUM WIND SPEED (mph)			MAXIMUM UNSUPPORTED WALL HEIGHT PER STORY (feet)	MINIMUM VERTICAL REINFORCEMENT-BAR SIZE AND SPACING (inches) ^{f, g}	
				Nominal ^h wall thickness (inches)	
Exposure Category				6	
				Top ⁱ	Side ⁱ
115			8	4@48	4@48
			9	4@48	5@41
			10	4@48	6@48
120			8	4@48	4@48
			9	4@48	5@38
			10	5@42	6@48
130	110		8	4@48	5@41
			9	5@44	6@48
			10	5@35	6@48
140	119	110	8	4@48	5@36
			9	5@38	6@48
			10	6@48	6@48
150	127	117	8	5@42	6@48

			9	6@48	6@48
			10	6@48	6@42
160	136	125	8	5@37	6@48
			9	6@48	6@45
			10	6@44	6@38

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s, 1 pound per square inch = 6.895 kPa, 1 square foot = 0.0929 m².

- a. Table is based on ASCE 7 components and cladding wind pressures for an enclosed building using a mean roof height of 35 feet, interior wall area 4, an effective wind area of 10 square feet, topographic factor, K_{zt} , equal to 1.0, and Risk Category II.
- b. Table is based on concrete with a minimum specified compressive strength of 2,500 psi.
- c. See Section R608.6.5 for location of reinforcement in wall.
- d. Deflection criterion is $L/240$, where L is the unsupported height of the wall in inches.
- e. Interpolation is not permitted.
- f. Where No. 4 reinforcing bars at a spacing of 48 inches are specified in the table as indicated by shaded cells, use of bars with a minimum yield strength of 40,000 psi or 60,000 psi is permitted.
- g. Other than for No. 4 bars spaced at 48 inches on center, table values are based on reinforcing bars with a minimum yield strength of 60,000 psi. Maximum spacings shown are the values calculated for the specified bar size. Where the bar used is Grade 60 and the size specified in the table, the actual spacing in the wall shall not exceed a whole-number multiple of 12 inches such as, 12, 24, 36 and 48, that is less than or equal to the tabulated spacing. Vertical reinforcement with a yield strength of less than 60,000 psi or bars of a different size than specified in the table are permitted in accordance with Section R608.5.4.7 and Table R608.5.4(2).
- h. See Table R608.3 for minimum core dimensions and maximum spacing of horizontal and vertical cores.
- i. "Top" means gravity load from roof or floor construction bears on top of wall. "Side" means gravity load from floor construction is transferred to wall from a wood ledger or cold-formed steel track bolted to side of wall. For nonload-bearing wall and where floor framing members span parallel to the wall, use of the "Top" bearing condition is permitted.

TABLE R608.6(4)

MINIMUM VERTICAL REINFORCEMENT FOR FLAT, WAFFLE- AND SCREEN-GRID ABOVE-GRADE WALLS DESIGNED CONTINUOUS WITH FOUNDATION STEM WALLS^{a, b, c, d, e, k, l}

MAXIMUM WIND SPEED (mph)			HEIGHT OFSTEM WALL ^{h, i} (feet)	MAXIMUM DESIGN LATERAL SOIL LOAD (psf/ft)	MAXIMUM UNSUPPORTED HEIGHT OF ABOVE-GRADE WALL (feet)	MINIMUM VERTICAL REINFORCEMENT-BAR SIZE AND SPACING (inches) ^{f, g}						
						Wall type and nominal thickness ^j (inches)						
Exposure Category						Flat				Waffle		Screen
						4	6	8	10	6	8	6
115			3	30	8	4@30	4@48	4@48	4@48	4@22	4@26	4@21
					10	4@23	5@43	4@48	4@48	4@17	4@20	4@16
				60	10	4@19	5@37	4@48	4@48	4@14	4@17	4@14

			6	30	10	DR	5@21	6@35	4@48	DR	4@10	DR
				60	10	DR	5@12	6@25	6@28	DR	DR	DR
120			3	30	8	4@28	4@48	4@48	4@48	4@21	4@48	4@20
					10	4@22	5@41	4@48	4@48	4@16	4@19	4@15
				60	10	4@18	5@35	4@48	4@48	4@14	4@17	4@13
			6	30	10	DR	5@21	6@35	4@48	DR	4@10	DR
				60	10	DR	5@12	6@25	6@28	DR	DR	DR
130	110		3	30	8	4@25	4@48	4@48	4@48	4@18	4@22	4@18
					10	4@19	5@36	4@48	4@48	4@14	4@17	4@13
			6	60	10	4@16	5@34	4@48	4@48	4@12	4@17	4@12
				30	10	DR	5@19	6@35	4@48	DR	4@9	DR
				60	10	DR	5@12	6@24	6@28	DR	DR	DR
140	119	110	3	30	8	4@22	5@42	4@48	4@48	4@16	4@20	4@16
					10	4@17	5@34	4@48	4@48	4@21	4@17	4@12
				60	10	4@15	5@34	4@48	4@48	4@11	4@17	4@10
			6	30	10	DR	5@18	6@35	6@35	DR	4@48	DR
				60	10	DR	5@11	6@23	6@28	DR	DR	DR
150	127	117	3	30	8	4@20	5@37	4@48	4@48	4@15	4@18	4@14
					10	4@15	5@34	4@48	4@48	4@11	4@17	4@11
				60	10	4@13	5@34	4@48	4@48	4@10	4@16	4@9
			6	30	10	DR	5@17	6@33	6@32	DR	4@8	DR
				60	10	DR	DR	6@22	6@28	DR	DR	DR
160	136	125	3	30	8	4@18	5@34	4@48	4@48	4@13	4@17	4@13
					10	4@13	5@34	4@48	4@48	4@10	4@16	4@9
				60	10	4@11	5@31	6@45	4@48	4@9	4@14	4@8
			6	30	10	DR	5@15	6@31	6@30	DR	4@7	DR
				60	10	DR	DR	6@21	6@27	DR	DR	DR

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s, 1 pound per square inch = 6.895 kPa, 1 square foot = 0.0929 m².

- Table is based on ASCE 7 components and cladding wind pressures for an enclosed building using a mean roof height of 35 feet, interior wall area 4, an effective wind area of 10 square feet, topographic factor, K_{zt} , equal to 1.0, and Risk Category II.
- Table is based on concrete with a minimum specified compressive strength of 2,500 psi.
- See Section R608.6.5 for location of reinforcement in wall.

- d. Deflection criterion is $L/240$, where L is the height of the wall in inches from the exterior finish ground level to the top of the above-grade wall.
- e. Interpolation is not permitted. For intermediate values of basic wind speed, heights of stem wall and above-grade wall, and design lateral soil load, use next higher value.
- f. Where No. 4 reinforcing bars at a spacing of 48 inches are specified in the table as indicated by shaded cells, use of bars with a minimum yield strength of 40,000 psi or 60,000 psi is permitted.
- g. Other than for No. 4 bars spaced at 48 inches on center, table values are based on reinforcing bars with a minimum yield strength of 60,000 psi. Maximum spacings shown are the values calculated for the specified bar size. In waffle and screen-grid walls where the bar used is Grade 60 and the size specified in the table, the actual spacing in the wall shall not exceed a whole-number multiple of 12 inches such as, 12, 24, 36 and 48, that is less than or equal to the tabulated spacing. Vertical reinforcement with a yield strength of less than 60,000 psi and/or bars of a different size than specified in the table are permitted in accordance with Section R608.5.4.7 and Table R608.5.4(2).
- h. Height of stem wall is the distance from the exterior finish ground level to the top of the slab-on-ground.
- i. Where the distance from the exterior finish ground level to the top of the slab-on-ground is equal to or greater than 4 feet, the stem wall shall be laterally supported at the top and bottom before backfilling. Where the wall is designed and constructed to be continuous with the above-grade wall, temporary supports bracing the top of the stem wall shall remain in place until the above-grade wall is laterally supported at the top by floor or roof construction.
- j. See Table R608.3 for tolerances on nominal thicknesses, and minimum core dimensions and maximum spacing of horizontal and vertical cores for waffle-and screen-grid walls.
- k. Tabulated values are applicable to construction where gravity loads bear on top of wall, and conditions where gravity loads from floor construction are transferred to wall from a wood ledger or cold-formed steel track bolted to side of wall. See Tables R608.6(1), R608.6(2) and R608.6(3).
- l. DR = Design Required.

R608.6.1 General

The minimum thickness of load-bearing and nonload-bearing above-grade walls and reinforcement shall be as set forth in the appropriate table in this section based on the type of wall form to be used. The wall shall be designed in accordance with ACI 318 where the wall or building is not within the limitations of Section R608.2, where design is required by the tables in this section or where the wall is not within the scope of the tables in this section.

Above-grade concrete walls shall be constructed in accordance with this section and Figure R608.6(1), R608.6(2), R608.6(3) or R608.6(4). Above-grade concrete walls that are continuous with stem walls and not laterally supported by the slab-on-ground shall be designed and constructed in accordance with this section. Concrete walls shall be supported on continuous foundation walls or slabs-on-ground that are monolithic with the footing in accordance with Section R403. The minimum length of solid wall without openings shall be in accordance with Section R608.7. Reinforcement around openings, including lintels, shall be in accordance with Section R608.8. Lateral support for above-grade walls in the out-of-plane direction shall be provided by connections to the floor framing system, if applicable, and to ceiling and roof framing systems in accordance with Section R608.9. The wall thickness shall be equal to or greater than the thickness of the wall in the *story* above.

R608.6.2 Wall Reinforcement for Wind

Vertical wall reinforcement for resistance to out-of-plane wind forces shall be determined from Table R608.6(1), R608.6(2), R608.6(3) or R608.6(4). For the design of nonload-bearing walls, in Tables R608.6(1), R608.6(2) and R608.6(3) use the appropriate column labeled "Top." (see Sections R608.7.2.2.2 and R608.7.2.2.3). There shall be a vertical bar at corners of exterior walls. Unless more horizontal reinforcement is required by Section R608.7.2.2.1, the minimum horizontal reinforcement shall be four No. 4 bars [Grade 40 (280 MPa)] placed as follows: top bar within 12 inches (305 mm) of the top of the wall, bottom bar within 12 inches (305 mm) of the finish floor and one bar each at approximately one-third and two-thirds of the wall height.

R608.6.3 Continuity of Wall Reinforcement Between Stories

Vertical reinforcement required by this section shall be continuous between elements providing lateral support for the wall. Reinforcement in the wall of the *story* above shall be continuous with the reinforcement in the wall of the *story* below, or the foundation

wall, if applicable. Lap splices, where required, shall comply with Section R608.5.4.3 and Figure R608.5.4(1). Where the above-grade wall is supported by a monolithic slab-on-ground and footing, dowel bars with a size and spacing to match the vertical above-grade concrete wall reinforcement shall be embedded in the monolithic slab-on-ground and footing the distance required to develop the dowel bar in tension in accordance with Section R608.5.4.4 and Figure R608.5.4(2) and lap-spliced with the above-grade wall reinforcement in accordance with Section R608.5.4.3 and Figure R608.5.4(1).

Exception: Where reinforcement in the wall above cannot be made continuous with the reinforcement in the wall below, the bottom of the reinforcement in the wall above shall be terminated in accordance with one of the following:

1. Extend below the top of the floor the distance required to develop the bar in tension in accordance with Section R608.5.4.4 and Figure R608.5.4(2).
2. Lap-spliced in accordance with Section R608.5.4.3 and Figure R608.5.4(1) with a dowel bar that extends into the wall below the distance required to develop the bar in tension in accordance with Section R608.5.4.4 and Figure R608.5.4(2).

Where a construction joint in the wall is located below the level of the floor and less than the distance required to develop the bar in tension, the distance required to develop the bar in tension shall be measured from the top of the concrete below the joint. See Section R608.5.5.

R608.6.4 Termination of Reinforcement

Where indicated in Items 1 through 3, vertical wall reinforcement in the top-most story with concrete walls shall be terminated with a 90-degree (1.57 rad) standard hook complying with Section R608.5.4.5 and Figure R608.5.4(3).

1. Vertical bars adjacent to door and window openings required by Section R608.8.1.2.
2. Vertical bars at the ends of required solid wall segments (see Section R608.7.2.2.2).
3. Vertical bars (other than end bars, see Item 2) used as shear reinforcement in required solid wall segments where the reduction factor for design strength, R_3 , used is based on the wall having horizontal and vertical shear reinforcement (see Section R608.7.2.2.3).

The bar extension of the hook shall be oriented parallel to the horizontal wall reinforcement and be within 4 inches (102 mm) of the top of the wall.

Horizontal reinforcement shall be continuous around the building corners by bending one of the bars and lap-splicing it with the bar in the other wall in accordance with Section R608.5.4.3 and Figure R608.5.4(1).

Exception: In lieu of bending horizontal reinforcement at corners, separate bent reinforcing bars shall be permitted provided that the bent bar is lap-spliced with the horizontal reinforcement in both walls in accordance with Section R608.5.4.3 and Figure R608.5.4(1).

In required solid wall segments where the reduction factor for design strength, R_3 , is based on the wall having horizontal and vertical shear reinforcement in accordance with Section R608.7.2.2.1, horizontal wall reinforcement shall be terminated with a standard hook complying with Section R608.5.4.5 and Figure R608.5.4(3) or in a lap-splice, except at corners where the reinforcement shall be continuous as required.

R608.6.5 Location of Reinforcement in Wall

Except for vertical reinforcement at the ends of required solid wall segments, which shall be located as required by Section R608.7.2.2.2, the location of the vertical reinforcement shall not vary from the center of the wall by more than the greater of 10 percent of the wall thickness and $\frac{3}{8}$ -inch (10 mm). Horizontal and vertical reinforcement shall be located to provide not less than the minimum cover required by Section R608.5.4.1.

R608.7 Solid Walls for Resistance to Lateral Forces

TABLE R608.7(1A)

UNREDUCED LENGTH, U_R , OF SOLID WALL REQUIRED IN EACH EXTERIOR ENDWALL FOR WIND PERPENDICULAR TO RIDGE ONE STORY OR TOP STORY OF TWO STORY^{a, c, d, e, f, g}

SIDEWALL LENGTH (feet)	ENDWALL LENGTH (feet)	ROOF SLOPE	UNREDUCED LENGTH, <i>U_R</i> , OF SOLID WALL REQUIRED IN ENDWALLS FOR WIND PERPENDICULAR TO RIDGE (feet)						
			Basic Wind Speed (mph) Exposure						
			115B	120B	130B	140B	150B	160B	Minimum ^b
					110C	119C	127C	136C	
						110D	117D	125D	
15	15	< 1:12	1.03	1.12	1.32	1.53	1.76	2.00	0.92
		5:12	1.43	1.56	1.83	2.12	2.43	2.77	1.15
		7:12	2.00	2.18	2.56	2.97	3.41	3.88	1.25
		12:12	3.20	3.48	4.09	4.74	5.44	6.19	1.54
	30	< 1:12	1.03	1.12	1.32	1.53	1.76	2.00	0.98
		5:12	1.43	1.56	1.83	2.12	2.43	2.77	1.43
		7:12	2.78	3.03	3.56	4.13	4.74	5.39	1.64
		12:12	5.17	5.63	6.61	7.67	8.80	10.01	2.21
	45	< 1:12	1.03	1.12	1.32	1.53	1.76	2.00	1.04
		5:12	1.43	1.56	1.83	2.12	2.43	2.77	1.72
		7:12	3.57	3.88	4.56	5.28	6.07	6.90	2.03
		12:12	7.15	7.78	9.13	10.59	12.16	13.84	2.89
	60	< 1:12	1.03	1.12	1.32	1.53	1.76	2.00	1.09
		5:12	1.43	1.56	1.83	2.12	2.43	2.77	2.01
		7:12	4.35	4.73	5.55	6.44	7.39	8.41	2.42
		12:12	9.12	9.93	11.66	13.52	15.52	17.66	3.57
30	15	< 1:12	1.84	2.01	2.35	2.73	3.13	3.57	1.82
		5:12	2.56	2.78	3.27	3.79	4.35	4.95	2.23
		7:12	3.61	3.93	4.61	5.34	6.13	6.98	2.42
		12:12	5.61	6.10	7.16	8.31	9.54	10.85	2.93
	30	< 1:12	1.84	2.01	2.35	2.73	3.13	3.57	1.93
		5:12	2.56	2.78	3.27	3.79	4.35	4.95	2.75
		7:12	4.92	5.35	6.28	7.29	8.37	9.52	3.12
		12:12	8.92	9.71	11.39	13.22	15.17	17.26	4.14
	45	< 1:12	1.84	2.01	2.35	2.73	3.13	3.57	2.03
		5:12	2.56	2.78	3.27	3.79	4.35	4.95	3.26
		7:12	6.23	6.78	7.96	9.23	10.60	12.06	3.82

		12:12	12.23	13.31	15.63	18.12	20.80	23.67	5.36
	60	< 1:12	1.84	2.01	2.35	2.73	3.13	3.57	2.14
		5:12	2.56	2.78	3.27	3.79	4.35	4.95	3.78
		7:12	7.54	8.21	9.64	11.17	12.83	14.60	4.52
		12:12	15.54	16.92	19.86	23.03	26.44	30.08	6.57
60	15	< 1:12	3.42	3.72	4.36	5.06	5.81	6.61	3.63
		5:12	4.75	5.17	6.06	7.03	8.07	9.19	4.40
		7:12	6.76	7.36	8.64	10.02	11.51	13.09	4.75
		12:12	10.35	11.27	13.23	15.34	17.61	20.04	5.71
	30	< 1:12	3.42	3.72	4.36	5.06	5.81	6.61	3.83
		5:12	4.75	5.17	6.06	7.03	8.07	9.19	5.37
		7:12	9.12	9.93	11.66	13.52	15.52	17.66	6.07
		12:12	16.30	17.75	20.83	24.16	27.73	31.55	8.00
	45	< 1:12	3.55	3.87	4.54	5.27	6.05	6.88	4.03
		5:12	4.94	5.37	6.31	7.31	8.40	9.55	6.34
		7:12	11.71	12.75	14.97	17.36	19.93	22.67	7.39
		12:12	22.70	24.71	29.00	33.64	38.62	43.94	10.29
	60	< 1:12	3.68	4.01	4.71	5.46	6.27	7.13	4.23
		5:12	5.11	5.57	6.54	7.58	8.70	9.90	7.31
		7:12	14.38	15.66	18.37	21.31	24.46	27.83	8.71
		12:12	29.30	31.90	37.44	43.42	49.85	56.72	12.57

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s, 1 pound-force per linear foot = 0.146 kN/m, 1 pound per square foot = 47.88 Pa.

- Tabulated lengths were derived by calculating design wind pressures in accordance with Figure 28.4-1 of ASCE 7 for a building with a mean roof height of 35 feet, topographic factor, K_{zt} , equal to 1.0, and Risk Category II. For wind perpendicular to the ridge, the effects of a 2-foot overhang on each endwall are included. The design pressures were used to calculate forces to be resisted by solid wall segments in each. The forces to be resisted by each wall line were then divided by the default design strength of 840 pounds per linear foot of length to determine the unreduced length, UR , of solid wall length required in each endwall. The actual mean roof height of the building shall not exceed the least horizontal dimension of the building.
- Tabulated lengths in the "minimum" column are based on the requirement of Section 28.4.4 of ASCE 7 that the main windforce-resisting system be designed for a minimum pressure of 16 psf multiplied by the wall area of the building and 8 psf multiplied by the roof area of the building projected onto a vertical plane normal to the assumed wind direction. Tabulated lengths in shaded cells are less than the "minimum" value. Where the minimum controls, it is permitted to be reduced in accordance with Notes c, d and e. See Section R608.7.1.1.
- For buildings with a mean roof height of less than 35 feet, tabulated lengths are permitted to be reduced by multiplying by the appropriate factor, R_1 , from Table R608.7(2). The reduced length shall be not less than the "minimum" value shown in the table.

- d. Tabulated lengths for "one story or top story of two story" are based on a floor-to-ceiling height of 10 feet. Tabulated lengths for "first story of two story" are based on floor-to-ceiling heights of 10 feet each for the first and second story. For floor-to-ceiling heights less than assumed, use the lengths in this table or Table R608.7 (1B) or (1C), or multiply the value in the table by the reduction factor, R_2 , from Table R608.7(3).
- e. Tabulated lengths are based on the default design shear strength of 840 pounds per linear foot of solid wall segment. The tabulated lengths are permitted to be reduced by multiplying by the applicable reduction factor for design strength, R_3 , from Table R608.7(4).
- f. The reduction factors, R_1 , R_2 and R_3 , in Tables R608.7(2), R608.7(3), and R608.7(4), respectively, are permitted to be compounded, subject to the limitations of Note b. However, the minimum number and minimum length of solid wall segments in each wall line shall comply with Sections R608.7.1 and R608.7.2.1, respectively.
- g. For intermediate values of sidewall length, endwall length, roof slope and basic wind speed, use the next higher value, or determine by interpolation.

TABLE R608.7(1B)

UNREDUCED LENGTH, U_R , OF SOLID WALL REQUIRED IN EACH EXTERIOR ENDWALL FOR WIND PERPENDICULAR TO RIDGE FIRST STORY OF TWO STORY^{a, c, d, e, f, g}

SIDEWALL LENGTH (feet)	ENDWALL LENGTH (feet)	ROOF SLOPE	UNREDUCED LENGTH, U_R , OF SOLID WALL REQUIRED IN ENDWALLS FOR WIND PERPENDICULAR TO RIDGE (feet)						
			Basic Wind Speed (mph) Exposure						
			115B	120B	130B	140B	150B	160B	Minimum ^b
					110C	119C	127C	136C	
						110D	117D	125D	
15	15	< 1:12	2.98	3.25	3.81	4.42	5.07	5.77	2.54
		5:12	4.13	4.50	5.28	6.12	7.03	8.00	2.76
		7:12	4.31	4.70	5.51	6.39	7.34	8.35	2.87
		12:12	5.51	6.00	7.04	8.16	9.37	10.66	3.15
	30	< 1:12	2.98	3.25	3.81	4.42	5.07	5.77	2.59
		5:12	4.13	4.50	5.28	6.12	7.03	8.00	3.05
		7:12	5.09	5.55	6.51	7.55	8.67	9.86	3.26
		12:12	7.48	8.15	9.56	11.09	12.73	14.49	3.83
	45	< 1:12	2.98	3.25	3.81	4.42	5.07	5.77	2.65
		5:12	4.13	4.50	5.28	6.12	7.03	8.00	3.34
		7:12	5.88	6.40	7.51	8.71	10.00	11.37	3.65
		12:12	9.46	10.30	12.09	14.02	16.09	18.31	4.51
	60	< 1:12	2.98	3.25	3.81	4.42	5.07	5.77	2.71
		5:12	4.13	4.50	5.28	6.12	7.03	8.00	3.63
		7:12	6.66	7.25	8.51	9.87	11.32	12.89	4.04

		12:12	11.43	12.45	14.61	16.94	19.45	22.13	5.19
30	15	< 1:12	5.32	5.79	6.80	7.89	9.05	10.30	5.06
		5:12	7.39	8.04	9.44	10.95	12.57	14.30	5.47
		7:12	7.94	8.65	10.15	11.77	13.51	15.37	5.65
		12:12	9.94	10.82	12.70	14.73	16.91	19.24	6.17
	30	< 1:12	5.32	5.79	6.80	7.89	9.05	10.30	5.16
		5:12	7.39	8.04	9.44	10.95	12.57	14.30	5.98
		7:12	9.25	10.07	11.82	13.71	15.74	17.91	6.35
		12:12	13.25	14.43	16.93	19.64	22.54	25.65	7.38
	45	< 1:12	5.32	5.79	6.80	7.89	9.05	10.30	5.27
		5:12	7.39	8.04	9.44	10.95	12.57	14.30	6.50
		7:12	10.56	11.50	13.50	15.65	17.97	20.45	7.06
		12:12	16.56	18.03	21.16	24.55	28.18	32.06	8.60
	60	< 1:12	5.32	5.79	6.80	7.89	9.05	10.30	5.38
		5:12	7.39	8.04	9.44	10.95	12.57	14.30	7.01
		7:12	11.87	12.93	15.17	17.60	20.20	22.98	7.76
		12:12	19.87	21.64	25.40	29.45	33.81	38.47	9.81
60	15	< 1:12	9.87	10.74	12.61	14.62	16.79	19.10	10.10
		5:12	13.71	14.93	17.52	20.32	23.33	26.54	10.87
		7:12	15.08	16.42	19.27	22.35	25.66	29.20	11.22
		12:12	18.67	20.33	23.86	27.67	31.77	36.14	12.19
	30	< 1:12	9.87	10.74	12.61	14.62	16.79	19.10	10.30
		5:12	13.71	14.93	17.52	20.32	23.33	26.54	11.85
		7:12	17.44	18.99	22.29	25.85	29.67	33.76	12.54
		12:12	24.62	26.81	31.46	36.49	41.89	47.66	14.48
	45	< 1:12	10.27	11.18	13.12	15.21	17.47	19.87	10.50
		5:12	14.26	15.52	18.22	21.13	24.26	27.60	12.82
		7:12	20.21	22.01	25.83	29.95	34.39	39.12	13.86
		12:12	31.20	33.97	39.87	46.23	53.07	60.39	16.76
	60	< 1:12	10.64	11.59	13.60	15.77	18.11	20.60	10.70
		5:12	14.77	16.09	18.88	21.90	25.14	28.60	13.79
		7:12	23.05	25.09	29.45	34.15	39.21	44.61	15.18
		12:12	37.97	41.34	48.52	56.27	64.60	73.49	19.05

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s, 1 pound force per linear foot = 0.146 kN/m, 1 pound per square foot = 47.88 Pa.

- a. Tabulated lengths were derived by calculating design wind pressures in accordance with Figure 28.4-1 of ASCE 7 for a building with a mean roof height of 35 feet, topographic factor, K_{zt} , equal to 1.0, and Risk Category II. For wind perpendicular to the ridge, the effects of a 2-foot overhang on each endwall are included. The design pressures were used to calculate forces to be resisted by solid wall segments in each endwall. The forces to be resisted by each wall line were then divided by the default design strength of 840 pounds per linear foot of length to determine the unreduced length, U_R , of solid wall length required in each endwall. The actual mean roof height of the building shall not exceed the least horizontal dimension of the building.
- b. Tabulated lengths in the "minimum" column are based on the requirement of Section 28.4.4 of ASCE 7 that the main windforce-resisting system be designed for a minimum pressure of 1016 psf multiplied by the wall area of the building and 8 psf multiplied by the roof area of the building projected onto a vertical plane normal to the assumed wind direction. Tabulated lengths in shaded cells are less than the "minimum" value. Where the minimum controls, it is permitted to be reduced in accordance with Notes c, d and e. See Section R608.7.1.1.
- c. For buildings with a mean roof height of less than 35 feet, tabulated lengths are permitted to be reduced by multiplying by the appropriate factor, R_1 , from Table R608.7(2). The reduced length shall be not less than the "minimum" value shown in the table.
- d. Tabulated lengths for "one story or top story of two story" are based on a floor-to-ceiling height of 10 feet. Tabulated lengths for "first story of two story" are based on floor-to-ceiling heights of 10 feet each for the first and second story. For floor-to-ceiling heights less than assumed, use the lengths in this table or Table R608.7(1A) or (1C), or multiply the value in the table by the reduction factor, R_2 , from Table R608.7(3).
- e. Tabulated lengths are based on the default design shear strength of 840 pounds per linear foot of solid wall segment. The tabulated lengths are permitted to be reduced by multiplying by the applicable reduction factor for design strength, R_3 , from Table R608.7(4).
- f. The reduction factors, R_1 , R_2 and R_3 , in Tables R608.7(2), R608.7(3), and R608.7(4), respectively, are permitted to be compounded, subject to the limitations of Note b. However, the minimum number and minimum length of solid wall segments in each wall line shall comply with Sections R608.7.1 and R608.7.2.1, respectively.
- g. For intermediate values of sidewall length, endwall length, roof slope and basic wind speed, use the next higher value, or determine by interpolation.

TABLE R608.7(1C)

UNREDUCED LENGTH, U_R , OF SOLID WALL REQUIRED IN EACH EXTERIOR SIDEWALL FOR WIND PARALLEL TO RIDGE^{a, c, d, e, f, g}

SIDEWALL LENGTH (feet)	ENDWALL LENGTH (feet)	ROOF SLOPE	UNREDUCED LENGTH, <i>U_R</i> , OF SOLID WALL REQUIRED IN SIDEWALLS FOR WIND PARALLEL TO RIDGE (feet)						
			Basic Wind Speed (mph) Exposure						Minimum ^b
			115B	120B	130B	140B	150B	160B	
					110C	119C	127C	136C	
						110D	117D	125D	
			One story or top story of two story						
< 30	15	< 1:12	1.08	1.18	1.39	1.61	1.84	2.10	0.90
		5:12	1.29	1.40	1.65	1.91	2.19	2.49	1.08
		7:12	1.38	1.50	1.76	2.04	2.35	2.67	1.17

		12:12	1.63	1.78	2.09	2.42	2.78	3.16	1.39	
	30	< 1:12	2.02	2.20	2.59	3.00	3.44	3.92	1.90	
		5:12	2.73	2.97	3.48	4.04	4.64	5.28	2.62	
		7:12	3.05	3.32	3.89	4.51	5.18	5.89	2.95	
		12:12	3.93	4.27	5.02	5.82	6.68	7.60	3.86	
	45	< 1:12	3.03	3.30	3.87	4.49	5.15	5.86	2.99	
		5:12	4.55	4.96	5.82	6.75	7.74	8.81	4.62	
		7:12	5.24	5.71	6.70	7.77	8.92	10.15	5.36	
		12:12	7.16	7.79	9.14	10.61	12.17	13.85	7.39	
	60	< 1:12	4.11	4.47	5.25	6.09	6.99	7.96	4.18	
		5:12	6.78	7.39	8.67	10.05	11.54	13.13	7.07	
		7:12	8.00	8.71	10.22	11.85	13.61	15.48	8.38	
		12:12	11.35	12.36	14.51	16.82	19.31	21.97	12.00	
	60	45	< 1:12	3.17	3.46	4.06	4.70	5.40	6.14	2.99
			5:12	4.75	5.18	6.07	7.04	8.09	9.20	4.62
7:12			5.47	5.96	6.99	8.11	9.31	10.59	5.36	
12:12			7.45	8.11	9.52	11.04	12.68	14.43	7.39	
60		< 1:12	4.41	4.81	5.64	6.54	7.51	8.54	4.18	
		5:12	7.22	7.86	9.23	10.70	12.29	13.98	7.07	
		7:12	8.50	9.25	10.86	12.59	14.46	16.45	8.38	
		12:12	12.02	13.09	15.36	17.81	20.45	23.27	12.00	
SIDEWALL LENGTH (feet)	ENDWALL LENGTH (feet)	ROOF SLOPE	UNREDUCED LENGTH, <i>U_R</i> , OF SOLID WALL REQUIRED IN SIDEWALLS FOR WIND PARALLEL TO RIDGE (feet)							
			Basic Wind Speed Exposure (mph)							
			115B	120B	130B	140B	150B	160B	Minimum ^b	
					110C	119C	127C	136C		
						110D	117D	125D		
			First story of two story							
< 30	15	< 1:12	3.03	3.30	3.88	4.49	5.16	5.87	2.52	
		5:12	3.24	3.52	4.14	4.80	5.51	6.26	2.70	
		7:12	3.33	3.62	4.25	4.93	5.66	6.44	2.79	
		12:12	3.58	3.90	4.58	5.31	6.10	6.94	3.01	
	30	< 1:12	5.50	5.99	7.03	8.16	9.36	10.65	5.14	
		5:12	6.21	6.76	7.93	9.20	10.56	12.01	5.86	

		7:12	6.52	7.10	8.34	9.67	11.10	12.63	6.19
		12:12	7.41	8.06	9.46	10.97	12.60	14.33	7.10
	45	< 1:12	8.00	8.71	10.22	11.85	13.61	15.48	7.85
		5:12	9.52	10.37	12.17	14.11	16.20	18.43	9.48
		7:12	10.21	11.12	13.05	15.14	17.38	19.77	10.21
		12:12	12.13	13.20	15.50	17.97	20.63	23.47	12.25
	60	< 1:12	10.56	11.50	13.50	15.65	17.97	20.44	10.65
		5:12	13.24	14.41	16.91	19.62	22.52	25.62	13.54
		7:12	14.45	15.73	18.46	21.41	24.58	27.97	14.85
		12:12	17.80	19.38	22.75	26.38	30.29	34.46	18.48
60	45	< 1:12	8.39	9.14	10.72	12.44	14.28	16.25	7.85
		5:12	9.97	10.86	12.74	14.78	16.97	19.30	9.48
		7:12	10.69	11.64	13.66	15.84	18.19	20.69	10.21
		12:12	12.67	13.80	16.19	18.78	21.56	24.53	12.25
	60	< 1:12	11.37	12.38	14.53	16.85	19.35	22.01	10.65
		5:12	14.18	15.44	18.12	21.02	24.13	27.45	13.54
		7:12	15.46	16.83	19.75	22.91	26.29	29.92	14.85
		12:12	18.98	20.66	24.25	28.13	32.29	36.74	18.48

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s, 1 pound force per linear foot = 0.146 kN/m, 1 pound per square foot = 47.88 Pa.

- a. Tabulated lengths were derived by calculating design wind pressures in accordance with Figure 28.4-1 of ASCE 7 for a building with a mean roof height of 35 feet, topographic factor, K_{zt} , equal to 1.0, and Risk Category II. The design pressures were used to calculate forces to be resisted by solid wall segments in each sidewall. The forces to be resisted by each wall line were then divided by the default design strength of 840 pounds per linear foot of length to determine the unreduced length, UR , of solid wall length required in each sidewall. The actual mean roof height of the building shall not exceed the least horizontal dimension of the building.
- b. Tabulated lengths in the "minimum" column are based on the requirement of Section 28.4.4 of ASCE 7 that the main windforce-resisting system be designed for a minimum pressure of 16 psf multiplied by the wall area of the building and 8 psf multiplied by the roof area of the building projected onto a vertical plane normal to the assumed wind direction. Tabulated lengths in shaded cells are less than the "minimum" value. Where the minimum controls, it is permitted to be reduced in accordance with Notes c, d and e. See Section R608.7.1.1.
- c. For buildings with a mean roof height of less than 35 feet, tabulated lengths are permitted to be reduced by multiplying by the appropriate factor, R_1 , from Table R608.7(2). The reduced length shall be not less than the "minimum" value shown in the table.
- d. Tabulated lengths for "one story or top story of two story" are based on a floor-to-ceiling height of 10 feet. Tabulated lengths for "first story of two story" are based on floor-to-ceiling heights of 10 feet each for the first and second story. For floor-to-ceiling heights less than assumed, use the lengths in this table or Table R608.7(1A) or (1B), or multiply the value in the table by the reduction factor, R_2 , from Table R608.7(3).

- e. Tabulated lengths are based on the default design shear strength of 840 pounds per linear foot of solid wall segment. The tabulated lengths are permitted to be reduced by multiplying by the applicable reduction factor for design strength, R_3 , from Table R608.7(4).
- f. The reduction factors, R_1 , R_2 and R_3 , in Tables R608.7(2), R608.7(3), and R608.7(4), respectively, are permitted to be compounded, subject to the limitations of Note b. However, the minimum number and minimum length of solid walls segments in each wall line shall comply with Sections R608.7.1 and R608.7.2.1, respectively.
- g. For intermediate values of sidewall length, endwall length, roof slope and basic wind speed, use the next higher value, or determine by interpolation.

TABLE R608.7(2)

REDUCTION FACTOR, R_1 , FOR BUILDINGS WITH MEAN ROOF HEIGHT LESS THAN 35 FEET^a

MEAN ROOF HEIGHT ^{b, c} (feet)	REDUCTION FACTOR R_1 , FOR MEAN ROOF HEIGHT		
	Exposure category		
	B	C	D
< 15	0.96	0.84	0.87
20	0.96	0.89	0.91
25	0.96	0.93	0.94
30	0.96	0.97	0.98
35	1.00	1.00	1.00

For SI: 1 foot = 304.8 mm, 1 degree = 0.0175 rad.

- a. See Section R608.7.1.1 and Note c to Table R608.7(1A) for application of reduction factors in this table. This reduction is not permitted for "minimum" values.
- b. For intermediate values of mean roof height, use the factor for the next greater height, or determine by interpolation.
- c. Mean roof height is the average of the roof eave height and height of the highest point on the roof surface, except that for roof slopes of less than or equal to $2\frac{1}{8}$:12 (10 degrees), the mean roof height is permitted to be taken as the roof eave height.

TABLE R608.7(3)

REDUCTION FACTOR, R_2 , FOR FLOOR-TO-CEILING WALL HEIGHTS LESS THAN 10 FEET^{a, b}

STORY UNDER CONSIDERATION	FLOOR-TO-CEILING HEIGHT ^c (feet)	ENDWALL LENGTH (feet)	ROOF SLOPE	REDUCTION FACTOR, <i>R</i> ₂
Endwalls—for wind perpendicular to ridge				
One story or top story of two story	8	15	< 5:12	0.83
			7:12	0.90
			12:12	0.94
		60	< 5:12	0.83

			7:12	0.95
			12:12	0.98
First story of two story	16 combined first and second story	15	< 5:12	0.83
			7:12	0.86
			12:12	0.89
		60	< 5:12	0.83
			7:12	0.91
			12:12	0.95
Sidewalls—for wind parallel to ridge				
One story or top story of two story	8	15	< 1:12	0.84
			5:12	0.87
			7:12	0.88
			12:12	0.89
		60	< 1:12	0.86
			5:12	0.92
			7:12	0.93
			12:12	0.95
First story of two story	16 combined first and second story	15	< 1:12	0.83
			5:12	0.84
			7:12	0.85
			12:12	0.86
		60	< 1:12	0.84
			5:12	0.87
			7:12	0.88
			12:12	0.90

For SI: 1 foot = 304.8 mm.

- See Section R608.7.1.1 and Note d to Table R608.7(1A) for application of reduction factors in this table.
- For intermediate values of endwall length, and/or roof slope, use the next higher value, or determine by interpolation.
- Tabulated values in Table R608.7(1A) and (1C) for "one story or top story of two story" are based on a floor-to-ceiling height of 10 feet. Tabulated values in Table R608.7(1B) and (1C) for "first story of two story" are based on floor-to-ceiling heights of 10 feet each for the first and second story. For floor to ceiling heights between those shown in this table and those assumed in Table R608.7(1A), (1B) or (1C), use the solid wall lengths in Table R608.7(1A), (1B) or (1C), or determine the reduction factor by interpolating between 1.0 and the factor shown in this table.

TABLE R608.7(4)

REDUCTION FACTOR FOR DESIGN STRENGTH, R_3 , FOR FLAT, WAFFLE- AND SCREEN-GRID WALLS^{a, c}

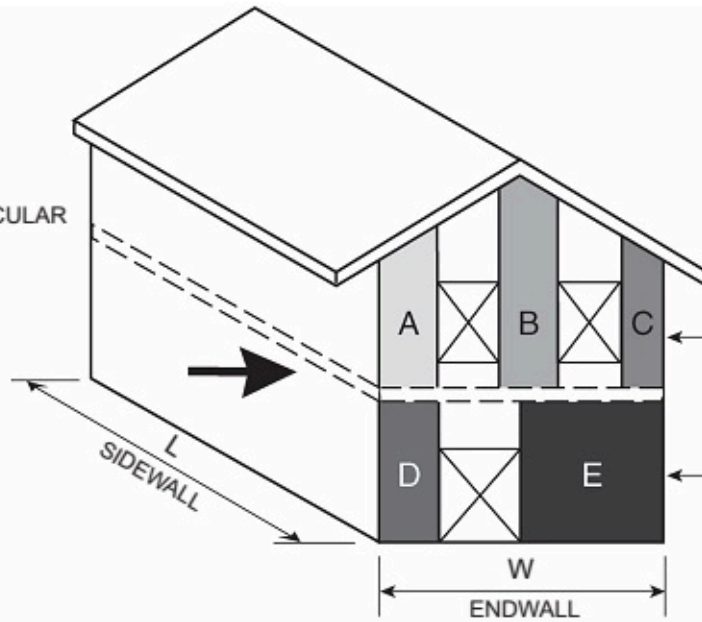
NOMINAL THICKNESS OF WALL (inches)	VERTICAL BARS AT EACH END OF SOLID WALL SEGMENT		VERTICAL REINFORCEMENT LAYOUT DETAIL [see Figure R608.7(2)]	REDUCTION FACTOR, R_3 , FOR LENGTH OF SOLID WALL			
				Horizontal and vertical shear reinforcement provided			
	Number of bars	Bar size		No		Yes ^d	
				40,000 ^b	60,000 ^b	40,000 ^b	60,000 ^b
Flat walls							
4	2	4	1	0.74	0.61	0.74	0.50
	3	4	2	0.61	0.61	0.52	0.27
	2	5	1	0.61	0.61	0.48	0.25
	3	5	2	0.61	0.61	0.26	0.18
6	2	4	3	0.70	0.48	0.70	0.48
	3	4	4	0.49	0.38	0.49	0.33
	2	5	3	0.46	0.38	0.46	0.31
	3	5	4	0.38	0.38	0.32	0.16
8	2	4	3	0.70	0.47	0.70	0.47
	3	4	5	0.47	0.32	0.47	0.32
	2	5	3	0.45	0.31	0.45	0.31
	4	4	6	0.36	0.28	0.36	0.25
	3	5	5	0.31	0.28	0.31	0.16
	4	5	6	0.28	0.28	0.24	0.12
10	2	4	3	0.70	0.47	0.70	0.47
	2	5	3	0.45	0.30	0.45	0.30
	4	4	7	0.36	0.25	0.36	0.25
	6	4	8	0.25	0.22	0.25	0.13
	4	5	7	0.24	0.22	0.24	0.12
	6	5	8	0.22	0.22	0.12	0.08
Waffle-grid walls ^e							
6	2	4	3	0.78	0.78	0.70	0.48
	3	4	4	0.78	0.78	0.49	0.25
	2	5	3	0.78	0.78	0.46	0.23
	3	5	4	0.78	0.78	0.24	0.16
8	2	4	3	0.78	0.78	0.70	0.47
	3	4	5	0.78	0.78	0.47	0.24

	2	5	3	0.78	0.78	0.45	0.23
	4	4	6	0.78	0.78	0.36	0.18
	3	5	5	0.78	0.78	0.23	0.16
	4	5	6	0.78	0.78	0.18	0.13
Screen-grid walls^e							
6	2	4	3	0.93	0.93	0.70	0.48
	3	4	4	0.93	0.93	0.49	0.25
	2	5	3	0.93	0.93	0.46	0.23
	3	5	4	0.93	0.93	0.24	0.16

For SI: 1 inch = 25.4 mm, 1,000 pounds per square inch = 6.895 MPa.

- See Note e to Table R608.7(1A) for application of adjustment factors in this table.
- Yield strength in pounds per square inch of vertical wall reinforcement at ends of solid wall segments.
- Values are based on concrete with a specified compressive strength, f'_c , of 2,500 psi. Where concrete with f'_c of not less than 3,000 psi is used, values in shaded cells are permitted to be decreased by multiplying by 0.91.
- Horizontal and vertical shear reinforcement shall be provided in accordance with Section R608.7.2.2.
- Each end of each solid wall segment shall have rectangular flanges. In the through-the-wall dimension, the flange shall be not less than $5\frac{1}{2}$ inches for 6-inch-nominal waffle- and screen-grid walls, and not less than $7\frac{1}{2}$ inches for 8-inch-nominal waffle-grid walls. In the in-plane dimension, flanges shall be long enough to accommodate the vertical reinforcement required by the layout detail selected from Figure R608.7(2) and provide the cover required by Section R608.5.4.1. If necessary to achieve the required dimensions, form material shall be removed or use of flat wall forms is permitted.

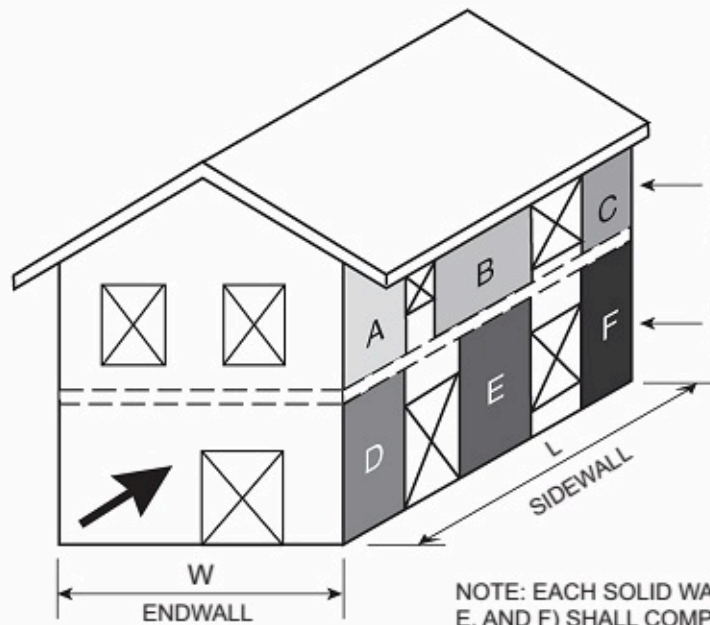
WIND
PERPENDICULAR
TO RIDGE



ONE STORY OR TOP STORY OF
TWO STORY
[TABLE R608.7(1B)]
SEE SECTION R608.7.1.1

FIRST STORY OF TWO STORY
[(TABLE R608.7(1A))]
SEE SECTION R608.7.1.1

WIND
PARALLEL
TO RIDGE



ONE STORY OR TOP STORY OF
TWO STORY
[TABLE R608.7(1C)]
SEE SECTION R608.7.1.1

FIRST STORY OF TWO STORY
[TABLE R608.7(1C)]
SEE SECTION R608.7.1.1

NOTE: EACH SOLID WALL SEGMENT (A, B, C, D, E, AND F) SHALL COMPLY WITH THE MINIMUM SOLID WALL SEGMENT LENGTH IN ORDER TO BE APPLICABLE TO THE MINIMUM SOLID WALL LENGTH EQUATIONS IN SECTION R608.7.1.1. SEE SECTION R608.7.2

FIGURE R608.7(1)

MINIMUM SOLID WALL LENGTH

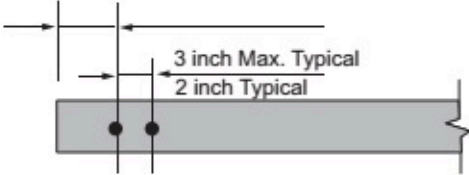
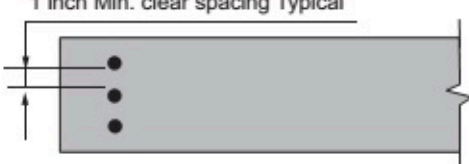
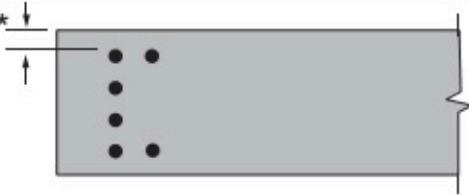
DETAIL NO.	NOM. WALL THICKNESS, IN.	REINFORCEMENT LAYOUT AT ENDS OF SOLID WALL SEGMENTS	NOTES
1	4		For SI: 1 inch = 25.4 mm. 1. See Table R608.7(4) for use of details.
2	4		2. Minimum length of solid wall segment and size and grade of reinforcement in each end of each solid wall segment shall be determined from Table R608.7(4).
3	6 8 10		3.. For minimum cover requirements, see Section R608.5.4.1.
4	6		4.. For details 3 - 8 where two or more bars are in the same row parallel to the end of the segment, place bars so that corner bars are as close to the sides of the wall segments as minimum cover requirements of Section R608.5.4.1 will permit.
5	8		
6	8		5.. For waffle- and screen-grid walls, each end of each solid wall segment shall have rectangular flanges. In the through-the-wall dimension, the flange shall be not less than 5½ inches for 6-inch nominal waffle- and screen-grid forms, and not less than 7½ inches for 8-inch nominal waffle-grid forms. In the in-plane dimension, flanges shall be long enough to accommodate the vertical reinforcement required by the layout detail selected and provide the cover required by Section R608.5.4.1. If necessary to achieve the required dimensions, form material shall be removed or flat wall forms are permitted. See Table R608.7(4), Note e.
7	10		
8	10	 * For minimum cover see Section R608.5.4.1	

FIGURE R608.7(2)

VERTICAL REINFORCEMENT LAYOUT DETAIL

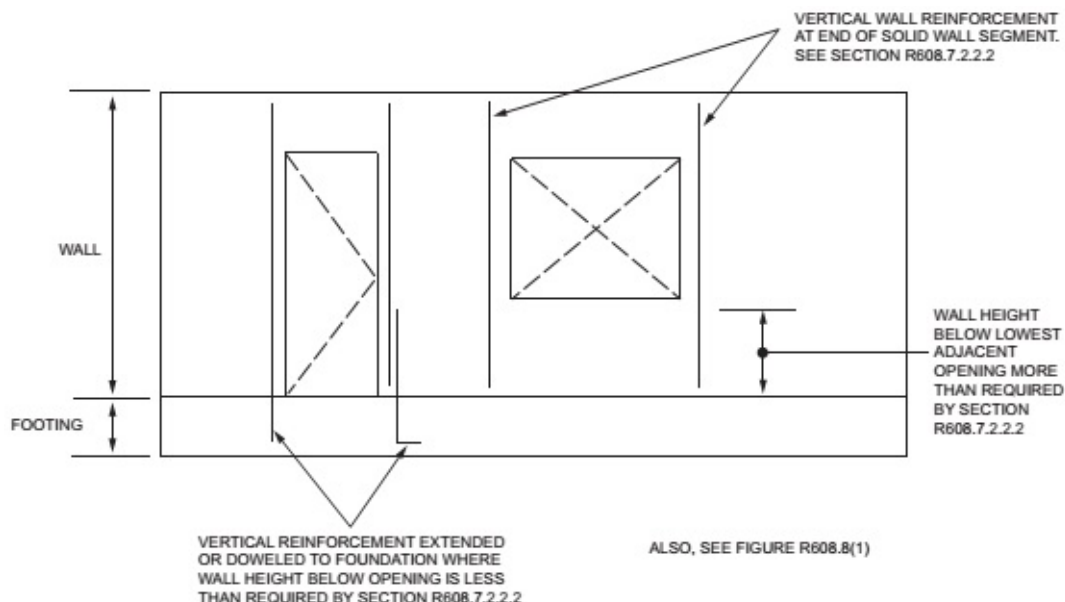


FIGURE R608.7(3)

VERTICAL WALL REINFORCEMENT ADJACENT TO WALL OPENINGS

R608.7.1 Length of Solid Wall

Each exterior wall line in each *story* shall have a total length of solid wall required by Section R608.7.1.1. A solid wall is a section of flat, waffle-grid or screen-grid wall, extending the full *story height* without openings or penetrations, except those permitted by Section R608.7.2. Solid wall segments that contribute to the total length of solid wall shall comply with Section R608.7.2.

R608.7.1.1 Length of Solid Wall for Wind

Buildings shall have solid walls in each exterior endwall line (the side of a building that is parallel to the span of the roof or floor framing) and sidewall line (the side of a building that is perpendicular to the span of the roof or floor framing) to resist lateral in-plane wind forces. The site-appropriate basic wind speed and exposure category shall be used in Tables R608.7(1A) through (1C) to determine the unreduced total length, *UR*, of solid wall required in each exterior endwall line and sidewall line. For buildings with a mean roof height of less than 35 feet (10 668 mm), the unreduced values determined from Tables R608.7(1A) through (1C) are permitted to be reduced by multiplying by the applicable factor, *R*₁, from Table R608.7(2); however, reduced values shall be not less than the minimum values in Tables R608.7(1A) through (1C). Where the floor-to-ceiling height of a *story* is less than 10 feet (3048 mm), the unreduced values determined from Tables R608.7(1A) through (1C), including minimum values, are permitted to be reduced by multiplying by the applicable factor, *R*₂, from Table R608.7(3). To account for different design strengths than assumed in determining the values in Tables R608.7(1A) through (1C), the unreduced lengths determined from Tables R608.7(1A) through (1C), including minimum values, are permitted to be reduced by multiplying by the applicable factor, *R*₃, from Table R608.7(4). The reductions permitted by Tables R608.7(2), R608.7(3) and R608.7(4) are cumulative.

The total length of solid wall segments, *TL*, in a wall line that comply with the minimum length requirements of Section R608.7.2.1 [see Figure R608.7(1)] shall be equal to or greater than the product of the unreduced length of solid wall from Tables R608.7(1A) through (1C), *UR* and the applicable reduction factors, if any, from Tables R608.7(2), R608.7(3) and R608.7(4) as indicated by Equation R6-1.

$$TL \geq R_1 \cdot R_2 \cdot R_3 \cdot UR$$

(Equation R6-1)

where:

- TL* = Total length of solid wall segments in a wall line that comply with Section R608.7.2.1 [see Figure R608.7(1)].
- R*₁ = 1.0 or reduction factor for mean roof height from Table R608.7(2).
- R*₂ = 1.0 or reduction factor for floor-to-ceiling wall height from Table R608.7(3).
- R*₃ = 1.0 or reduction factor for design strength from Table R608.7(4).
- UR* = Unreduced length of solid wall from Tables R608.7(1A) through (1C).

The total length of solid wall in a wall line, TL , shall be not less than that provided by two solid wall segments complying with the minimum length requirements of Section R608.7.2.1.

To facilitate determining the required wall thickness, wall type, number and *grade* of vertical bars at each end of each solid wall segment, and whether shear reinforcement is required, use of Equation R6-2 is permitted.

$$R \leq \frac{TL}{R_1 \cdot R_2 \cdot UR} \quad \text{(Equation R6-2)}$$

After determining the maximum permitted value of the reduction factor for design strength, R_3 , in accordance with Equation R6-2, select a wall type from Table R608.7(4) with R_3 less than or equal to the value calculated.

R608.7.2 Solid Wall Segments

Solid wall segments that contribute to the required length of solid wall shall comply with this section. Reinforcement shall be provided in accordance with Section R608.7.2.2 and Table R608.7(4). Solid wall segments shall extend the full story-height without openings, other than openings for the utilities and other building services passing through the wall. In flat walls and waffle-grid walls, such openings shall have an area of less than 30 square inches (19 355 mm²) without any dimension exceeding 6¹/₄ inches (159 mm), and shall not be located within 6 inches (152 mm) of the side edges of the solid wall segment. In screen-grid walls, such openings shall be located in the portion of the solid wall segment between horizontal and vertical cores of concrete and opening size and location are not restricted provided there is not any concrete removed.

R608.7.2.1 Minimum Length of Solid Wall Segment and Maximum Spacing

Only solid wall segments equal to or greater than 24 inches (610 mm) in length shall be included in the total length of solid wall required by Section R608.7.1. In addition, not more than two solid wall segments equal to or greater than 24 inches (610 mm) in length and less than 48 inches (1219 mm) in length shall be included in the required total length of solid wall. The maximum clear opening width shall be 18 feet (5486 mm). See Figure R608.7(1).

R608.7.2.2 Reinforcement in Solid Wall Segments

R608.7.2.2.1 Horizontal Shear Reinforcement

Where reduction factors for design strength, R_3 , from Table R608.7(4) based on horizontal and vertical shear reinforcement being provided are used, solid wall segments shall have horizontal reinforcement consisting of minimum No. 4 bars. Horizontal shear reinforcement shall be the same grade of steel required for the vertical reinforcement at the ends of solid wall segments by Section R608.7.2.2.2.

The spacing of horizontal reinforcement shall not exceed the smaller of one-half the length of the solid wall segment, minus 2 inches (51 mm), and 18 inches (457 mm). Horizontal shear reinforcement shall terminate in accordance with Section R608.6.4.

R608.7.2.2.2 Vertical Reinforcement

Vertical reinforcement applicable to the reduction factor(s) for design strength, R_3 , from Table R608.7(4) that is used, shall be located at each end of each solid wall segment in accordance with the applicable detail in Figure R608.7(2). The No. 4 vertical bar required on each side of an opening by Section R608.8.1.2 is permitted to be used as reinforcement at the ends of solid wall segments where installed in accordance with the applicable detail in Figure R608.7(2). There shall be not less than two No. 4 bars at each end of solid wall segments located as required by the applicable detail in Figure R608.7(2). One of the bars at each end of solid wall segments shall be deemed to meet the requirements for vertical wall reinforcement required by Section R608.6.

The vertical wall reinforcement at each end of each solid wall segment shall be developed below the bottom of the adjacent wall opening [see Figure R608.7(3)] by one of the following methods:

1. Where the wall height below the bottom of the adjacent opening is equal to or greater than 22 inches (559 mm) for No. 4 or 28 inches (711 mm) for No. 5 vertical wall reinforcement, reinforcement around openings in accordance with Section R608.8.1 shall be sufficient.
2. Where the wall height below the bottom of the adjacent opening is less than required by Item 1 above, the vertical wall reinforcement adjacent to the opening shall extend into the footing far enough to develop the bar in tension in accordance with Section R608.5.4.4 and Figure R608.5.4(2), or shall be lap-spliced with a dowel that is embedded in the footing far enough to develop the dowel-bar in tension.

R608.7.2.2.3 Vertical Shear Reinforcement

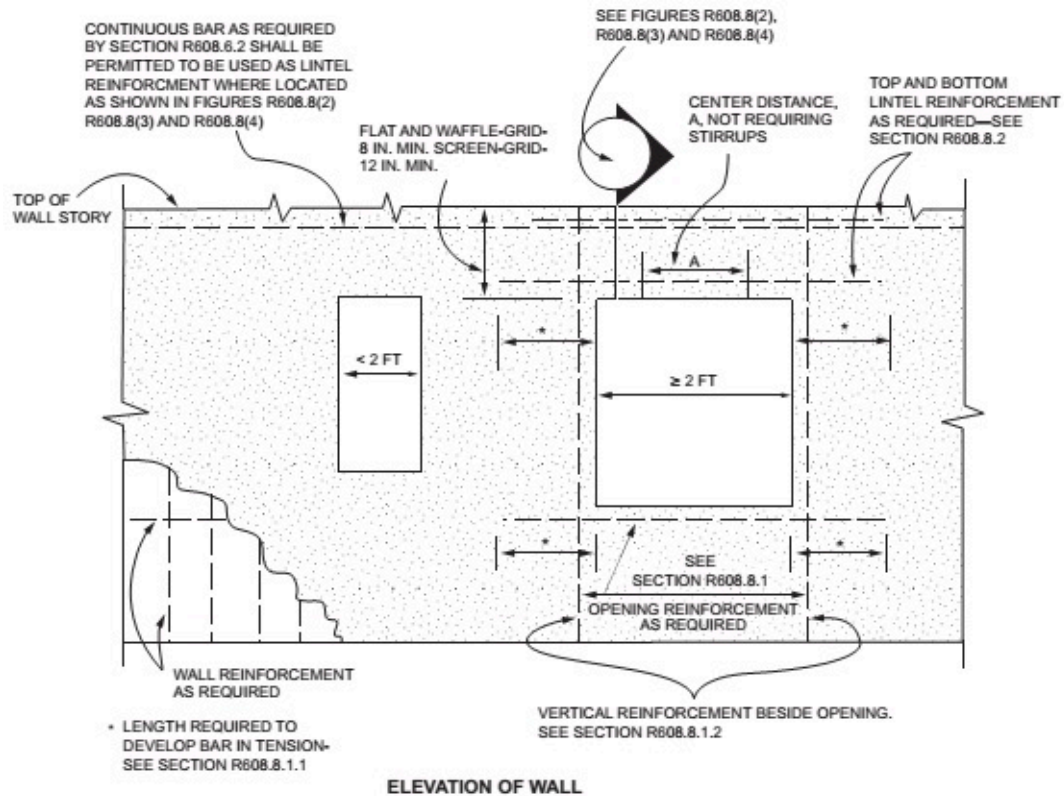
Where reduction factors for design strength, R_3 , from Table R608.7(4) based on horizontal and vertical shear reinforcement being provided are used, solid wall segments shall have vertical reinforcement consisting of minimum No. 4 bars. Vertical shear reinforcement shall be the same grade of steel required by Section R608.7.2.2.2 for the vertical reinforcement at the ends of solid wall segments. The spacing of vertical reinforcement throughout the length of the segment shall not exceed the smaller of one third the length of the segment, and 18 inches (457 mm). Vertical shear reinforcement shall be continuous between stories in accordance with Section R608.6.3, and shall terminate in accordance with Section R608.6.4. Vertical shear reinforcement required by this section is permitted to be used for vertical reinforcement required by Table R608.6(1), R608.6(2), R608.6(3) or R608.6(4), whichever is applicable.

R608.7.2.3 Solid Wall Segments at Corners

At all interior and exterior corners of exterior walls, a solid wall segment shall extend the full height of each wall story. The segment shall have the length required to develop the horizontal reinforcement above and below the adjacent opening in tension in accordance with Section R608.5.4.4. For an exterior corner, the limiting dimension is measured on the outside of the wall, and for an interior corner the limiting dimension is measured on the inside of the wall. See Section R608.8.1. The length of a segment contributing to the required length of solid wall shall comply with Section R608.7.2.1.

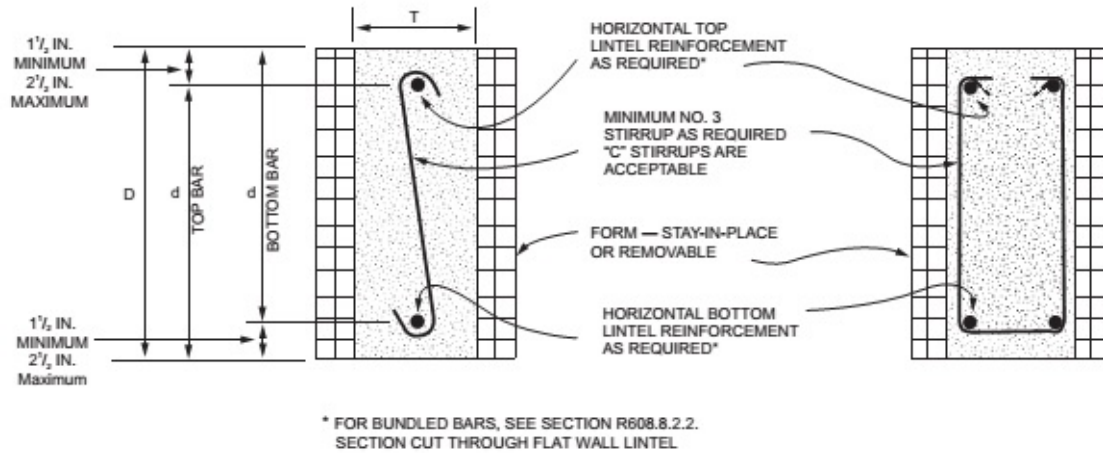
The end of a solid wall segment complying with the minimum length requirements of Section R608.7.2.1 shall be located not more than 6 feet (1829 mm) from each corner.

R608.8 Requirements for Lintels and Reinforcement Around Openings



For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

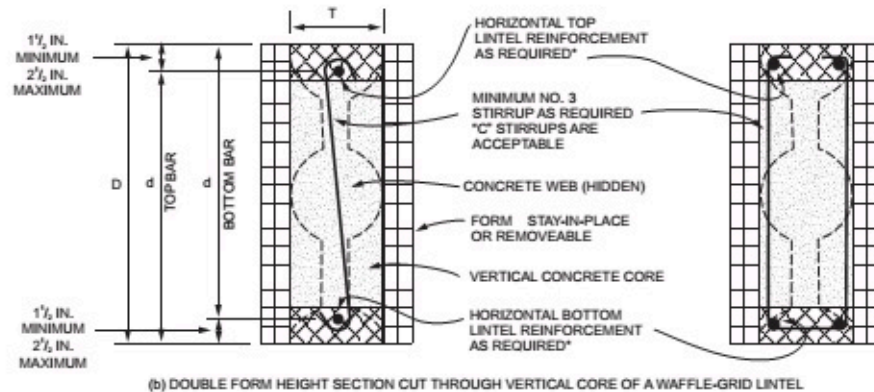
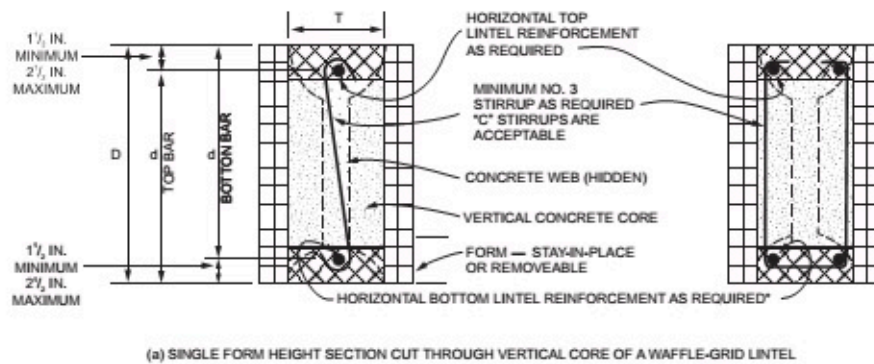
FIGURE R608.8(1)
REINFORCEMENT OF OPENINGS



For SI: 1 inch = 25.4 mm.

FIGURE R608.8(2)

LINTEL FOR FLAT WALLS



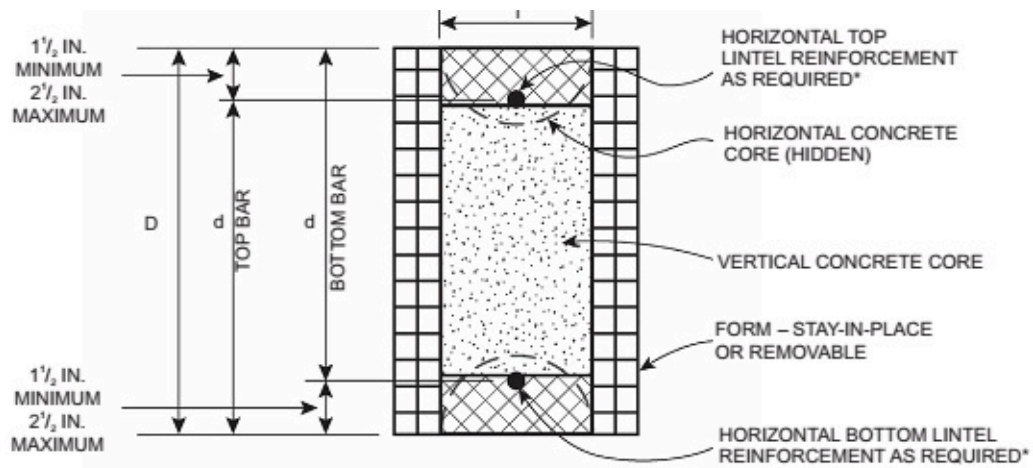
*FOR BUNDLED BARS, SEE SECTION R608.8.2.2.

NOTE: CROSS-HATCHING REPRESENTS THE AREA IN WHICH FORM MATERIAL SHALL BE REMOVED, IF NECESSARY, TO CREATE FLANGES CONTINUOUS THE LENGTH OF THE LINTEL. FLANGES SHALL HAVE A MINIMUM THICKNESS OF 3 IN., AND A MINIMUM WIDTH OF 5 IN. AND 7 IN. IN 6 IN. NOMINAL AND 8 IN. NOMINAL WAFFLE-GRID WALLS, RESPECTIVELY. SEE NOTE a TO TABLES R608.8(6) AND R608.8(10).

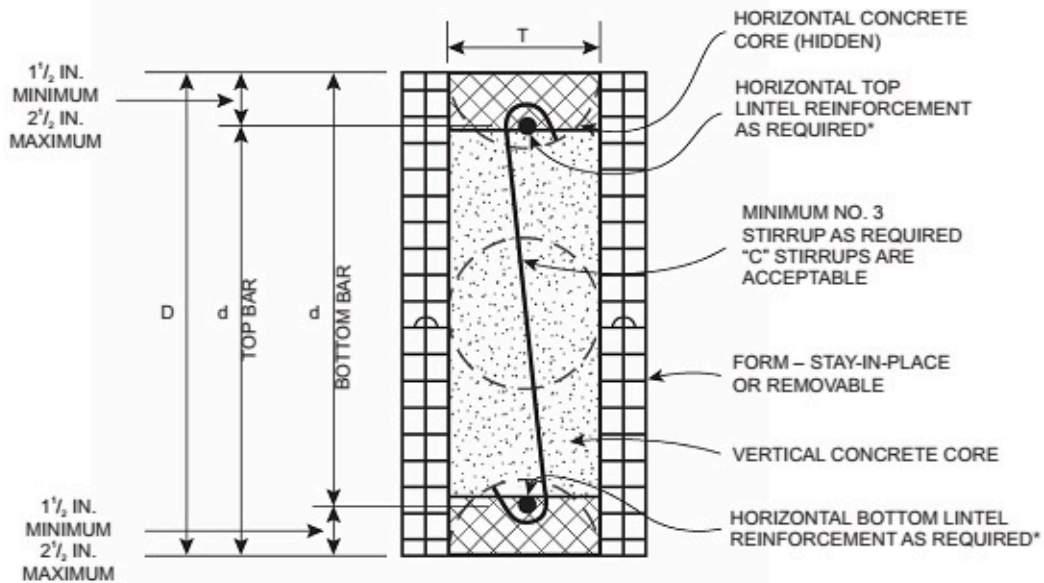
For SI: 1 inch = 25.4 mm.

FIGURE R608.8(3)

LINTELS FOR WAFFLE-GRID WALLS



(a) SINGLE FORM HEIGHT SECTION CUT THROUGH VERTICAL CORE OF A SCREEN-GRID LINTEL



(b) DOUBLE FORM HEIGHT SECTION CUT THROUGH VERTICAL CORE OF A SCREEN-GRID LINTEL

*FOR BUNDLED BARS, SEE SECTION R608.8.2.2

NOTE: CROSS-HATCHING REPRESENTS THE AREA IN WHICH FORM MATERIAL SHALL BE REMOVED, IF NECESSARY, TO CREATE FLANGES CONTINUOUS THE LENGTH OF THE LINTEL. FLANGES SHALL HAVE A MINIMUM THICKNESS OF 2.5 IN. AND A MINIMUM WIDTH OF 5 IN. SEE NOTE a TO TABLES R608.8(8) AND R608.8(10).

For SI: 1 inch = 25.4 mm.

FIGURE R608.8(4)

LINTELS FOR SCREEN-GRID WALLS

TABLE R608.8(1)

LINTEL DESIGN LOADING CONDITIONS^{a, b, d}

DESCRIPTION OF LOADS AND OPENINGS ABOVE INFLUENCING DESIGN OF LINTEL			DESIGN LOAD CONDITION ^c
Opening in wall of top story of two-story building, or first story of one-story building			
Wall supporting loads from roof, including attic floor, if applicable, and	Top of lintel equal to or less than W/2 below top of wall		2
	Top of lintel greater than W/2 below top of wall		NLB
Wall not supporting loads from roof or attic floor			NLB
Opening in wall of first story of two-story building where wall immediately above is of concrete construction, or opening in basement wall of one-story building where wall immediately above is of concrete construction			
LB ledger board mounted to side of wall with bottom of ledger less than or equal to W/2 above top of lintel, and	Top of lintel greater than W/2 below bottom of opening in story above		1
	Top of lintel less than or equal to W/2 below bottom of opening in story above, and	Opening is entirely within the footprint of the opening in the story above	1
		Opening is partially within the footprint of the opening in the story above	4
LB ledger board mounted to side of wall with bottom of ledger more than W/2 above top of lintel			NLB
NLB ledger board mounted to side of wall with bottom of ledger less than or equal to W/2 above top of lintel, or no ledger board, and	Top of lintel greater than W/2 below bottom of opening in story above		NLB
	Top of lintel less than or equal to W/2 below bottom of opening in story above, and	Opening is entirely within the footprint of the opening in the story above	NLB
		Opening is partially within the footprint of the opening in the story above	1
Opening in basement wall of two-story building where walls of two stories above are of concrete construction			
LB ledger board mounted to side of wall with bottom of ledger less than or equal to W/2 above top of lintel, and	Top of lintel greater than W/2 below bottom of opening in story above		1
	Top of lintel less than or equal to W/2 below bottom of opening in story above, and	Opening is entirely within the footprint of the opening in the story above	1
		Opening is partially within the footprint of the opening in the story above	5
LB ledger board mounted to side of wall with bottom of ledger more than W/2 above top of lintel			NLB
NLB ledger board mounted to side of wall with bottom of ledger less than or equal to W/2 above top of lintel, or no ledger board, and	Top of lintel greater than W/2 below bottom of opening in story above		NLB
	Top of lintel less than or equal to W/2 below bottom of opening in story above, and	Opening is entirely within the footprint of the opening in the story above	NLB
		Opening is partially within the footprint of the opening in the story above	1
Opening in wall of first story of two-story building where wall immediately above is of light-framed construction, or opening in basement wall of one-story building where wall immediately above is of light-framed construction			

Wall supporting loads from roof, second floor and top-story wall of light-framed construction, and	Top of lintel equal to or less than W/2 below top of wall	3
	Top of lintel greater than W/2 below top of wall	NLB
Wall not supporting loads from roof or second floor		NLB

- a. LB means load bearing, NLB means nonload bearing, and W means width of opening.
- b. Footprint is the area of the wall below an opening in the story above, bounded by the bottom of the opening and vertical lines extending downward from the edges of the opening.
- c. For design loading condition "NLB" see Tables R608.8(9) and R608.8(10). For all other design loading conditions, see Tables R608.8(2) through R608.8(8).
- d. A NLB ledger board is a ledger attached to a wall that is parallel to the span of the floor, roof or ceiling framing that supports the edge of the floor, ceiling or roof.

TABLE R608.8(2)

MAXIMUM ALLOWABLE CLEAR SPANS FOR 4-INCH-NOMINAL THICK FLAT LINTELS IN LOAD-BEARING WALLS^{a, b, c, d, e, f, m}

ROOF CLEAR SPAN 40 FEET AND FLOOR CLEAR SPAN 32 FEET

LINTEL DEPTH, <i>D</i> ^g (inches)	NUMBER OF BARS AND BAR SIZE IN TOP AND BOTTOM OF LINTEL	STEEL YIELD STRENGTH ^h , <i>f_y</i> (psi)	DESIGN LOADING CONDITION DETERMINED FROM TABLE R608.8(1)				
			1	2	3	4	5
			Maximum clear span of lintel (feet - inches)				
8	Span without stirrups ^{i,j}		3-2	3-4	2-6	2-1	2-0
	1-#4	40,000	5-2	5-5	4-3	3-7	2-9
		60,000	6-2	6-5	5-1	4-2	2-11
	1-#5	40,000	6-3	6-7	5-2	4-2	2-11
		60,000	DR	DR	DR	DR	DR
	Center distance <i>A</i> ^{k,l}		1-1	1-2	0-9	0-6	0-4
12	Span without stirrups ^{i,j}		3-4	3-7	2-11	2-6	2-2
	1-#4	40,000	6-7	7-0	5-7	4-9	3-8
		60,000	7-11	8-6	6-9	5-9	4-5
	1-#5	40,000	8-1	8-8	6-10	5-10	4-6
		60,000	9-8	10-4	8-2	6-11	4-10
	2-#4 1-#6	40,000	9-1	9-8	7-8	6-6	4-10
		60,000	DR	DR	DR	DR	DR
	Center distance <i>A</i> ^{k,l}		1-8	1-11	1-3	0-11	0-6
16	Span without stirrups ^{i,j}		4-7	5-0	4-0	3-7	3-1
	1-#4	40,000	6-8	7-3	5-9	4-11	3-10

			60,000	9-3	10-1	8-0	6-10	5-4
	1-#4		40,000	9-6	10-4	8-2	6-11	5-5
			60,000	11-5	12-5	9-10	8-4	6-6
	2-#4 1-#6		40,000	10-7	11-7	9-2	7-9	6-1
			60,000	12-9	13-10	11-0	9-4	6-9
	2-#5		40,000	13-0	14-1	11-2	9-2	6-6
			60,000	DR	DR	DR	DR	DR
	Center distance ^{k,l}			2-3	2-8	1-8	1-3	0-9
20	Span without stirrups ^{i,j}			5-9	6-5	5-2	4-7	3-11
	1-#4		40,000	7-5	8-2	6-6	5-7	4-4
			60,000	9-0	10-0	7-11	6-9	5-3
	1-#5		40,000	9-2	10-2	8-1	6-11	5-4
			60,000	12-9	14-2	11-3	9-7	7-5
	2-#4 1-#6		40,000	11-10	13-2	10-5	8-11	6-11
			60,000	14-4	15-10	12-7	10-9	8-4
	2-#5		40,000	14-7	16-2	12-9	10-6	7-7
			60,000	17-5	19-2	15-3	12-4	8-8
	2-#6		40,000	16-4	18-11	13-3	10-6	7-7
			60,000	DR	DR	DR	DR	DR
	Center distance A ^{k,l}			2-9	3-5	2-2	1-7	0-11
24	Span without stirrups ^{i,j}			6-11	7-9	6-3	5-7	4-9
	1-#4		40,000	8-0	9-0	7-2	6-2	4-9
			60,000	9-9	11-0	8-9	7-6	5-10
	1-#5		40,000	10-0	11-3	8-11	7-7	5-11
			60,000	13-11	15-8	12-5	10-7	8-3
	2-#4 1-#6		40,000	12-11	14-6	11-6	9-10	7-8
			60,000	15-7	17-7	13-11	11-11	9-3
	2-#5		40,000	15-11	17-11	14-3	11-9	8-7
			60,000	19-1	21-6	17-1	14-0	9-11
	2-#6		40,000	17-7	21-1	14-10	11-9	8-7
			60,000	DR	DR	DR	DR	DR
	Center distance A ^{k,l}			3-3	4-1	2-7	1-11	1-2

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

- a. See Table R608.3 for tolerances permitted from nominal thickness.
- b. Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Note j.
- c. Table values are based on uniform loading. See Section R608.8.2 for lintels supporting concentrated loads.
- d. Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.
- e. Reserved.
- f. DR indicates design required.
- g. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- h. Stirrups shall be fabricated from reinforcing bars with the same yield strength as that used for the main longitudinal reinforcement.
- i. Allowable clear span without stirrups applicable to all lintels of the same depth, D . Top and bottom reinforcement for lintels without stirrups shall be not less than the least amount of reinforcement required for a lintel of the same depth and loading condition with stirrups. All other spans require stirrups spaced at not more than $d/2$.
- j. Where concrete with a minimum specified compressive strength of 3,000 psi is used, clear spans for lintels without stirrups shall be permitted to be multiplied by 1.05. If the increased span exceeds the allowable clear span for a lintel of the same depth and loading condition with stirrups, the top and bottom reinforcement shall be equal to or greater than that required for a lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups that has been increased.
- k. Center distance, A , is the center portion of the clear span where stirrups are not required. This is applicable to all longitudinal bar sizes and steel yield strengths.
- l. Where concrete with a minimum specified compressive strength of 3,000 psi is used, center distance, A , shall be permitted to be multiplied by 1.10.
- m. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel clear spans in the table greater than 18 feet are shown for interpolation and information only.

TABLE R608.8(3)

MAXIMUM ALLOWABLE CLEAR SPANS FOR 6-INCH-NOMINAL THICK FLAT LINTELS IN LOAD-BEARING WALLS^{a, b, c, d, e, f, m}

ROOF CLEAR SPAN 40 FEET AND FLOOR CLEAR SPAN 32 FEET

LINTEL DEPTH, <i>D</i> ^g (inches)	NUMBER OF BARS AND BAR SIZE IN TOP AND BOTTOM OF LINTEL	STEEL YIELD STRENGTH ^h , <i>f_y</i> (psi)	DESIGN LOADING CONDITION DETERMINED FROM TABLE R608.8(1)				
			1	2	3	4	5
			Maximum clear span of lintel (feet - inches)				
8	Span without stirrups ^{i, j}		4-2	4-8	3-3	2-6	2-0
	1-#4	40,000	5-1	5-5	4-3	3-6	2-8
		60,000	6-2	6-7	5-2	4-2	3-3
	1-#5	40,000	6-3	6-8	5-3	4-3	3-3
		60,000	7-6	8-0	6-4	5-1	3-8
	2-#4 1-#6	40,000	7-0	7-6	5-11	4-9	3-8

		60,000	DR	DR	DR	DR	DR
	Center distance $A^{k,l}$		1-7	1-10	1-2	0-9	0-5
12	Span without stirrups ^{i,j}		4-2	4-8	3-6	2-11	2-5
	1-#4	40,000	5-7	6-1	4-10	3-11	3-0
		60,000	7-9	8-6	6-9	5-6	4-3
	1-#5	40,000	7-11	8-8	6-11	5-7	4-4
		60,000	9-7	10-6	8-4	6-9	5-2
	2-#4 1-#6	40,000	8-11	9-9	7-9	6-3	4-10
		60,000	10-8	11-9	9-4	7-6	5-10
	2-#5	40,000	10-11	12-0	9-6	7-8	5-6
		60,000	12-11	14-3	11-3	9-0	6-1
	2-#6	40,000	12-9	14-0	11-1	8-1	5-6
		60,000	DR	DR	DR	DR	DR
	Center distance $A^{k,l}$		2-6	3-0	1-10	1-3	0-9
16	Span without stirrups ^{i,j}		5-7	6-5	4-11	4-0	3-4
	1-#4	40,000	6-5	7-2	5-9	4-8	3-7
		60,000	7-10	8-9	7-0	5-8	4-4
	1-#5	40,000	7-11	8-11	7-1	5-9	4-5
		60,000	11-1	12-6	9-11	8-0	6-2
	2-#4 1-#6	40,000	10-3	11-7	9-2	7-6	5-9
		60,000	12-5	14-0	11-1	9-0	7-0
	2-#5	40,000	12-8	14-3	11-4	9-2	6-9
		60,000	15-2	17-1	13-7	11-0	7-11
	2-#6	40,000	14-11	16-9	13-4	9-8	6-9
		60,000	DR	DR	DR	DR	DR
	Center distance $A^{k,l}$		3-3	4-1	2-7	1-9	1-0
20	Span without stirrups ^{i,j}		6-11	8-2	6-3	5-2	4-4
	1-#5	40,000	8-9	10-1	8-0	6-6	5-1
		60,000	10-8	12-3	9-9	8-0	6-2
	2-#4 1-#6	40,000	9-11	11-4	9-1	7-4	5-8
		60,000	13-9	15-10	12-8	10-3	7-11
	2-#5	40,000	14-0	16-2	12-11	10-6	7-11
		60,000	16-11	19-6	15-6	12-7	9-1

	2-#6	40,000	16-7	19-1	15-3	11-3	7-11
		60,000	19-11	22-10	18-3	13-2	9-1
	Center distance $A^{k,l}$		3-11	5-2	3-3	2-2	1-4
24	Span without stirrups ^{i,j}		8-2	9-10	7-8	6-4	5-3
	1-#5	40,000	9-5	11-1	8-10	7-3	5-7
		60,000	11-6	13-6	10-9	8-9	6-10
	2-#4 1-#6	40,000	10-8	12-6	10-0	8-2	6-4
		60,000	12-11	15-2	12-2	9-11	7-8
	2-#5	40,000	15-2	17-9	14-3	11-7	9-0
		60,000	18-4	21-6	17-3	14-0	10-4
	2-#6	40,000	18-0	21-1	16-11	12-9	9-2
		60,000	21-7	25-4	20-4	14-9	10-4
	Center distance $A^{k,l}$		4-6	6-2	4-0	2-8	1-7

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pounds per square foot = 0.0479 kPa, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

- a. See Table R608.3 for tolerances permitted from nominal thickness.
- b. Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Note j.
- c. Table values are based on uniform loading. See Section R608.8.2 for lintels supporting concentrated loads.
- d. Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.
- e. Reserved.
- f. DR indicates design required.
- g. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- h. Stirrups shall be fabricated from reinforcing bars with the same yield strength as that used for the main longitudinal reinforcement.
- i. Allowable clear span without stirrups applicable to all lintels of the same depth, D . Top and bottom reinforcement for lintels without stirrups shall be not less than the least amount of reinforcement required for a lintel of the same depth and loading condition with stirrups. All other spans require stirrups spaced at not more than $d/2$.
- j. Where concrete with a minimum specified compressive strength of 3,000 psi is used, clear spans for lintels without stirrups shall be permitted to be multiplied by 1.05. If the increased span exceeds the allowable clear span for a lintel of the same depth and loading condition with stirrups, the top and bottom reinforcement shall be equal to or greater than that required for a lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups that has been increased.
- k. Center distance, A , is the center portion of the clear span where stirrups are not required. This is applicable to all longitudinal bar sizes and steel yield strengths.
- l. Where concrete with a minimum specified compressive strength of 3,000 psi is used, center distance, A , shall be permitted to be multiplied by 1.10.

m. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel clear spans in the table greater than 18 feet are shown for interpolation and information only.

TABLE R608.8(4)

MAXIMUM ALLOWABLE CLEAR SPANS FOR 8-INCH-NOMINAL THICK FLAT LINTELS IN LOAD-BEARING WALLS^{a, b, c, d, e, f, m}

ROOF CLEAR SPAN 40 FEET AND FLOOR CLEAR SPAN 32 FEET

LINTEL DEPTH, <i>D</i> ^g (inches)	NUMBER OF BARS AND BAR SIZE IN TOP AND BOTTOM OF LINTEL	STEEL YIELD STRENGTH ^h , <i>f_y</i> (psi)	DESIGN LOADING CONDITION DETERMINED FROM TABLE R608.8(1)				
			1	2	3	4	5
			Maximum clear span of lintel (feet - inches)				
8	Span without stirrups ^{i,j}		4-4	4-9	3-9	2-10	2-1
	1-#4	40,000	4-4	4-9	3-9	2-11	2-3
		60,000	6-1	6-7	5-3	4-0	3-1
	1-#5	40,000	6-2	6-9	5-4	4-1	3-2
		60,000	7-5	8-1	6-5	4-11	3-9
	2-#4 1-#6	40,000	6-11	7-6	6-0	4-7	3-6
		60,000	8-3	9-0	7-2	5-6	4-2
	2-#5	40,000	8-5	9-2	7-3	5-7	4-2
		60,000	DR	DR	DR	DR	DR
	Center distance A ^{k,l}		2-1	2-6	1-6	0-11	0-6
12	Span without stirrups ^{i,j}		4-10	5-8	4-2	3-2	2-7
	1-#4	40,000	5-5	6-1	4-10	3-9	2-10
		60,000	6-7	7-5	5-11	4-7	3-6
	1-#5	40,000	6-9	7-7	6-0	4-8	3-7
		60,000	9-4	10-6	8-4	6-6	5-0
	2-#4 1-#6	40,000	8-8	9-9	7-9	6-0	4-7
		60,000	10-6	11-9	9-5	7-3	5-7
	2-#5	40,000	10-8	12-0	9-7	7-5	5-6
		60,000	12-10	14-5	11-6	8-11	6-7
	2-#6	40,000	12-7	14-2	11-3	8-3	5-6
		60,000	DR	DR	DR	DR	DR
	Center distance A ^{k,l}		3-2	4-0	2-6	1-6	0-11
16	Span without stirrups ^{i,j}		6-5	7-9	5-10	4-5	3-7
	1-#4	40,000	6-2	7-1	5-8	4-5	3-5
		60,000	7-6	8-8	6-11	5-5	4-2

	1-#5	40,000	7-8	8-10	7-1	5-6	4-3
		60,000	9-4	10-9	8-7	6-8	5-2
	2-#4 1-#6	40,000	8-8	10-0	8-0	6-2	4-9
		60,000	12-0	13-11	11-2	8-8	6-8
	2-#5	40,000	12-3	14-2	11-4	8-10	6-9
		60,000	14-10	17-2	13-8	10-8	7-11
	2-#6	40,000	14-6	16-10	13-5	10-1	6-11
		60,000	17-5	20-2	16-1	11-10	7-11
	Center distance ^{k, l}		4-1	5-5	3-6	2-1	1-3
20	Span without stirrups ^{i, j}		7-10	9-10	7-5	5-8	4-7
	1-#5	40,000	8-4	9-11	8-0	6-3	4-9
		60,000	10-2	12-1	9-9	7-7	5-10
	2-#4 1-#6	40,000	9-5	11-3	9-0	7-0	5-5
		60,000	11-6	13-8	11-0	8-7	6-7
	2-#5	40,000	11-9	13-11	11-2	8-9	6-8
		60,000	16-4	19-5	15-7	12-2	9-3
	2-#6	40,000	16-0	19-0	15-3	11-10	8-3
		60,000	19-3	22-11	18-5	13-7	9-3
Center distance A ^{k, l}		4-10	6-10	4-5	2-8	1-7	
24	Span without stirrups ^{i, j}		9-2	11-9	8-11	6-11	5-7
	1-#5	40,000	8-11	10-10	8-9	6-10	5-3
		60,000	10-11	13-3	10-8	8-4	6-5
	2-#4 1-#6	40,000	10-1	12-3	9-11	7-9	6-0
		60,000	12-3	15-0	12-1	9-5	7-3
	2-#5	40,000	12-6	15-3	12-4	9-7	7-5
		60,000	17-6	21-3	17-2	13-5	10-4
	2-#6	40,000	17-2	20-11	16-10	13-2	9-7
		60,000	20-9	25-3	20-4	15-4	10-7
Center distance A ^{k, l}		5-6	8-1	5-3	3-3	1-11	

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa, Grade 40 = 280 MPa; Grade 60 = 420 MPa.

Note: Top and bottom reinforcement for lintels without stirrups shown in shaded cells shall be equal to or greater than that required for lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups.

a. See Table R608.3 for tolerances permitted from nominal thickness.

- b. Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Note j.
- c. Table values are based on uniform loading. See Section R608.8.2 for lintels supporting concentrated loads.
- d. Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.
- e. Reserved.
- f. DR indicates design required.
- g. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- h. Stirrups shall be fabricated from reinforcing bars with the same yield strength as that used for the main longitudinal reinforcement.
- i. Allowable clear span without stirrups applicable to all lintels of the same depth, D . Top and bottom reinforcement for lintels without stirrups shall be not less than the least amount of reinforcement required for a lintel of the same depth and loading condition with stirrups. All other spans require stirrups spaced at not more than $d/2$.
- j. Where concrete with a minimum specified compressive strength of 3,000 psi is used, clear spans for lintels without stirrups shall be permitted to be multiplied by 1.05. If the increased span exceeds the allowable clear span for a lintel of the same depth and loading condition with stirrups, the top and bottom reinforcement shall be equal to or greater than that required for a lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups that has been increased.
- k. Center distance, A , is the center portion of the clear span where stirrups are not required. This is applicable to all longitudinal bar sizes and steel yield strengths.
- l. Where concrete with a minimum specified compressive strength of 3,000 psi is used, center distance, A , shall be permitted to be multiplied by 1.10.
- m. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel clear spans in the table greater than 18 feet are shown for interpolation and information only.

TABLE R608.8(5)

MAXIMUM ALLOWABLE CLEAR SPANS FOR 10-INCH-NOMINAL THICK FLAT LINTELS IN LOAD-BEARING WALLS^{a, b, c, d, e, f, m}

ROOF CLEAR SPAN 40 FEET AND FLOOR CLEAR SPAN 32 FEET

LINTEL DEPTH, <i>D</i> ^g (inches)	NUMBER OF BARS AND BAR SIZE IN TOP AND BOTTOM OF LINTEL	STEEL YIELD STRENGTH ^h , <i>f_y</i> (psi)	DESIGN LOADING CONDITION DETERMINED FROM TABLE R608.8(1)				
			1	2	3	4	5
			Maximum clear span of lintel (feet - inches)				
8	Span without stirrups ^{i,j}		6-0	7-2	4-10	3-1	2-3
	1-#4	40,000	4-3	4-9	3-9	2-9	2-1
		60,000	5-11	6-7	5-3	3-10	2-11
	1-#5	40,000	6-1	6-9	5-4	3-11	3-0
		60,000	7-4	8-1	6-5	4-9	3-7
	2-#4 1-#6	40,000	6-10	7-6	6-0	4-5	3-4
		60,000	8-2	9-1	7-2	5-4	4-1
	2-#5	40,000	8-4	9-3	7-4	5-5	4-1

		60,000	9-11	11-0	8-9	6-6	4-8
	2-#6	40,000	9-9	10-10	8-7	6-4	4-1
		60,000	DR	DR	DR	DR	DR
	Center distance $A^{k,l}$		2-6	3-1	1-11	1-1	0-7
12	Span without stirrups ^{i,j}		5-5	6-7	4-10	3-5	2-8
	1-#4	40,000	5-3	6-0	4-10	3-7	2-9
		60,000	6-5	7-4	5-10	4-4	3-4
	1-#5	40,000	6-6	7-6	6-0	4-5	3-5
		60,000	7-11	9-1	7-3	5-5	4-2
	2-#4 1-#6	40,000	7-4	8-5	6-9	5-0	3-10
		60,000	10-3	11-9	9-5	7-0	5-4
	2-#5	40,000	10-5	12-0	9-7	7-2	5-5
		60,000	12-7	14-5	11-6	8-7	6-6
	2-#6	40,000	12-4	14-2	11-4	8-5	5-7
		60,000	14-9	17-0	13-6	10-0	6-6
	Center distance $A^{k,l}$		3-9	4-11	3-2	1-9	1-0
16	Span without stirrups ^{i,j}		7-1	9-0	6-8	4-9	3-9
	1-#4	40,000	5-11	7-0	5-8	4-3	3-3
		60,000	7-3	8-7	6-11	5-2	3-11
	1-#5	40,000	7-4	8-9	7-0	5-3	4-0
		60,000	9-0	10-8	8-7	6-5	4-11
	2-#4 1-#6	40,000	8-4	9-11	7-11	5-11	4-6
		60,000	10-2	12-0	9-8	7-3	5-6
	2-#5	40,000	10-4	12-3	9-10	7-4	5-8
		60,000	14-4	17-1	13-8	10-3	7-10
	2-#6	40,000	14-1	16-9	13-5	10-1	7-0
		60,000	17-0	20-2	16-2	12-0	8-0
	Center distance ^{k,l}		4-9	6-8	4-4	2-5	1-5
20	Span without stirrups ^{i,j}		8-7	11-4	8-5	6-1	4-10
	1-#4	40,000	6-5	7-10	6-4	4-9	3-8
		60,000	7-10	9-7	7-9	5-10	4-5
	1-#5	40,000	8-0	9-9	7-11	5-11	4-6
		60,000	9-9	11-11	9-8	7-3	5-6

	2-#4 1-#6	40,000	9-0	11-1	8-11	6-9	5-2
		60,000	11-0	13-6	10-11	8-2	6-3
	2-#5	40,000	11-3	13-9	11-1	8-4	6-5
		60,000	15-8	19-2	15-6	11-8	8-11
	2-#6	40,000	15-5	18-10	15-2	11-5	8-6
		60,000	18-7	22-9	18-5	13-10	9-5
	Center distance $A^{k,l}$		5-7	8-4	5-5	3-1	1-10
24	Span without stirrups ^{i,j}		9-11	13-7	10-2	7-5	5-10
	1-#5	40,000	8-6	10-8	8-8	6-6	5-0
		60,000	10-5	13-0	10-7	8-0	6-1
	2-#4 1-#6	40,000	9-7	12-1	9-9	7-5	5-8
		60,000	11-9	14-9	11-11	9-0	6-11
	2-#5	40,000	12-0	15-0	12-2	9-2	7-1
		60,000	14-7	18-3	14-10	11-2	8-7
	2-#6	40,000	14-3	17-11	14-7	11-0	8-5
		60,000	19-11	25-0	20-3	15-3	10-10
	Center distance $A^{k,l}$		6-3	9-11	6-6	3-9	2-2

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

Note: Top and bottom reinforcement for lintels without stirrups shown in shaded cells shall be equal to or greater than that required for lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups.

- See Table R608.3 for tolerances permitted from nominal thickness.
- Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Note j.
- Table values are based on uniform loading. See Section R608.8.2 for lintels supporting concentrated loads.
- Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.
- Reserved.
- DR indicates design required.
- Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- Stirrups shall be fabricated from reinforcing bars with the same yield strength as that used for the main longitudinal reinforcement.
- Allowable clear span without stirrups applicable to all lintels of the same depth, D . Top and bottom reinforcement for lintels without stirrups shall be not less than the least amount of reinforcement required for a lintel of the same depth and loading condition with stirrups. All other spans require stirrups spaced at not more than $d/2$.
- Where concrete with a minimum specified compressive strength of 3,000 psi is used, clear spans for lintels without stirrups shall be permitted to be multiplied by 1.05. If the increased span exceeds the allowable clear span for a lintel of the

same depth and loading condition with stirrups, the top and bottom reinforcement shall be equal to or greater than that required for a lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups that has been increased.

k. Center distance, A , is the center portion of the clear span where stirrups are not required. This is applicable to all longitudinal bar sizes and steel yield strengths.

l. Where concrete with a minimum specified compressive strength of 3,000 psi is used, center distance, A , shall be permitted to be multiplied by 1.10.

m. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel clear spans in the table greater than 18 feet are shown for interpolation and information only.

TABLE R608.8(6)

MAXIMUM ALLOWABLE CLEAR SPANS FOR 6-INCH-THICK WAFFLE-GRID LINTELS IN LOAD-BEARING WALLS^{a, b, c, d, e, f, o}

MAXIMUM ROOF CLEAR SPAN 40 FEET AND MAXIMUM FLOOR SPAN 32 FEET

LINTEL DEPTH, <i>D</i> ^g (inches)	NUMBER OF BARS AND BAR SIZE IN TOP AND BOTTOM OF LINTEL	STEEL YIELD STRENGTH ^h , <i>f_y</i> (psi)	DESIGN LOADING CONDITION DETERMINED FROM TABLE R608.8(1)				
			1	2	3	4	5
			Maximum clear span of lintel (feet - inches)				
8 ⁱ	Span without stirrups ^{k, l}		2-7	2-9	2-1	2-0	2-0
	1-#4	40,000	5-2	5-5	4-3	3-3	2-4
		60,000	5-9	6-3	4-3	3-3	2-4
	1-#5	40,000	5-9	6-3	4-3	3-3	2-4
		60,000	5-9	6-3	4-3	3-3	2-4
	2-#4 1-#6	40,000	5-9	6-3	4-3	3-3	2-4
		60,000	DR	DR	DR	DR	DR
	Center distance <i>A</i> ^{m, n}		0-9	0-10	0-6	0-5	STL
12 ⁱ	Span without stirrups ^{k, l}		2-11	3-1	2-7	2-4	2-1
	1-#4	40,000	5-9	6-2	4-10	4-1	3-2
		60,000	8-0	8-7	6-9	5-5	3-11
	1-#5	40,000	8-1	8-9	6-11	5-5	3-11
		60,000	9-1	10-3	7-0	5-5	3-11
	2-#4 1-#6	40,000	9-1	9-9	7-0	5-5	3-11
	Center distance <i>A</i> ^{m, n}		1-3	1-5	0-11	0-8	STL
	Span without stirrups ^{k, l}		4-0	4-4	3-7	3-3	2-10
16 ⁱ	1-#4	40,000	6-7	7-3	5-9	4-10	3-9
		60,000	8-0	8-10	7-0	5-11	4-7
	1-#5	40,000	8-2	9-0	7-2	6-0	4-8

			60,000	11-5	12-6	9-9	7-7	5-6
	2-#4 1-#6		40,000	10-7	11-7	9-3	7-7	5-6
			60,000	12-2	14-0	9-9	7-7	5-6
	2-#5		40,000	12-2	14-2	9-9	7-7	5-6
			60,000	DR	DR	DR	DR	DR
	Center distance $A^{m,n}$			1-8	2-0	1-3	0-11	STL
20 ⁱ	Span without stirrups ^{k,l}			5-0	5-6	4-7	4-1	3-8
	1-#4		40,000	7-2	8-2	6-6	5-6	4-3
			60,000	8-11	9-11	7-11	6-8	5-2
	1-#5		40,000	9-1	10-2	8-1	6-10	5-4
			60,000	12-8	14-2	11-3	9-6	7-1
	2-#4 1-#6		40,000	10-3	11-5	9-1	7-8	6-0
			60,000	14-3	15-11	12-5	9-9	7-1
	2-#5		40,000	14-6	16-3	12-1	9-6	6-11
			60,000	DR	DR	DR	DR	DR
	Center distance $A^{m,n}$			2-0	2-6	1-7	1-1	STL
24w ^j	Span without stirrups ^{k,l}			6-0	6-8	5-7	5-0	4-6
	1-#4		40,000	7-11	9-0	7-2	6-0	4-8
			60,000	9-8	10-11	8-9	7-4	5-9
	1-#5		40,000	9-10	11-2	8-11	7-6	5-10
			60,000	12-0	13-7	10-10	9-2	7-2
	2-#4 1-#6		40,000	11-1	12-7	10-1	8-6	6-7
			60,000	15-6	17-7	14-0	11-10	8-7
	2-#5		40,000	15-6	17-11	13-4	10-7	7-10
			60,000	DR	DR	DR	DR	DR
	Center distance $A^{m,n}$			2-4	3-0	1-11	1-4	STL

For SI: 1 inch = 25.4 mm, 1 pound per square foot = 0.0479 kPa, 1 foot = 304.8 mm, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

- Where lintels are formed with waffle-grid forms, form material shall be removed, if necessary, to create top and bottom flanges of the lintel that are not less than 3 inches in depth (in the vertical direction), are not less than 5 inches in width for 6-inch-nominal waffle-grid forms and not less than 7 inches in width for 8-inch-nominal waffle-grid forms. See Figure R608.8(3). Flat form lintels shall be permitted in place of waffle-grid lintels. See Tables R608.8(2) through R608.8(5).
- See Table R608.3 for tolerances permitted from nominal thicknesses and minimum dimensions and spacing of cores.
- Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Notes I and n. Table values are based on uniform loading. See Section R608.8.2 for lintels supporting concentrated loads.
- Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.

- e. Reserved.
- f. DR indicates design required. STL - stirrups required throughout lintel.
- g. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- h. Stirrups shall be fabricated from reinforcing bars with the same yield strength as that used for the main longitudinal reinforcement.
- i. Lintels less than 24 inches in depth with stirrups shall be formed from flat-walls forms [see Tables R608.8(2) through R608.8(5)], or, if necessary, form material shall be removed from waffle-grid forms so as to provide the required cover for stirrups. Allowable spans for lintels formed with flat-wall forms shall be determined from Tables R608.8(2) through R608.8(5).
- j. Where stirrups are required for 24-inch deep lintels, the spacing shall not exceed 12 inches on center.
- k. Allowable clear span without stirrups applicable to all lintels of the same depth, D . Top and bottom reinforcement for lintels without stirrups shall be not less than the least amount of reinforcement required for a lintel of the same depth and loading condition with stirrups. All other spans require stirrups spaced at not more than $d/2$.
- l. Where concrete with a minimum specified compressive strength of 3,000 psi is used, clear spans for lintels without stirrups shall be permitted to be multiplied by 1.05. If the increased span exceeds the allowable clear span for a lintel of the same depth and loading condition with stirrups, the top and bottom reinforcement shall be equal to or greater than that required for a lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups that has been increased.
- m. Center distance, A , is the center portion of the span where stirrups are not required. This is applicable to all longitudinal bar sizes and steel yield strengths.
- n. Where concrete with a minimum specified compressive strength of 3,000 psi is used, center distance, A , shall be permitted to be multiplied by 1.10.
- o. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel spans in the table greater than 18 feet are shown for interpolation and information only.

TABLE R608.8(7)

MAXIMUM ALLOWABLE CLEAR SPANS FOR 8-INCH-THICK WAFFLE-GRID LINTELS IN LOAD-BEARING WALLS^{a, b, c, d, e, f, o}

MAXIMUM ROOF CLEAR SPAN 40 FEET AND MAXIMUM FLOOR CLEAR SPAN 32 FEET

LINTEL DEPTH, <i>D</i> ^g (inches)	NUMBER OF BARS AND BAR SIZE IN TOP AND BOTTOM OF LINTEL	STEEL YIELD STRENGTH ^h , <i>f_y</i> (psi)	DESIGN LOADING CONDITION DETERMINED FROM TABLE R608.8(1)				
			1	2	3	4	5
			Maximum clear span of lintel (feet - inches)				
8 ⁱ	Span with stirrups ^{k, l}		2-6	2-9	2-1	2-0	2-0
	1-#4	40,000	4-5	4-9	3-9	3-0	2-3
		60,000	5-6	6-2	4-3	3-1	2-3
	1-#5	40,000	5-6	6-2	4-3	3-1	2-3
	Center distance <i>A</i> ^{m, n}		0-9	0-10	0-6	0-4	STL
12 ⁱ	Span without stirrups ^{k, l}		2-10	3-1	2-7	2-3	2-0
	1-#4	40,000	5-7	6-1	4-10	3-11	3-0

		60,000	6-9	7-5	5-11	4-9	3-8
	1-#5	40,000	6-11	7-7	6-0	4-10	3-9
		60,000	8-8	10-1	7-0	5-2	3-9
	2-#4 1-#6	40,000	8-8	9-10	7-0	5-2	3-9
		60,000	8-8	10-1	7-0	5-2	3-9
	Center distance $A^{m,n}$		1-2	1-5	0-11	0-7	STL
16 ⁱ	Span without stirrups ^{k,l}		3-10	4-3	3-7	3-2	2-10
	1-#4	40,000	6-5	7-2	5-9	4-8	3-7
		60,000	7-9	8-9	7-0	5-8	4-4
	1-#5	40,000	7-11	8-11	7-1	5-9	4-5
		60,000	9-8	10-11	8-8	7-0	5-2
	2-#4 1-#6	40,000	9-0	10-1	8-0	6-6	5-0
		60,000	11-5	13-10	9-8	7-2	5-2
	Center distance $A^{m,n}$		1-6	1-11	1-3	0-10	STL
20 ⁱ	Span without stirrups ^{k,l}		4-10	5-5	4-7	4-0	3-7
	1-#4	40,000	7-0	8-1	6-5	5-3	4-1
		60,000	8-7	9-10	7-10	6-5	4-11
	1-#5	40,000	8-9	10-1	8-0	6-6	5-1
		60,000	10-8	12-3	9-10	8-0	6-2
	2-#4 1-#6	40,000	9-10	11-4	9-1	7-4	5-8
		60,000	12-0	13-10	11-0	9-0	6-8
	2-#5	40,000	12-3	14-1	11-3	8-11	6-6
		60,000	14-0	17-6	12-3	9-1	6-8
	Center distance $A^{m,n}$		1-10	2-5	1-7	1-0	STL
24 ^j	Span without stirrups ^{k,l}		5-9	6-7	5-6	4-11	4-5
	1-#4	40,000	7-6	8-10	7-1	5-9	4-6
		60,000	9-2	10-9	8-8	7-1	5-6
	1-#5	40,000	9-5	11-0	8-10	7-2	5-7
		60,000	11-5	13-5	10-9	8-9	6-10
	2-#4 1-#6	40,000	10-7	12-5	10-0	8-1	6-3
		60,000	12-11	15-2	12-2	9-11	7-8
	2-#5	40,000	13-2	15-6	12-5	9-11	7-5
		60,000	16-3	21-0	14-10	11-1	8-1

	2-#6	40,000	14-4	18-5	13-2	9-11	7-5
	Center distance A ^{m, n}		2-1	2-11	1-10	1-3	STL

For SI: 1 inch = 25.4 mm, 1 pound per square foot = 0.0479 kPa, 1 foot = 304.8 mm, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

- a. Where lintels are formed with waffle-grid forms, form material shall be removed, if necessary, to create top and bottom flanges of the lintel that are not less than 3 inches in depth (in the vertical direction), are not less than 5 inches in width for 6-inch-nominal waffle-grid forms and not less than 7 inches in width for 8-inch-nominal waffle-grid forms. See Figure R608.8(3). Flat form lintels shall be permitted in lieu of waffle-grid lintels. See Tables R608.8(2) through R608.8(5).
- b. See Table R608.3 for tolerances permitted from nominal thicknesses and minimum dimensions and spacing of cores.
- c. Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Notes l and n. Table values are based on uniform loading. See Section R608.8.2 for lintels supporting concentrated loads.
- d. Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.
- e. Reserved.
- f. DR indicates design required. STL - stirrups required throughout lintel.
- g. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- h. Stirrups shall be fabricated from reinforcing bars with the same yield strength as that used for the main longitudinal reinforcement.
- i. Lintels less than 24 inches in depth with stirrups shall be formed from flat-walls forms [see Tables R608.8(2) through R608.8(5)], or, if necessary, form material shall be removed from waffle-grid forms so as to provide the required cover for stirrups. Allowable spans for lintels formed with flat-wall forms shall be determined from Tables R608.8(2) through R608.8(5).
- j. Where stirrups are required for 24-inch deep lintels, the spacing shall not exceed 12 inches on center.
- k. Allowable clear span without stirrups applicable to all lintels of the same depth, D . Top and bottom reinforcement for lintels without stirrups shall be not less than the least amount of reinforcement required for a lintel of the same depth and loading condition with stirrups. All other spans require stirrups spaced at not more than $d/2$.
- l. Where concrete with a minimum specified compressive strength of 3,000 psi is used, clear spans for lintels without stirrups shall be permitted to be multiplied by 1.05. If the increased span exceeds the allowable clear span for a lintel of the same depth and loading condition with stirrups, the top and bottom reinforcement shall be equal to or greater than that required for a lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups that has been increased.
- m. Center distance, A , is the center portion of the span where stirrups are not required. This is applicable to all longitudinal bar sizes and steel yield strengths.
- n. Where concrete with a minimum specified compressive strength of 3,000 psi is used, center distance, A , shall be permitted to be multiplied by 1.10.
- o. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel spans in the table greater than 18 feet are shown for interpolation and information only.

TABLE R608.8(8)

MAXIMUM ALLOWABLE CLEAR SPANS FOR 6-INCH-THICK SCREEN-GRID LINTELS IN LOAD-BEARING WALLS^{a, b, c, d, e, f, p}

ROOF CLEAR SPAN 40 FEET AND FLOOR CLEAR SPAN 32 FEET

LINTEL DEPTH, <i>D</i> ^g (inches)	NUMBER OF BARS AND BAR SIZE IN TOP AND BOTTOM OF LINTEL	STEEL YIELD STRENGTH ^h , <i>f_y</i> (psi)	DESIGN LOADING CONDITION DETERMINED FROM TABLE R608.8(1)				
			1	2	3	4	5
			Maximum clear span of lintel (feet - inches)				
12 ^{i,j}	Span without stirrups		2-9	2-11	2-5	2-3	2-0
16 ^{i,j}	Span without stirrups		3-9	4-0	3-5	3-1	2-9
20 ^{i,j}	Span without stirrups		4-9	5-1	4-4	4-0	3-7
24 ^k	Span without stirrups ^{l,m}		5-8	6-3	5-3	4-10	4-4
	1-#4	40,000	7-11	9-0	7-2	6-1	4-9
		60,000	9-9	11-0	8-9	7-5	5-9
	1-#5	40,000	9-11	11-2	8-11	7-7	5-11
		60,000	12-1	13-8	10-10	9-3	7-2
	2-#4 1-#6	40,000	11-2	12-8	10-1	8-7	6-8
		60,000	15-7	17-7	13-4	10-8	7-11
	2-#5	40,000	14-11	18-0	12-10	10-3	7-8
		60,000	DR	DR	DR	DR	DR
	Center distance <i>A</i> ^{n,o}		2-0	2-6	1-7	1-2	STL

For SI: 1 inch = 25.4 mm, 1 pound per square foot = 0.0479 kPa, 1 foot = 304.8 mm, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

- a. Where lintels are formed with screen-grid forms, form material shall be removed if necessary to create top and bottom flanges of the lintel that are not less than 5 inches in width and not less than 2.5 inches in depth (in the vertical direction). See Figure R608.8(4). Flat form lintels shall be permitted in lieu of screen-grid lintels. See Tables R608.8(2) through R608.8(5).
- b. See Table R608.3 for tolerances permitted from nominal thickness and minimum dimensions and spacings of cores.
- c. Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Notes m and o. Table values are based on uniform loading. See Section R608.7.2.1 for lintels supporting concentrated loads.
- d. Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.
- e. Reserved.
- f. DR indicates design required. STL indicates stirrups required throughout lintel.
- g. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- h. Stirrups shall be fabricated from reinforcing bars with the same yield strength as that used for the main longitudinal reinforcement.
- i. Stirrups are not required for lintels less than 24 inches in depth fabricated from screen-grid forms. Top and bottom reinforcement shall consist of a No. 4 bar having a yield strength of 40,000 psi or 60,000 psi.
- j. Lintels between 12 and 24 inches in depth with stirrups shall be formed from flat-wall forms [see Tables R608.8(2) through R608.8(5)], or form material shall be removed from screen-grid forms to provide a concrete section comparable to that required for a flat wall. Allowable spans for flat lintels with stirrups shall be determined from Tables R608.8(2) through R608.8(5).

- k. Where stirrups are required for 24-inch deep lintels, the spacing shall not exceed 12 inches on center.
- l. Allowable clear span without stirrups applicable to all lintels of the same depth, D . Top and bottom reinforcement for lintels without stirrups shall be not less than the least amount of reinforcement required for a lintel of the same depth and loading condition with stirrups. All other spans require stirrups spaced at not more than 12 inches.
- m. Where concrete with a minimum specified compressive strength of 3,000 psi is used, clear spans for lintels without stirrups shall be permitted to be multiplied by 1.05. If the increased span exceeds the allowable clear span for a lintel of the same depth and loading condition with stirrups, the top and bottom reinforcement shall be equal to or greater than that required for a lintel of the same depth and loading condition that has an allowable clear span that is equal to or greater than that of the lintel without stirrups that has been increased.
- n. Center distance, A , is the center portion of the span where stirrups are not required. This is applicable to all longitudinal bar sizes and steel yield strengths.
- o. Where concrete with a minimum specified compressive strength of 3,000 psi is used, center distance, A , shall be permitted to be multiplied by 1.10.
- p. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel spans in the table greater than 18 feet are shown for interpolation and information only.

TABLE R608.8(9)

MAXIMUM ALLOWABLE CLEAR SPANS FOR FLAT LINTELS WITHOUT STIRRUPS IN NONLOAD-BEARING WALLS^{a, b, c, d, e, g, h}

LINTEL DEPTH, D^f (inches)	NUMBER OF BARS AND BAR SIZE	STEEL YIELD STRENGTH, f_y (psi)	NOMINAL WALL THICKNESS (inches)							
			4		6		8		10	
			Lintel Supporting							
			Concrete Wall	Light-framed Gable	Concrete Wall	Light-framed Gable	Concrete Wall	Light-framed Gable	Concrete Wall	Light-framed Gable
			Maximum Clear Span of Lintel (feet - inches)							
8	1-#4	40,000	10-11	11-5	9-7	11-2	7-10	9-5	7-3	9-2
		60,000	12-5	11-7	10-11	13-5	9-11	13-2	9-3	12-10
	1-#5	40,000	12-7	11-7	11-1	13-8	10-1	13-5	9-4	13-1
		60,000	DR	DR	12-7	16-4	11-6	14-7	10-9	14-6
	2-#4 1-#6	40,000	DR	DR	12-0	15-3	10-11	15-0	10-2	14-8
		60,000	DR	DR	DR	DR	12-2	15-3	11-7	15-3
	2-#5	40,000	DR	DR	DR	DR	12-7	16-7	11-9	16-7
		60,000	DR	DR	DR	DR	DR	DR	13-3	16-7
	2-#6	40,000	DR	DR	DR	DR	DR	DR	13-2	17-8
		60,000	DR	DR	DR	DR	DR	DR	DR	DR
12	1-#4	40,000	11-5	9-10	10-6	12-0	9-6	11-6	8-9	11-1
		60,000	11-5	9-10	11-8	13-3	10-11	14-0	10-1	13-6
	1-#5	40,000	11-5	9-10	11-8	13-3	11-1	14-4	10-3	13-9

		60,000	11-5	9-10	11-8	13-3	11-10	16-0	11-9	16-9
	2-#4 1-#6	40,000	DR	DR	11-8	13-3	11-10	16-0	11-2	15-6
		60,000	DR	DR	11-8	13-3	11-10	16-0	11-11	18-4
	2-#5	40,000	DR	DR	11-8	13-3	11-10	16-0	11-11	18-4
		60,000	DR	DR	11-8	13-3	11-10	16-0	11-11	18-4
16	1-#4	40,000	13-6	13-0	11-10	13-8	10-7	12-11	9-11	12-4
		60,000	13-6	13-0	13-8	16-7	12-4	15-9	11-5	15-0
	1-#5	40,000	13-6	13-0	13-10	17-0	12-6	16-1	11-7	15-4
		60,000	13-6	13-0	13-10	17-1	14-0	19-7	13-4	18-8
	2-#4 1-#6	40,000	13-6	13-0	13-10	17-1	13-8	18-2	12-8	17-4
		60,000	13-6	13-0	13-10	17-1	14-0	20-3	14-1	—
	2-#5	40,000	13-6	13-0	13-10	17-1	14-0	20-3	14-1	—
		60,000	DR	DR	13-10	17-1	14-0	20-3	14-1	—
20	1-#4	40,000	14-11	15-10	13-0	14-10	11-9	13-11	10-10	13-2
		60,000	15-3	15-10	14-11	18-1	13-6	17-0	12-6	16-2
	1-#5	40,000	15-3	15-10	15-2	18-6	13-9	17-5	12-8	16-6
		60,000	15-3	15-10	15-8	20-5	15-9	—	14-7	20-1
	2-#4 1-#6	40,000	15-3	15-10	15-8	20-5	14-11	—	13-10	—
		60,000	15-3	15-10	15-8	20-5	15-10	—	15-11	—
	2-#5	40,000	15-3	15-10	15-8	20-5	15-10	—	15-11	—
		60,000	15-3	15-10	15-8	20-5	15-10	—	15-11	—
24	1-#4	40,000	16-1	17-1	13-11	15-10	12-7	14-9	11-8	13-10
		60,000	16-11	18-5	16-1	19-3	14-6	18-0	13-5	17-0
	1-#5	40,000	16-11	18-5	16-3	19-8	14-9	18-5	13-8	17-4
		60,000	16-11	18-5	17-4	—	17-0	—	15-8	—
	2-#4 1-#6	40,000	16-11	18-5	17-4	—	16-1	—	14-10	—
		60,000	16-11	18-5	17-4	—	17-6	—	17-1	—
	2-#5	40,000	16-11	18-5	17-4	—	17-6	—	17-4	—
		60,000	16-11	18-5	17-4	—	17-6	—	17-8	—

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

- See Table R608.3 for tolerances permitted from nominal thickness.
- Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Note e.
- Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ inch, whichever is less.

- d. Linear interpolation between lintels depths, D , is permitted provided the two cells being used to interpolate are shaded.
- e. Where concrete with a minimum specified compressive strength of 3,000 psi is used, spans in cells that are shaded shall be permitted to be multiplied by 1.05.
- f. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.
- g. DR indicates design required.
- h. The maximum clear opening width between two solid wall segments shall be 18 feet. See Section R608.7.2.1. Lintel spans in the table greater than 18 feet are shown for interpolation and information purposes only.

TABLE R608.8(10)

MAXIMUM ALLOWABLE CLEAR SPANS FOR WAFFLE-GRID AND SCREEN-GRID LINTELS WITHOUT STIRRUPS IN NONLOAD-BEARING WALLS^{c, d, e, f, g}

LINTEL DEPTH ^h , <i>D</i> (inches)	FORM TYPE AND NOMINAL WALL THICKNESS (inches)					
	6-inch Waffle-grid ^a		8-inch Waffle-grid ^a		6-inch Screen-grid ^b	
	Lintel supporting					
	Concrete Wall	Light-framed Gable	Concrete Wall	Light-framed Gable	Concrete Wall	Light-framed Gable
	Maximum Clear Span of Lintel (feet - inches)					
8	10-3	8-8	8-8	8-3	—	—
12	9-2	7-6	7-10	7-1	8-8	6-9
16	10-11	10-0	9-4	9-3	—	—
20	12-5	12-2	10-7	11-2	—	—
24	13-9	14-2	11-10	12-11	13-0	12-9

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, Grade 40 = 280 MPa, Grade 60 = 420 MPa.

- a. Where lintels are formed with waffle-grid forms, form material shall be removed, if necessary, to create top and bottom flanges of the lintel that are not less than 3 inches in depth (in the vertical direction), are not less than 5 inches in width for 6-inch waffle-grid forms and not less than 7 inches in width for 8-inch waffle-grid forms. See Figure R608.8(3). Flat form lintels shall be permitted in lieu of waffle-grid lintels. See Tables R608.8(2) through R608.8(5).
- b. Where lintels are formed with screen-grid forms, form material shall be removed if necessary to create top and bottom flanges of the lintel that are not less than 5 inches in width and not less than 2.5 inches in depth (in the vertical direction). See Figure R608.8(4). Flat form lintels shall be permitted in lieu of screen-grid lintels. See Tables R608.8(2) through R608.8(5).
- c. See Table R608.3 for tolerances permitted from nominal thickness and minimum dimensions and spacing of cores.
- d. Table values are based on concrete with a minimum specified compressive strength of 2,500 psi. See Note g.
- e. Deflection criterion is $L/240$, where L is the clear span of the lintel in inches, or $1/2$ -inch, whichever is less.
- f. Top and bottom reinforcement shall consist of a No. 4 bar having a minimum yield strength of 40,000 psi.
- g. Where concrete with a minimum specified compressive strength of 3,000 psi is used, spans in shaded cells shall be permitted to be multiplied by 1.05.

- h. Lintel depth, D , is permitted to include the available height of wall located directly above the lintel, provided that the increased lintel depth spans the entire length of the lintel.

R608.8.1 Reinforcement Around Openings

Reinforcement shall be provided around openings in walls equal to or greater than 2 feet (610 mm) in width in accordance with this section and Figure R608.8(1), in addition to the minimum wall reinforcement required by Sections R404.1.3, R608.6 and R608.7. Vertical wall reinforcement required by this section is permitted to be used as reinforcement at the ends of solid wall segments required by Section R608.7.2.2.2 provided it is located in accordance with Section R608.8.1.2. Wall openings shall have a minimum depth of concrete over the width of the opening of 8 inches (203 mm) in flat walls and waffle-grid walls, and 12 inches (305 mm) in screen-grid walls. Wall openings in waffle-grid and screen-grid walls shall be located such that not less than one-half of a vertical core occurs along each side of the opening.

R608.8.1.1 Horizontal Reinforcement

Lintels complying with Section R608.8.2 shall be provided above wall openings equal to or greater than 2 feet (610 mm) in width.

Exception: Continuous horizontal wall reinforcement placed within 12 inches (305 mm) of the top of the wall story as required in Sections R404.1.2.2 and R608.6.2 is permitted in lieu of top or bottom lintel reinforcement required by Section R608.8.2 provided that the continuous horizontal wall reinforcement meets the location requirements specified in Figures R608.8(2), R608.8(3), and R608.8(4) and the size requirements specified in Tables R608.8(2) through R608.8(10).

Openings equal to or greater than 2 feet (610 mm) in width shall have not less than one No. 4 bar placed within 12 inches (305 mm) of the bottom of the opening. See Figure R608.8(1).

Horizontal reinforcement placed above and below an opening shall extend beyond the edges of the opening the dimension required to develop the bar in tension in accordance with Section R608.5.4.4.

R608.8.1.2 Vertical Reinforcement

Not less than one No. 4 bar [Grade 40 (280 MPa)] shall be provided on each side of openings equal to or greater than 2 feet (610 mm) in width. The vertical reinforcement required by this section shall extend the full height of the wall story and shall be located within 12 inches (305 mm) of each side of the opening. The vertical reinforcement required on each side of an opening by this section is permitted to serve as reinforcement at the ends of solid wall segments in accordance with Section R608.7.2.2.2, provided it is located as required by the applicable detail in Figure R608.7(2). Where the vertical reinforcement required by this section is used to satisfy the requirements of Section R608.7.2.2.2 in waffle- and screen-grid walls, a concrete flange shall be created at the ends of the solid wall segments in accordance with Table R608.7(4), Note e. In the top-most story, the reinforcement shall terminate in accordance with Section R608.6.4.

R608.8.2 Lintels

Lintels shall be provided over all openings equal to or greater than 2 feet (610 mm) in width. Lintels with uniform loading shall conform to Sections R608.8.2.1 and R608.8.2.2, or Section R608.8.2.3. Lintels supporting concentrated loads, such as from roof or floor beams or girders, shall be designed in accordance with ACI 318.

R608.8.2.1 Lintels Designed for Gravity Load-Bearing Conditions

Where a lintel will be subjected to gravity load condition 1 through 5 of Table R608.8(1), the clear span of the lintel shall not exceed that permitted by Tables R608.8(2) through R608.8(8). The maximum clear span of lintels with and without stirrups in flat walls shall be determined in accordance with Tables R608.8(2) through R608.8(5), and constructed in accordance with Figure R608.8(2). The maximum clear span of lintels with and without stirrups in waffle-grid walls shall be determined in accordance with Tables R608.8(6) and R608.8(7), and constructed in accordance with Figure R608.8(3). The maximum clear span of lintels with and without stirrups in screen-grid walls shall be determined in accordance with Table R608.8(8), and constructed in accordance with Figure R608.8(4).

Where required by the applicable table, No. 3 stirrups shall be installed in lintels at a maximum spacing of $d/2$ where d equals the depth of the lintel, D , less the cover of the concrete as shown in Figures R608.8(2) through R608.8(4). The smaller value of d computed for the top and bottom bar shall be used to determine the maximum stirrup spacing. Where stirrups are required in a lintel with a single bar or two bundled bars in the top and bottom, they shall be fabricated like the letter "c" or "s" with 135-degree (2.36 rad) standard hooks at each end that comply with Section R608.5.4.5 and Figure R608.5.4(3) and installed as shown in Figures R608.8(2) through R608.8(4). Where two bars are required in the top and bottom of the lintel and the bars are not bundled, the bars shall be separated by not less than 1 inch (25 mm). The free end of the stirrups shall be fabricated with 90- or 135-degree (1.57 or 2.36 rad) standard hooks that comply with Section R608.5.4.5 and Figure R608.5.4(3) and installed as shown in Figures R608.8(2)

and R608.8(3). For flat, waffle-grid and screen-grid lintels, stirrups are not required in the center distance, A, portion of spans in accordance with Figure R608.8(1) and Tables R608.8(2) through R608.8(8). See Section R608.8.2.2, Item 5, for requirement for stirrups through out lintels with bundled bars.

R608.8.2.2 Bundled Bars in Lintels

It is permitted to bundle two bars in contact with each other in lintels if all of the following are observed:

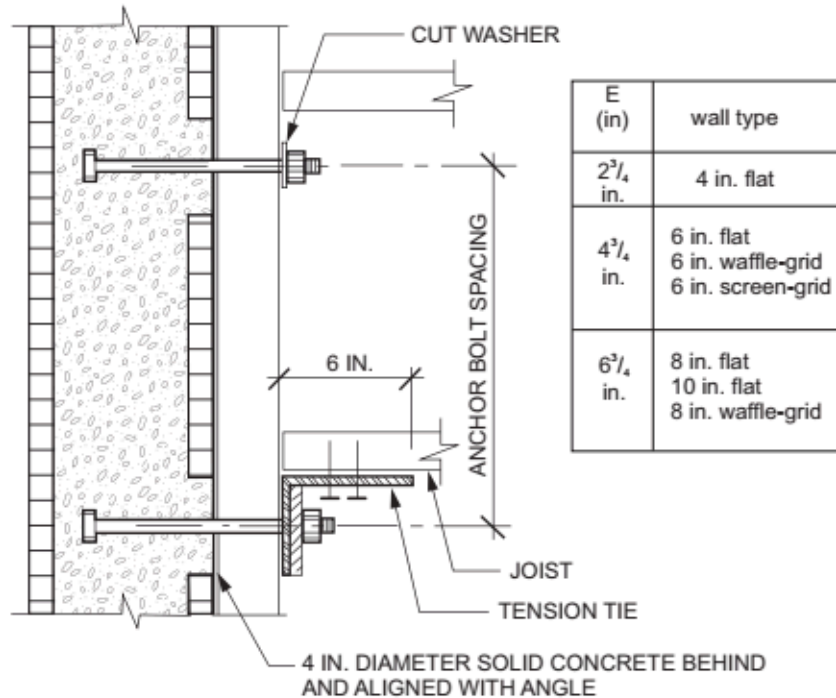
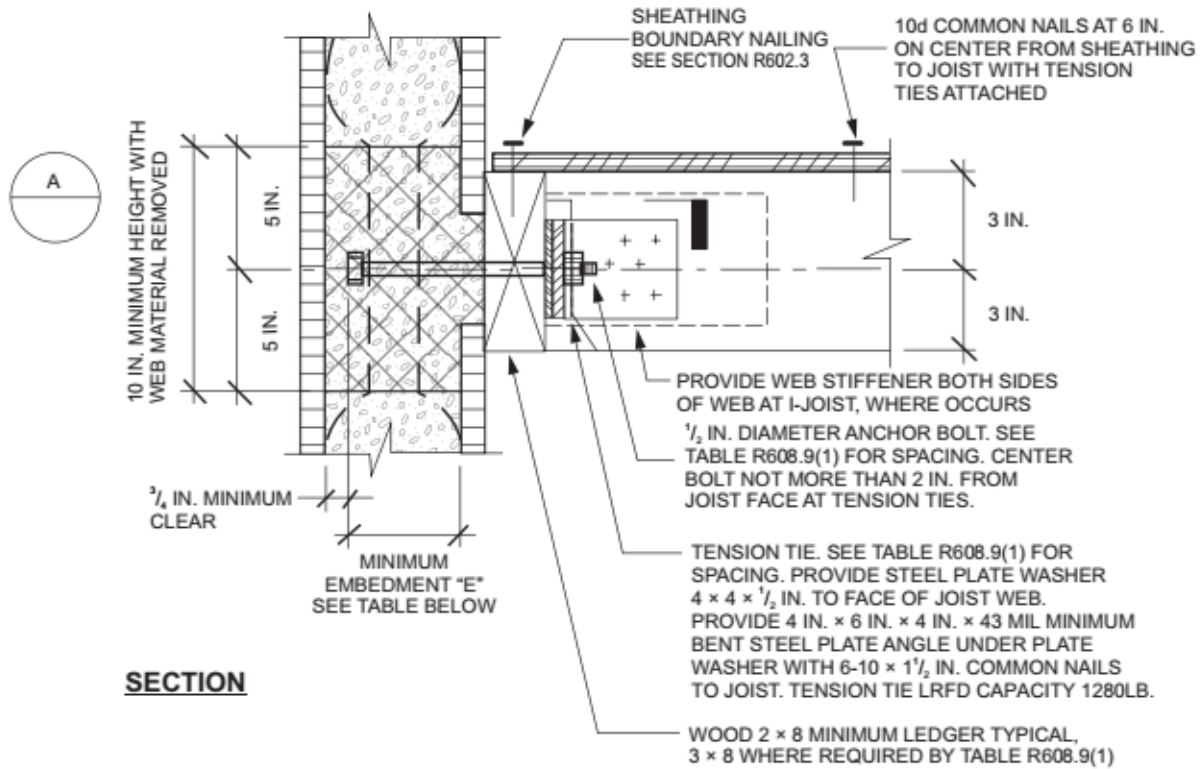
1. Bars equal to or less than No. 6 are bundled.
2. Where the wall thickness is not sufficient to provide not less than 3 inches (76 mm) of clear space beside bars (total on both sides) oriented horizontally in a bundle, the bundled bars shall be oriented in a vertical plane.
3. Where vertically oriented bundled bars terminate with standard hooks to develop the bars in tension beyond the support (see Section R608.5.4.4), the hook extensions shall be staggered to provide not less than 1 inch (25 mm) clear spacing between the extensions.
4. Bundled bars shall not be lap spliced within the lintel span and the length on each end of the lintel that is required to develop the bars in tension.
5. Bundled bars shall be enclosed within stirrups throughout the length of the lintel. Stirrups and the installation thereof shall comply with Section R608.8.2.1.

R608.8.2.3 Lintels Without Stirrups Designed for Nonload-Bearing Conditions

The maximum clear span of lintels without stirrups designed for nonload-bearing conditions of Table R608.8(1).1 shall be determined in accordance with this section. The maximum clear span of lintels without stirrups in flat walls shall be determined in accordance with Table R608.8(9), and the maximum clear span of lintels without stirrups in walls of waffle-grid or screen-grid construction shall be determined in accordance with Table R608.8(10).

R608.9 Requirements for Connections—General

Concrete walls shall be connected to footings, floors, ceilings and roofs in accordance with this section.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(1)

WOOD-FRAMED FLOOR TO SIDE OF CONCRETE WALL, FRAMING PERPENDICULAR

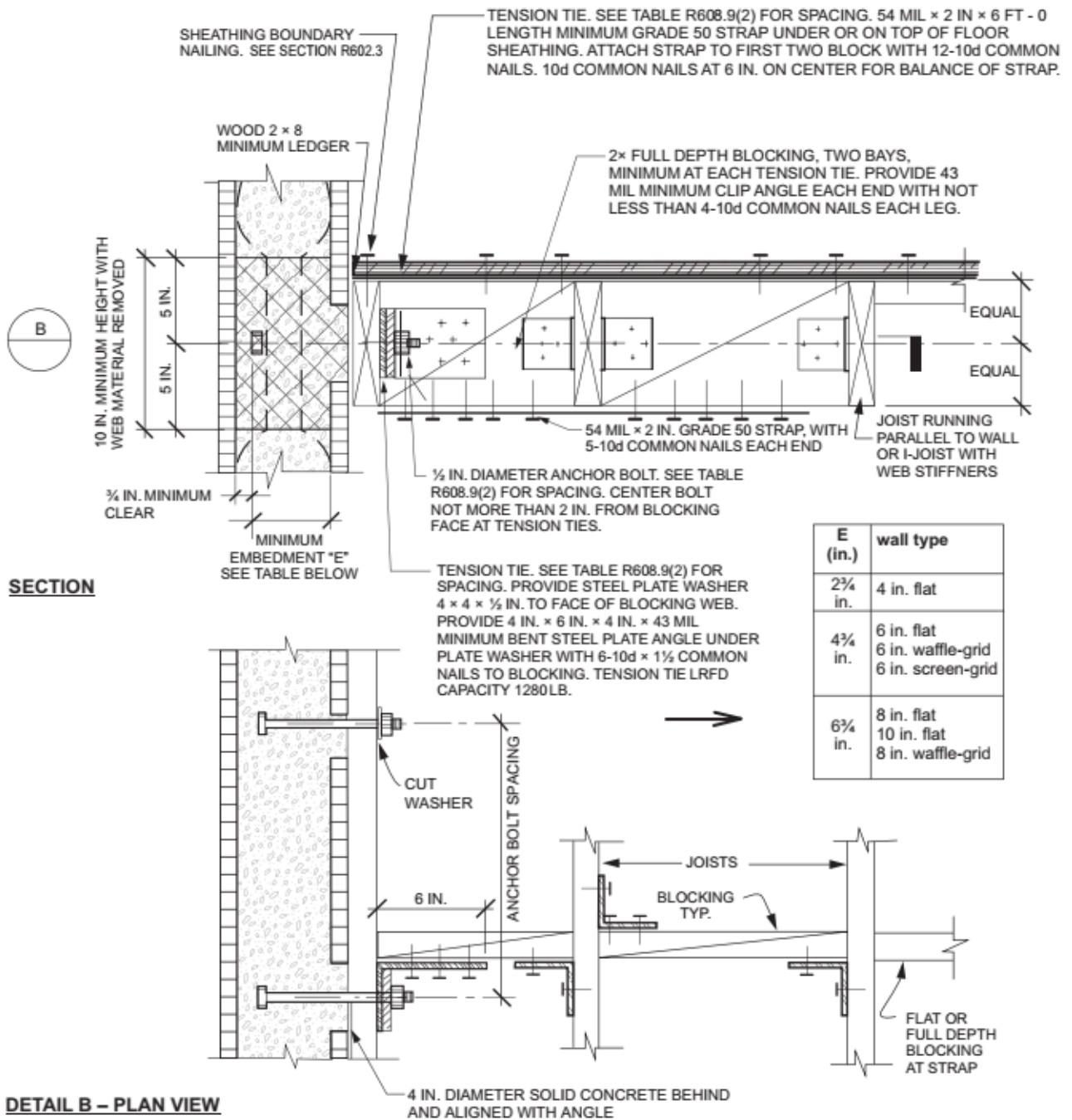
TABLE R608.9(1)

WOOD-FRAMED FLOOR TO SIDE OF CONCRETE WALL, FRAMING PERPENDICULAR^{a, b}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph)					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						
12	24						
12	36						
12	48						
16	16						
16	32						
16	48						
19.2	19.2						
19.2	38.4						

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(1). Use of this detail is permitted where a cell is not shaded and prohibited where shaded.
- Wall design per other provisions of Section R608 is required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(2)

WOOD-FRAMED FLOOR TO SIDE OF CONCRETE WALL, FRAMING PARALLEL

TABLE R608.9(2)

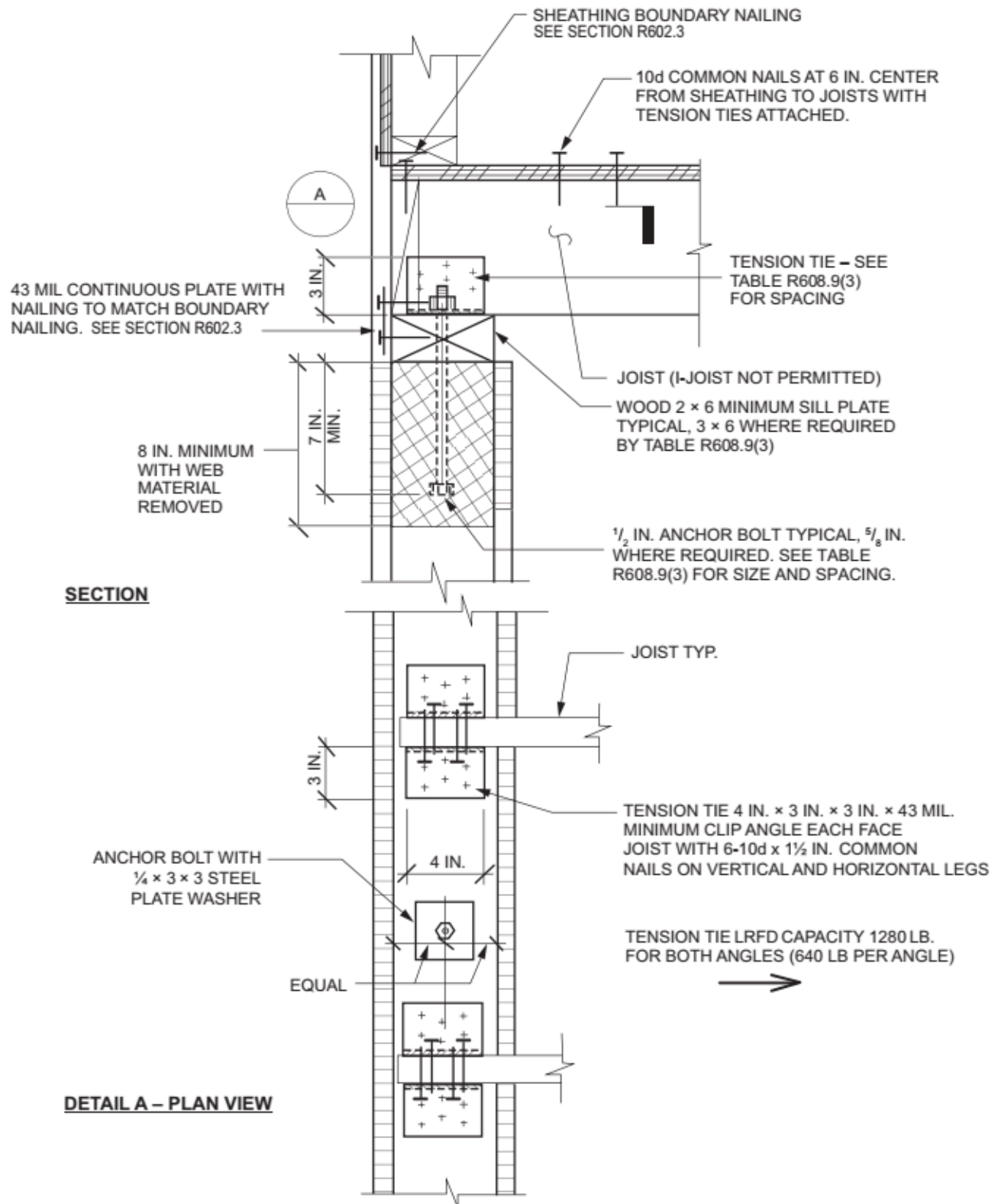
WOOD-FRAMED FLOOR TO SIDE OF CONCRETE WALL, FRAMING PARALLEL^{a, b}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115b	120B	130B	140B	150B	160B
				110C	119C	127C	136C

					110D	117D	125D
12	12						
12	24						
12	36						
12	48						
16	16						
16	32						
16	48						
19.2	19.2						
19.2	38.4						
24	24						
24	48						

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(2). Use of this detail is permitted where a cell is not shaded and prohibited where shaded.
- Wall design per other provisions of Section R608 is required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(3)

WOOD-FRAMED FLOOR TO TOP OF CONCRETE WALL FRAMING, PERPENDICULAR

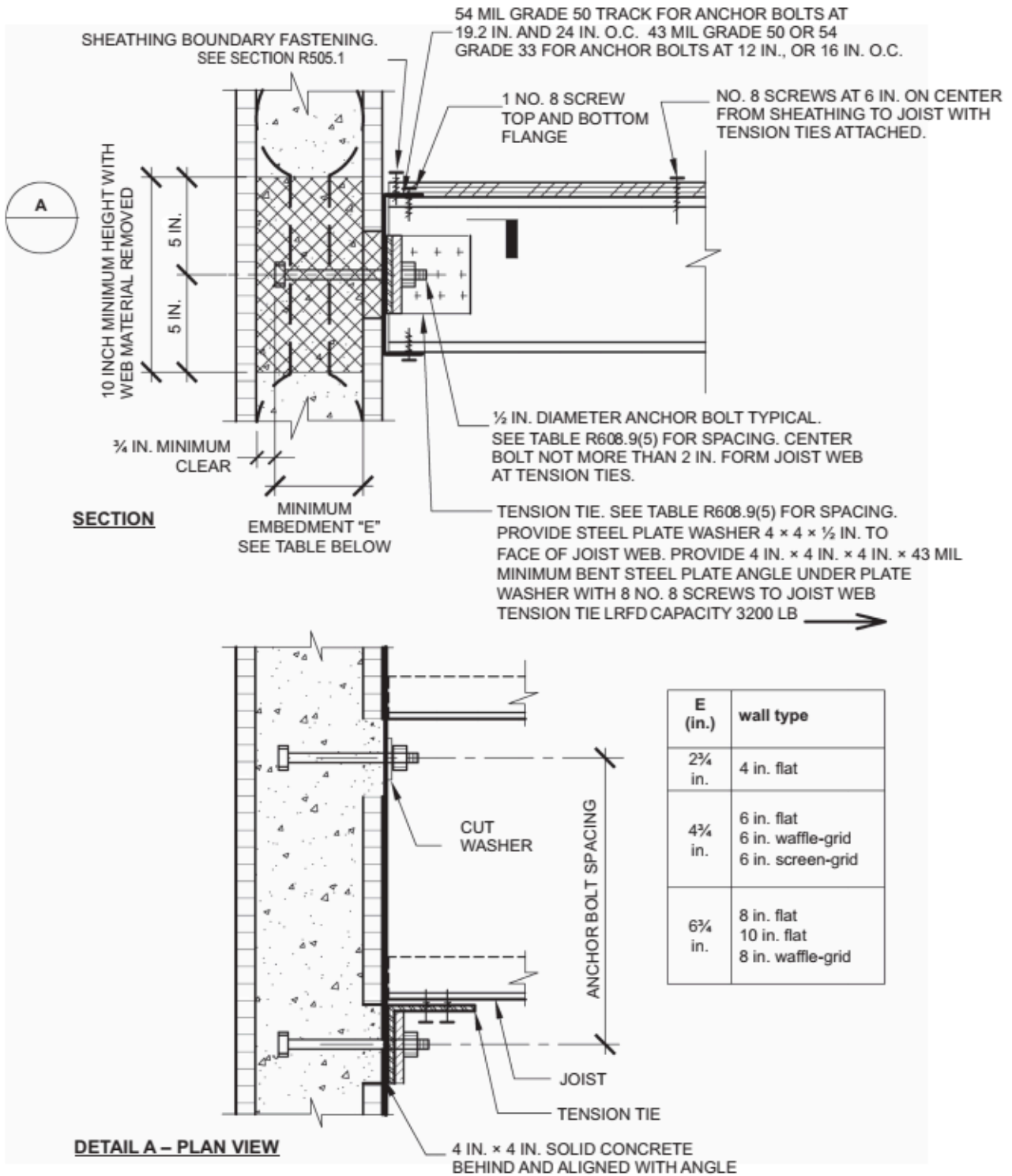
TABLE R608.9(3)

WOOD-FRAMED FLOOR TO TOP OF CONCRETE WALL, FRAMING PERPENDICULAR^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						6
12	24					6	6
12	36					6	6
12	48				6	6	6
16	16					6	6A
16	32				6	6	6A
16	48			6	6	6	6A
19.2	19.2				6A	6A	6B
19.2	38.4			6	6A	6A	6B
24	24			6A	6B	6B	6B
24	48		6	6A	6B	6B	8B

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(3). Use of this detail is permitted where cell is not shaded.
- Wall design per other provisions in Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.
- Numbers 6 and 8 indicate minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(3). For the remainder of the wall, see Note b.
- Letter "A" indicates that a minimum nominal 3 × 6 sill plate is required. Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt and a minimum nominal 3 × 6 sill plate are required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(4)

WOOD-FRAMED FLOOR TO TOP OF CONCRETE WALL, FRAMING PARALLEL

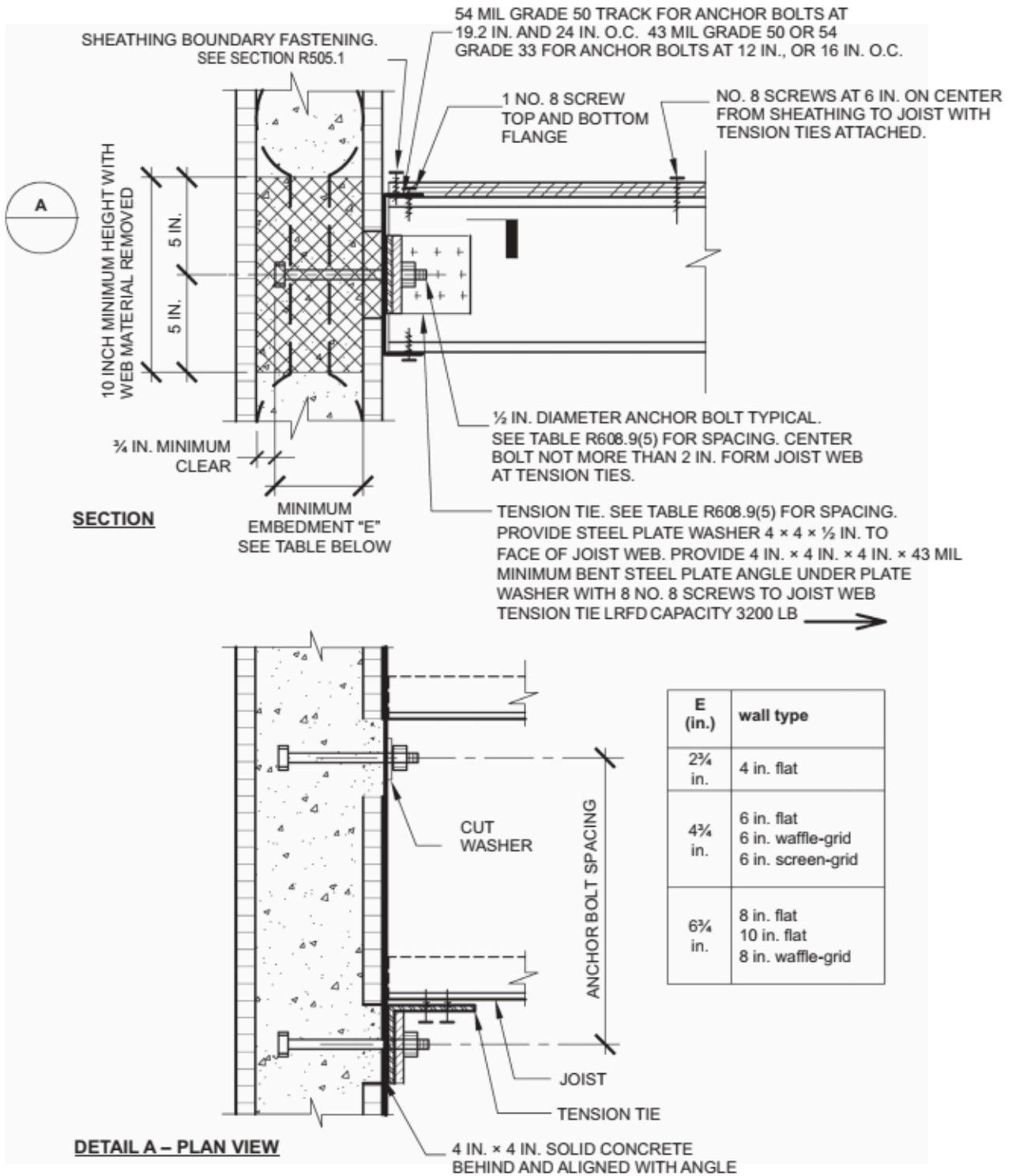
TABLE R608.9(4)

WOOD-FRAMED FLOOR TO TOP OF CONCRETE WALL, FRAMING PARALLEL^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						6
12	24					6	6
12	36					6	6
12	48				6	6	6
16	16					6	6A
16	32				6	6	6A
16	48			6	6	6	6A
19.2	19.2				6A	6A	6B
19.2	38.4			6	6A	6A	6B
24	24			6A	6B	6B	6B
24	48		6	6A	6B	6B	8B

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(4). Use of this detail is permitted where a cell is not shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.
- Numbers 6 and 8 indicate minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(4). For the remainder of the wall, see Note b.
- Letter "A" indicates that a minimum nominal 3 × 6 sill plate is required. Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt and a minimum nominal 3 × 6 sill plate are required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(5)

COLD-FORMED STEEL FLOOR TO SIDE OF CONCRETE WALL, FRAMING PERPENDICULAR

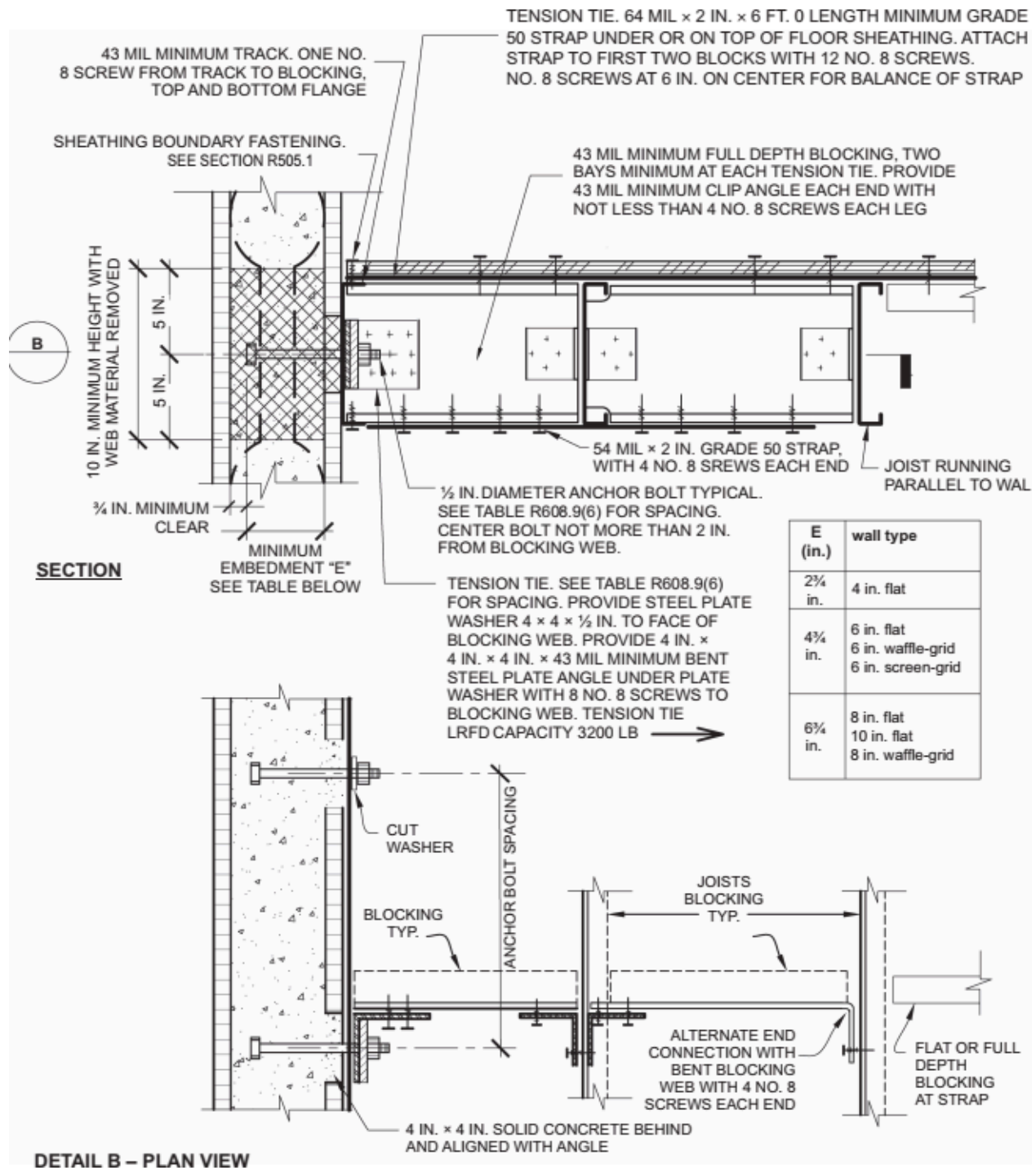
TABLE R608.9(5)

COLD-FORMED STEEL-FRAMED FLOOR TO SIDE OF CONCRETE WALL, FRAMING PERPENDICULAR^{a, b, c}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						
12	24						
12	36						
12	48						
16	16						
16	32						
16	48						
19.2	19.2						
19.2	38.4						
24	24						
24	48						

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.4470 m/s.

- This table is for use with the detail in Figure R608.9(5). Use of this detail is permitted where a cell is not shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(6)

COLD-FORMED STEEL FLOOR TO SIDE OF CONCRETE WALL, FRAMING PARALLEL

TABLE R608.9(6)

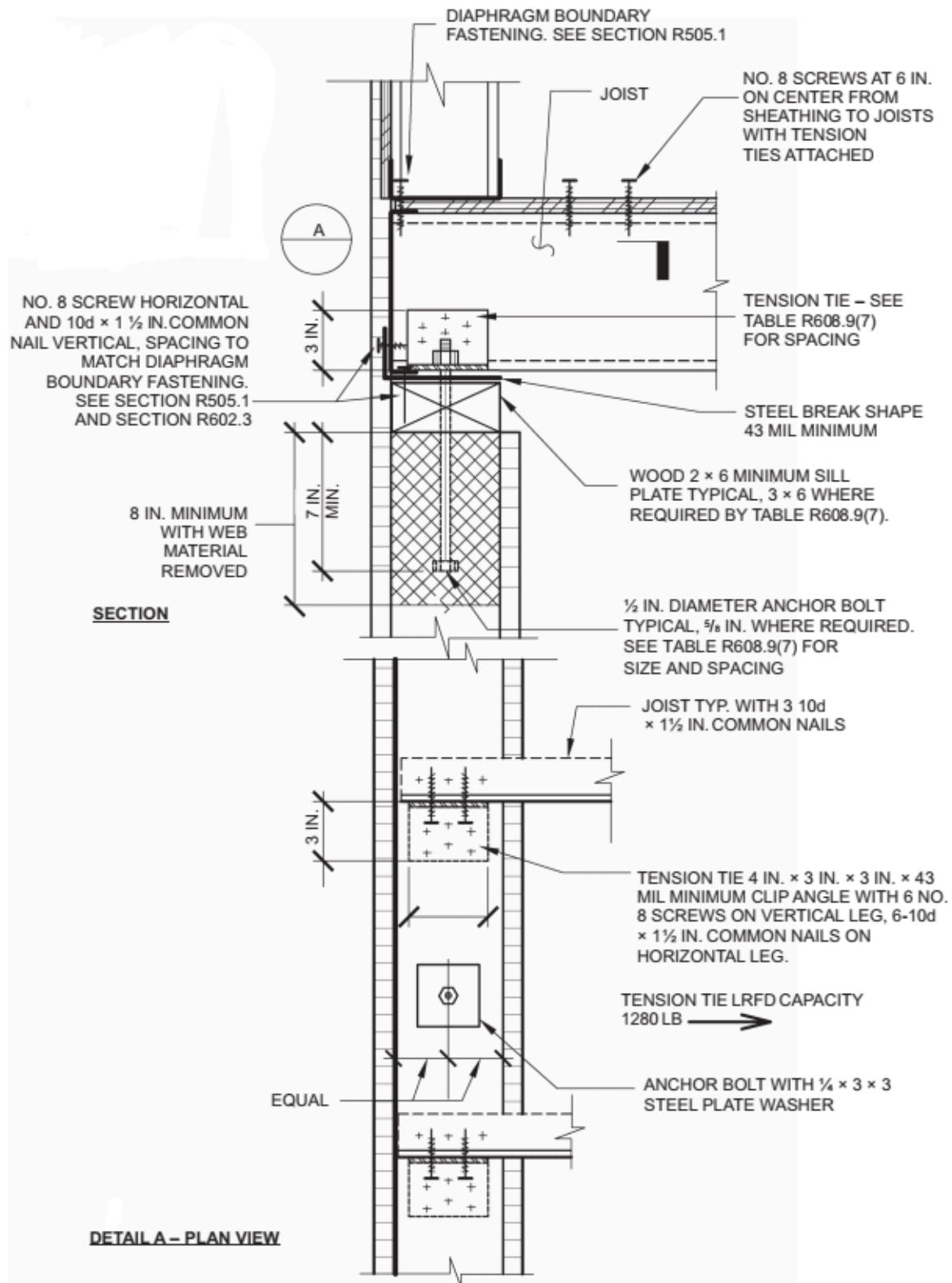
COLD-FORMED STEEL-FRAMED FLOOR TO SIDE OF CONCRETE WALL, FRAMING PARALLEL^{a, b, c}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115B	120B	130B	140B	150B	160B

				110C	119C	127C	136C
					110D	117D	125D
12	12						
12	24						
12	36						
12	48						
16	16						
16	32						
16	48						
19.2	19.2						
19.2	38.4						
24	24						
24	48						

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(6). Use of this detail is permitted where a cell is not shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(7)

COLD-FORMED STEEL FLOOR TO TOP OF CONCRETE WALL, FRAMING PERPENDICULAR

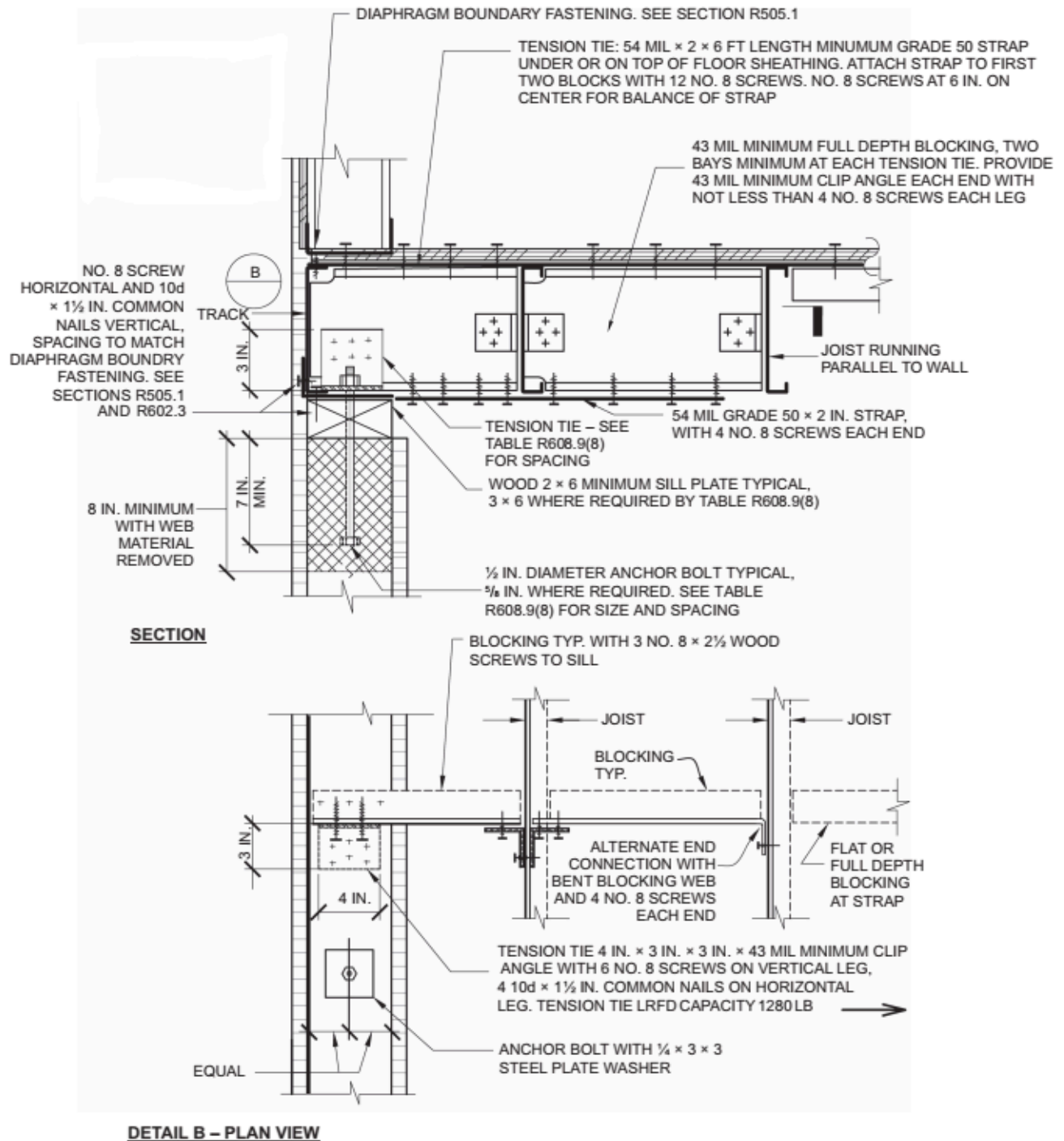
TABLE R608.9(7)

COLD-FORMED STEEL-FRAMED FLOOR TO TOP OF CONCRETE WALL, FRAMING PERPENDICULAR^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED AND WIND EXPOSURE CATEGORY (mph)					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						6
12	24					6	6
16	16					6	6A
16	32				6	6	6A
19.2	19.2				6A	6A	6B
19.2	38.4			6	6A	6A	6B
24	24			6A	6B	6B	6B

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(7). Use of this detail is permitted where a cell is not shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.
- Number 6 indicates minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(7). For the remainder of the wall, see Note b.
- Letter "A" indicates that a minimum nominal 3 × 6 sill plate is required. Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt and a minimum nominal 3 × 6 sill plate are required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(8)

COLD-FORMED STEEL FLOOR TO TOP OF CONCRETE WALL, FRAMING PARALLEL

TABLE R608.9(8)

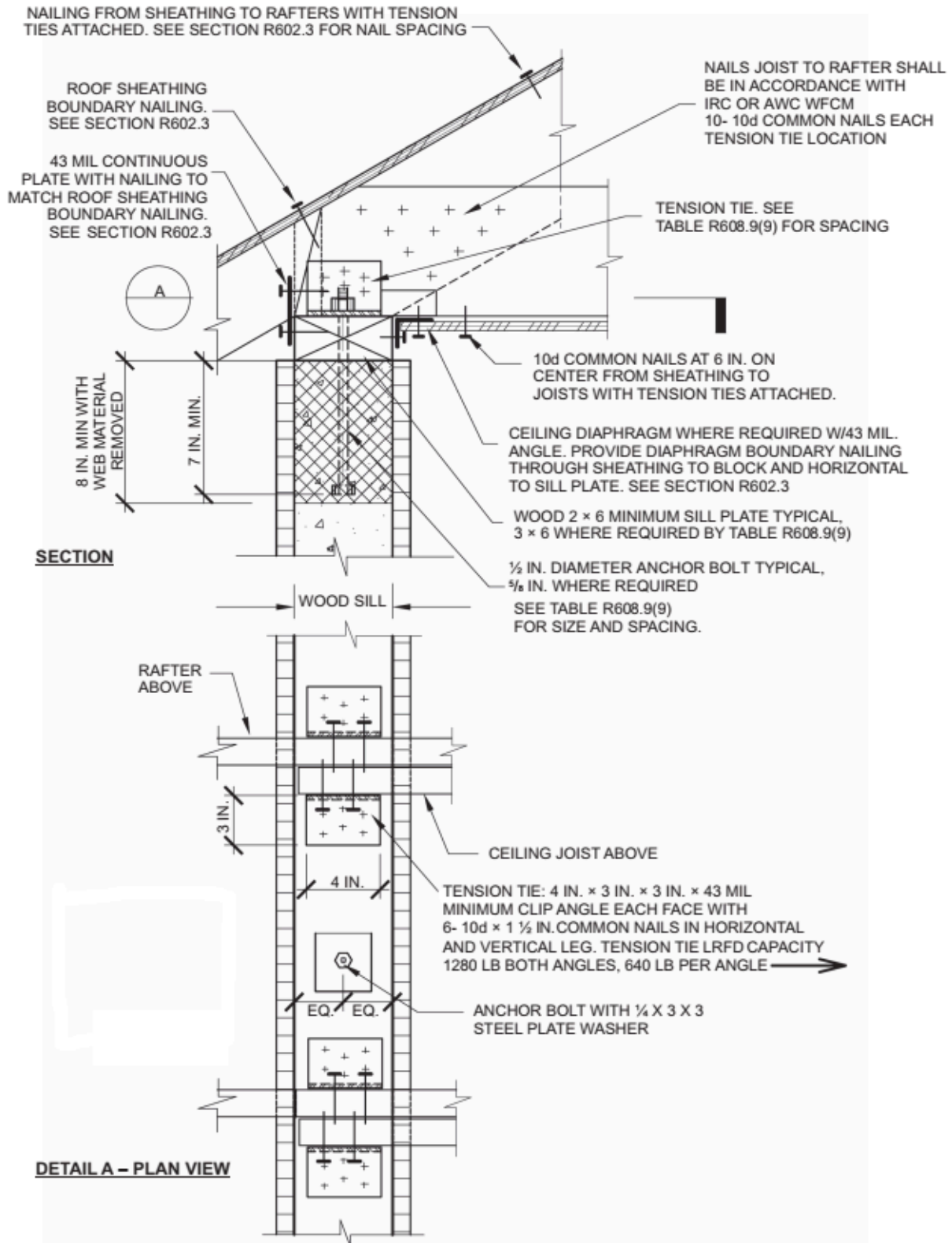
COLD-FORMED STEEL-FRAMED FLOOR TO TOP OF CONCRETE WALL, FRAMING PARALLEL^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED AND WIND EXPOSURE CATEGORY (mph)
---------------------------------	---------------------------------	--

		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						6
12	24					6	6
16	16					6	6A
16	32				6	6	6A
19.2	19.2				6A	6A	6B
19.2	38.4			6	6A	6A	6B
24	24			6A	6B	6B	6B

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(8). Use of this detail is permitted where a cell is not shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.
- Number 6 indicates minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(8). For the remainder of the wall, see Note b.
- Letter "A" indicates that a minimum nominal 3 × 6 sill plate is required. Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt and a minimum nominal 3 × 6 sill plate are required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(9)

WOOD-FRAMED ROOF TO TOP OF CONCRETE WALL, FRAMING PERPENDICULAR

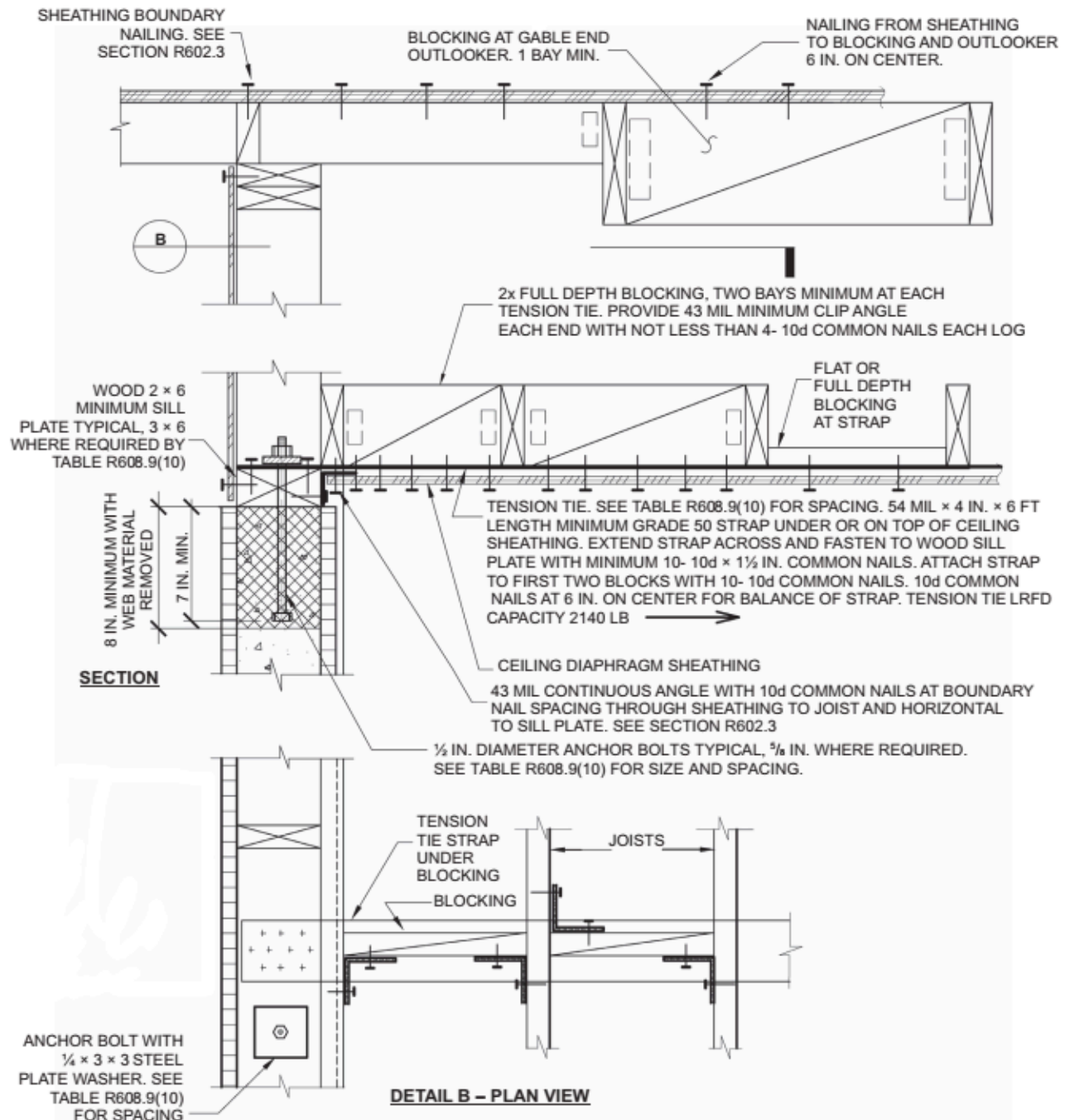
TABLE R608.9(9)

WOOD-FRAMED ROOF TO TOP OF CONCRETE WALL, FRAMING PERPENDICULAR^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						6
12	24						6
12	36					6	6
12	48				6	6	6
16	16					6	6
16	32					6	6
16	48				6	6	6
19.2	19.2					6	6
19.2	38.4				6	6	
24	24				6		
24	48			6	8B		

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(9). Use of this detail is permitted where cell a is not shaded, prohibited where shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.
- Numbers 6 and 8 indicate minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(9). For the remainder of the wall, see Note b.
- Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt and a minimum nominal 3 × 6 sill plate are required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(10)

WOOD-FRAMED ROOF TO TOP OF CONCRETE WALL, FRAMING PARALLEL

TABLE R608.9(10)

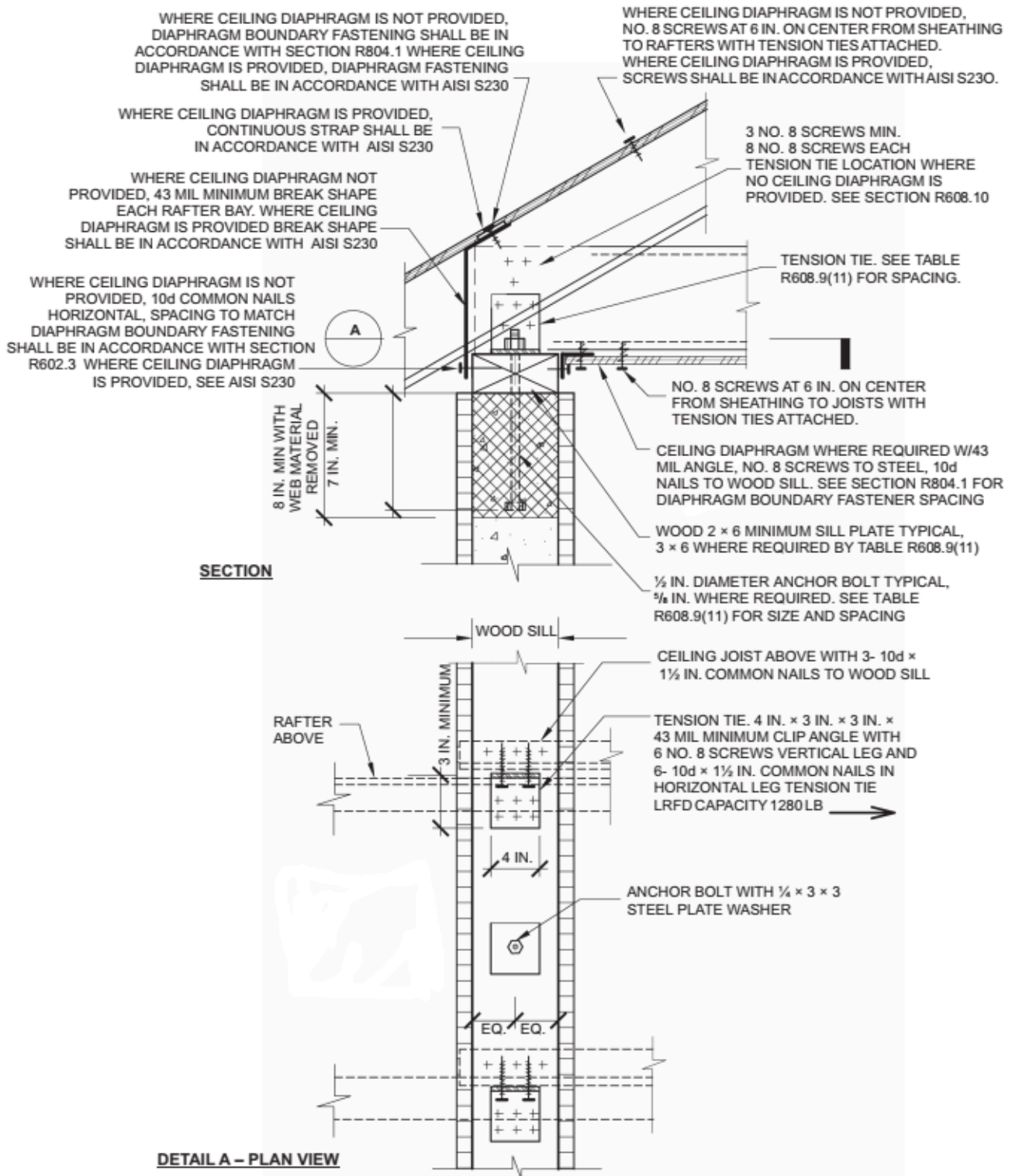
WOOD-FRAMED ROOF TO TOP OF CONCRETE WALL, FRAMING PARALLEL^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C

					110D	117D	125D
12	12						6
12	24						6
12	36					6	6
12	48				6	6	6
16	16					6	6
16	32					6	6
16	48				6	6	6
19.2	19.2					6	6
19.2	38.4				6	6	
24	24				6		
24	48			6	8B		

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(10). Use of this detail is permitted where a cell is not shaded, prohibited where shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in cells that do not contain a number.
- Numbers 6 and 8 indicate minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(10). For the remainder of the wall, see Note b.
- Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt and a minimum nominal 3 × 6 sill plate are required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(11)

COLD-FORMED STEEL ROOF TO TOP OF CONCRETE WALL, FRAMING PERPENDICULAR

TABLE R608.9(11)

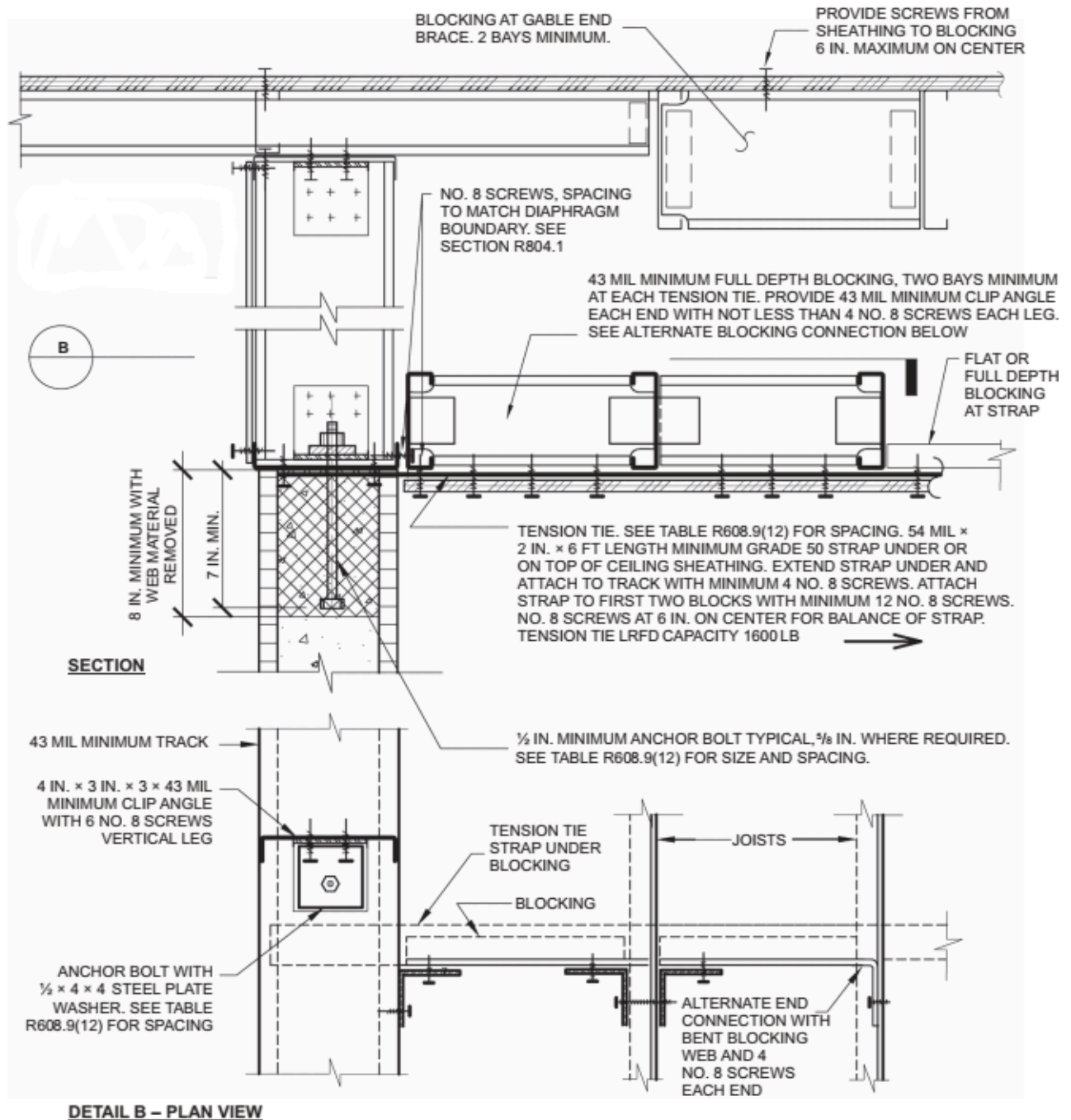
WOOD-FRAMED ROOF TO TOP OF CONCRETE WALL, FRAMING PERPENDICULAR^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY
---------------------------------	---------------------------------	---

		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D
12	12						6
12	24						6
16	16					6	6
16	32					6	6
19.2	19.2					6	6
19.2	38.4				6	6	6
24	24				6	6A	6B

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(11). Use of this detail is permitted where a cell is not shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in unshaded cells that do not contain a number.
- Number 6 indicates minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(11). For the remainder of the wall, see Note b.
- Letter "A" indicates that a minimum nominal 3 × 6 sill plate is required. Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt and a minimum nominal 3 × 6 sill plate are required.



For SI: 1 mil = 0.0254 mm, 1 inch = 25.4 mm, 1 pound-force = 4.448 N.

FIGURE R608.9(12)

COLD-FORMED STEEL ROOF TO TOP OF CONCRETE WALL, FRAMING PARALLEL

TABLE R608.9(12)

COLD-FORMED STEEL ROOF TO TOP OF CONCRETE WALL, FRAMING PARALLEL^{a, b, c, d, e}

ANCHOR BOLT SPACING (inches)	TENSION TIE SPACING (inches)	BASIC WIND SPEED (mph) AND WIND EXPOSURE CATEGORY					
		115B	120B	130B	140B	150B	160B
				110C	119C	127C	136C
					110D	117D	125D

12	12						6
12	24						6
16	16					6	6
16	32					6	6
19.2	19.2					6	6
19.2	38.4				6	6	6
24	24				6	6	6B

For SI: 1 inch = 25.4 mm, 1 mile per hour = 0.447 m/s.

- This table is for use with the detail in Figure R608.9(12). Use of this detail is permitted where a cell is not shaded.
- Wall design per other provisions of Section R608 is required.
- For wind design, minimum 4-inch-nominal wall is permitted in cells that do not contain a number.
- Number 6 indicates minimum permitted nominal wall thickness in inches necessary to develop required strength (capacity) of connection. As a minimum, this nominal thickness shall occur in the portion of the wall indicated by the cross-hatching in Figure R608.9(12). For the remainder of the wall, see Note b.
- Letter "B" indicates that a $\frac{5}{8}$ -inch-diameter anchor bolt is required.

R608.9.1 Connections Between Concrete Walls and Light-Framed Floor, Ceiling and Roof Systems

Connections between concrete walls and light-framed floor, ceiling and roof systems using the prescriptive details of Figures R608.9(1) through R608.9(12) shall comply with this section and Sections R608.9.2 and R608.9.3.

R608.9.1.1 Anchor Bolts

Anchor bolts used to connect light-framed floor, ceiling and roof systems to concrete walls in accordance with Figures R608.9(1) through R608.9(12) shall have heads, or shall be rods with threads on both ends with a hex or square nut on the end embedded in the concrete. Bolts and threaded rods shall comply with Section R608.5.2.2. Anchor bolts with J- or L-hooks shall not be used where the connection details in these figures are used.

R608.9.1.2 Removal of Stay-in-Place Form Material at Bolts

Holes in stay-in-place forms for installing bolts for attaching face-mounted wood ledger boards to the wall shall be not less than 4 inches (102 mm) in diameter for forms not greater than $1\frac{1}{2}$ inches (38 mm) in thickness, and increased 1 inch (25 mm) in diameter for each $\frac{1}{2}$ -inch (12.7 mm) increase in form thickness. Holes in stay-in-place forms for installing bolts for attaching face-mounted cold-formed steel tracks to the wall shall be not less than 4 inches (102 mm) square. The wood ledger board or steel track shall be in direct contact with the concrete at each bolt location.

Exception: A vapor retarder or other material less than or equal to $\frac{1}{16}$ inch (1.6 mm) in thickness is permitted to be installed between the wood ledger or cold-formed track and the concrete.

R608.9.2 Connections Between Concrete Walls and Light-Framed Floor Systems

Connections between concrete walls and light-framed floor systems shall be in accordance with one of the following:

- For floor systems of wood frame construction, the provisions of Section R608.9.1 and the prescriptive details of Figures R608.9(1) through R608.9(4), where permitted by the tables accompanying those figures. Portions of connections of wood-framed floor systems not noted in the figures shall be in accordance with Section R502, or AWC WFCM, if applicable. Wood framing members shall be of a species having a specific gravity equal to or greater than 0.42.
- For floor systems of cold-formed steel construction, the provisions of Section R608.9.1 and the prescriptive details of Figures R608.9(5) through R608.9(8), where permitted by the tables accompanying those figures. Portions of connections of cold-formed-steel framed floor systems not noted in the figures shall be in accordance with Section R505, or AISI S230, if applicable.

3. Proprietary connectors selected to resist loads and load combinations in accordance with Appendix A (ASD) or Appendix B (LRFD) of PCA 100.
4. An engineered design using loads and load combinations in accordance with Appendix A (ASD) or Appendix B (LRFD) of PCA 100.
5. An engineered design using loads and material design provisions in accordance with this code, or in accordance with ASCE 7, ACI 318, and AWC NDS for wood frame construction or AISI S100 for cold-formed steel frame construction.

R608.9.3 Connections Between Concrete Walls and Light-Framed Ceiling and Roof Systems

Connections between concrete walls and light-framed ceiling and roof systems shall be in accordance with one of the following:

1. For ceiling and roof systems of wood frame construction, the provisions of Section R608.9.1 and the prescriptive details of Figures R608.9(9) and R608.9(10), where permitted by the tables accompanying those figures. Portions of connections of wood-framed ceiling and roof systems not noted in the figures shall be in accordance with Section R802, or AWC WFCM, if applicable. Wood framing members shall be of a species having a specific gravity equal to or greater than 0.42.
2. For ceiling and roof systems of cold-formed-steel construction, the provisions of Section R608.9.1 and the prescriptive details of Figures R608.9(11) and R608.9(12), where permitted by the tables accompanying those figures. Portions of connections of cold-formed-steel framed ceiling and roof systems not noted in the figures shall be in accordance with Section R804, or AISI S230, if applicable.
3. Proprietary connectors selected to resist loads and load combinations in accordance with Appendix A (ASD) or Appendix B (LRFD) of PCA 100.
4. An engineered design using loads and load combinations in accordance with Appendix A (ASD) or Appendix B (LRFD) of PCA 100.
5. An engineered design using loads and material design provisions in accordance with this code, or in accordance with ASCE 7, ACI 318, and AWC NDS for wood-frame construction or AISI

R608.10 Floor, Roof and Ceiling Diaphragms

Floors and roofs in buildings with exterior walls of concrete shall be designed and constructed as diaphragms. Where gable-end walls occur, ceilings shall be designed and constructed as diaphragms. The design and construction of floors, roofs and ceilings of wood framing or cold-formed-steel framing serving as diaphragms shall comply with the applicable requirements of this code, or AWC WFCM or AISI S230, if applicable. Wood framing members shall be of a species having a specific gravity equal to or greater than 0.42.

Section R609 Exterior Windows and Doors

R609.1 General

This section prescribes performance and construction requirements for exterior windows and doors installed in walls. Windows and doors shall be installed and flashed in accordance with the fenestration manufacturer's written instructions. Window and door openings shall be flashed in accordance with Section R703.4. Written installation instructions shall be provided by the fenestration manufacturer for each window or door.

R609.2 Performance

Exterior windows and doors shall be designed to resist the design wind loads specified in Table R301.2(2) adjusted for height and exposure in accordance with Table R301.2(3) or determined in accordance with ASCE 7 using the allowable stress design load combinations of ASCE 7. Design wind loads for exterior glazing not part of a labeled assembly shall be permitted to be determined in accordance with Chapter 24 of the *Florida Building Code, Building*.

R609.2.1 Custom Doors

Custom (one-of-a-kind) exterior door assemblies shall be tested by an approved testing laboratory or be designed and engineered in accordance with accepted engineering practices by a Florida registered design professional. Signed and sealed copies of the rational analysis and calculations shall be provided to the building official upon permit application.

R609.3 Testing and Labeling

Exterior windows and sliding doors shall be tested by an *approved* independent laboratory, and bear a *label* identifying manufacturer, performance characteristics and *approved* inspection agency to indicate compliance with AAMA/WDMA/CSA 101/I.S.2/A440 or TAS 202

(HVHZ shall comply with TAS 202 and ASTM E1300). Exterior side-hinged doors shall be tested and *labeled* as conforming to AAMA/WDMA/CSA 101/I.S.2/A440 or ANSI/WMA 100, or comply with Section R609.5. Exterior windows and doors shall be labeled with a permanent label, marking, or etching providing traceability to the manufacturer and product. The following shall also be required either on a permanent label or on a temporary supplemental label applied by the manufacturer: information identifying the manufacturer, the product model/series number, positive and negative design pressure rating, product maximum size tested, impact-resistance rating if applicable, Florida product approval number or Miami-Dade product approval number applicable test standard(s), and approved product certification agency, testing laboratory, evaluation entity or Miami-Dade product approval.

The labels are limited to one design pressure rating per reference standard. The temporary supplemental label shall remain on the window or door until final approval by the building official.

Exceptions:

1. Door assemblies installed in nonhabitable areas where the door assembly and area are designed to accept water infiltration need not be tested for water infiltration.
2. Door assemblies installed where the overhang (OH) ratio is equal to or more than 1 need not be tested for water infiltration. The overhang ratio shall be calculated by the following equation:

OH ratio = OH Length/OH Height

Where:

OH length = The horizontal measure of how far an overhang over a door projects out from door surface.

OH height = The vertical measure of the distance from the door sill to the bottom of the overhang over a door.

3. Decorative glazed openings.
4. Pass-through windows for serving from a single-family kitchen, where protected by a roof overhang of 5 feet (1.5 m) or more shall be exempted from the requirements of the water infiltration test.

R609.3.1 Comparative Analysis

Structural wind load design pressures for window and door units different than the size tested in accordance with Section R609.3 shall be permitted to be different than the design value of the tested unit where determined in accordance with one of the following comparative analysis methods:

1. Structural wind load design pressures for window and door units other than the size tested in accordance with Section R609.3 shall be permitted to be different than the design value of the tested unit provided such different pressures are determined by accepted engineering analysis. All components of the alternate size assembly shall be the same as the tested or labeled assembly; however, lineal components shall be permitted to vary in length compared to the tested or labeled assembly.

Exceptions:

1. Operable windows and doors rated in this manner shall comply with the following:
 - a. For windows and doors (other than sliding or bi-fold), the frame area of the alternate size unit shall not exceed the frame area of the tested approved unit.
 - b. For sliding or bi-fold doors, the panel area of the alternate size unit shall not exceed the panel area of the tested approved unit.
 - c. Shall vary from the tested approved unit only in width, height or load requirements.
 - d. Shall not exceed 100 percent of the proportional deflection for fiber stress of the intermediate members of the approved unit.

- e. Shall not exceed 100 percent of the concentrated load at the juncture of the intermediate members and the frame of the approved unit.
- f. Shall not exceed the air and water infiltration resistance of the tested approved unit.
- g. Shall not exceed the maximum cyclic pressure of the tested approved unit when tested per TAS 201 and TAS 203 or ASTM E1886 and ASTM E1996 where applicable.

2. Nonoperable windows and doors rated in this manner shall comply with the following:

- a. The frame area of the alternate size unit shall not exceed the frame area of the tested approved unit.
- b. Shall vary from the tested approved unit only in width, height or load requirements.
- c. The maximum uniform load distribution (ULD) of any side shall be equal to the uniform load carried by the side divided by the length of the side.
- d. The ULD of any member shall not exceed the ULD of the corresponding member of the tested approved unit.
- e. The ULD of each member shall be calculated in accordance with standard engineering analysis.
- f. Shall not exceed the air and water infiltration resistance of the tested approved unit.
- g. Shall not exceed the maximum cyclic pressure of the tested approved unit when tested per TAS 201 and TAS 203 or ASTM E1886 and ASTM E1996 where applicable.

2. In accordance with WDMA I.S.11 or AAMA 2502.

R609.4 Garage Doors

Garage doors shall be tested for determination of structural performance under uniform static air pressure difference in accordance with ANSI/DASMA 108, ASTM E330 Procedure A, or TAS 202. For garage doors tested in accordance with ASTM E330, acceptance criteria shall be in accordance with ANSI/DASMA 108. (HVHZ shall comply with TAS 202.) Design pressures shall be determined from Table 301.2(4) or ASCE 7. The design pressures, as determined from ASCE 7, are permitted to be multiplied by 0.6.

R609.4.1 Garage Door Labeling

Garage doors shall be labeled with a permanent label provided by the garage door manufacturer. The label shall identify the garage door manufacturer, the garage door model/series number, the positive and negative design pressure rating, indicate impact rated if applicable, the installation instruction drawing reference number, the Florida product approval or Miami-Dade product approval number if applicable, and the applicable test standards. The required garage door components for an approved garage door assembly may be indicated using a checklist form on the label. If a checklist format is used on the label, the door installer or the garage door manufacturer shall mark the selected components on the checklist that are required to assemble an approved garage door system. The installation instructions shall be provided and available on the job site.

R609.5 Other Exterior Window and Door Assemblies

Exterior windows and door assemblies not included within the scope of Section R609.3 or R609.4 shall be tested in accordance with ASTM E330. Glass in assemblies covered by this exception shall comply with Section R308.5.

R609.6 Wind-Borne Debris Protection

Protection of exterior glazed openings in buildings located in wind-borne debris regions shall be in accordance with Section R301.2.1.2.

R609.6.1 Fenestration Testing and Labeling

Fenestration shall be tested by an *approved* independent laboratory, listed by an *approved* entity, and bear a *label* identifying manufacturer, performance characteristics, and *approved* inspection agency to indicate compliance with the requirements of the following specification(s):

1. ASTM E1886 and ASTM E1996; or
2. AAMA 506.

R609.7 Anchorage Methods

The methods cited in this section apply only to anchorage of window and glass door assemblies to the main force-resisting system.

R609.7.1 Anchoring Requirements

Window and glass door assemblies shall be anchored in accordance with the published manufacturer's recommendations to achieve the design pressure specified. Substitute anchoring systems used for substrates not specified by the fenestration manufacturer shall provide equal or greater anchoring performance as demonstrated by accepted engineering practice.

R609.7.2 Anchorage Details

Products shall be anchored in accordance with the minimum requirements illustrated in Figures R609.7.2(1), R609.7.2(2), R609.7.2(3), R609.7.2(4), R609.7.2(5), R609.7.2(6), R609.7.2(7) and R609.7.2(8).

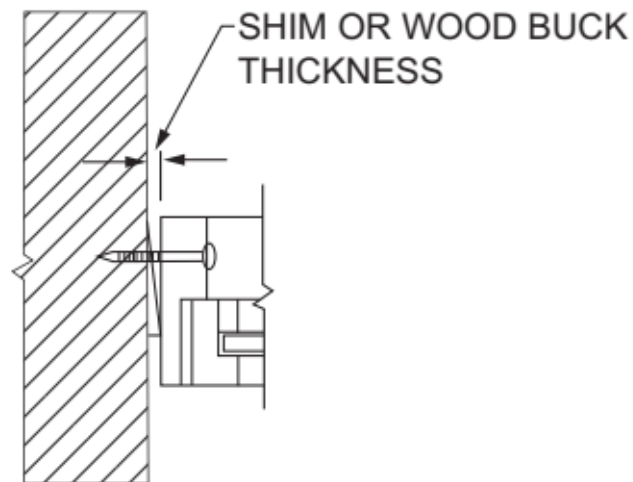


FIGURE R609.7.2(1)
THROUGH THE FRAME

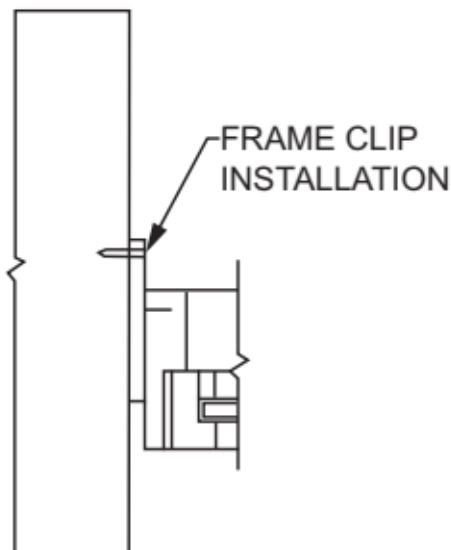


FIGURE R609.7.2(2)

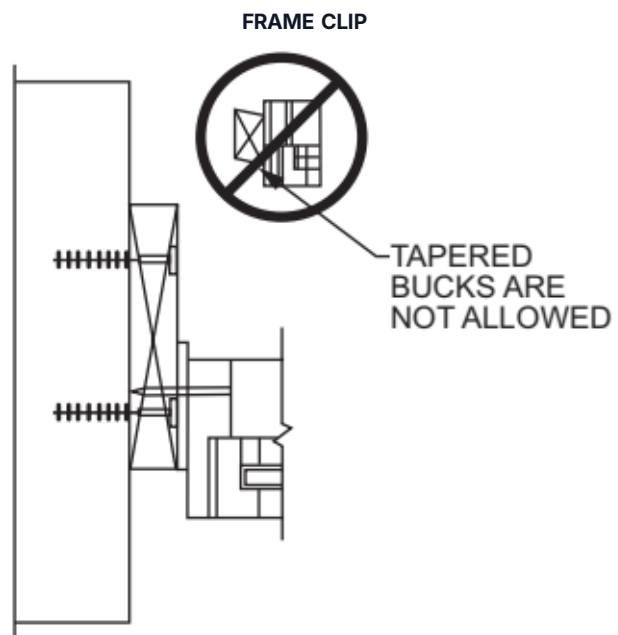


FIGURE R609.7.2(3)
THROUGH THE FRAME

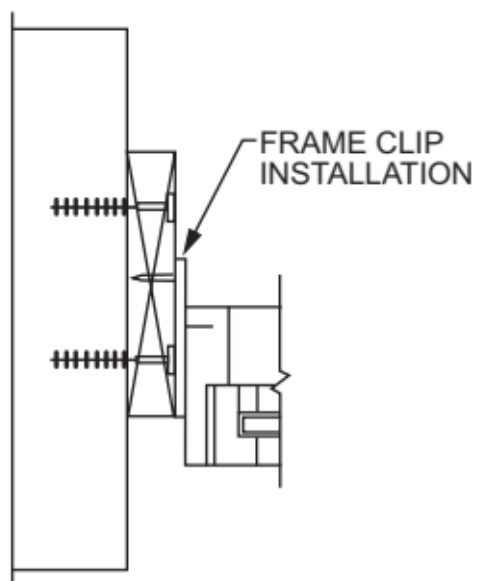


FIGURE R609.7.2(4)
FRAME CLIP

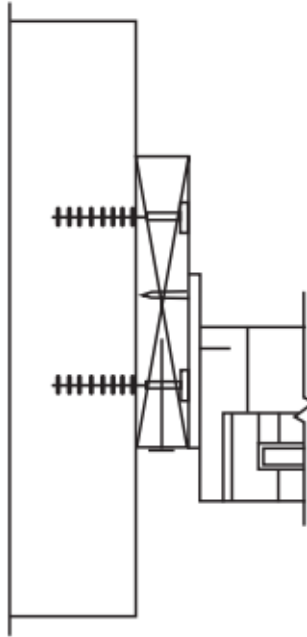


FIGURE R609.7.2(5)
THROUGH THE FLANGE

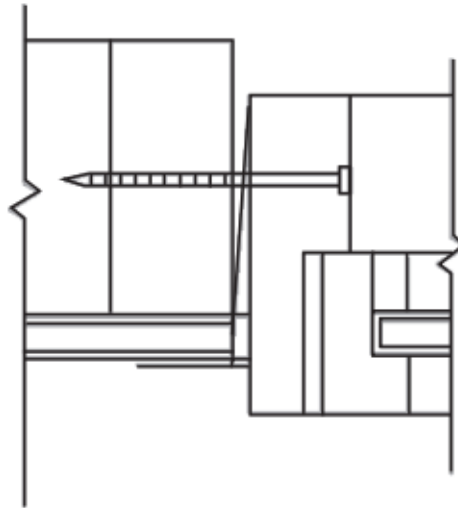


FIGURE R609.7.2(6)
THROUGH THE FLANGE

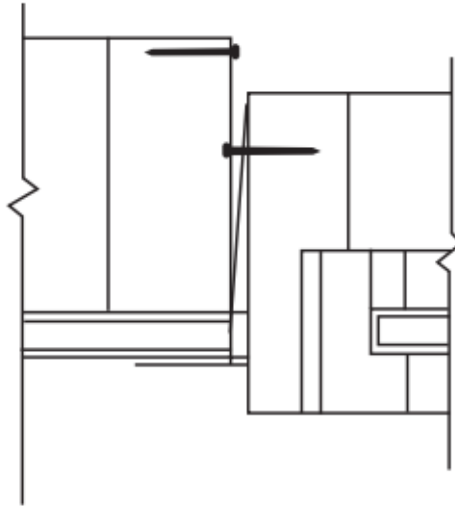


FIGURE R609.7.2(7)

FRAME CLIP

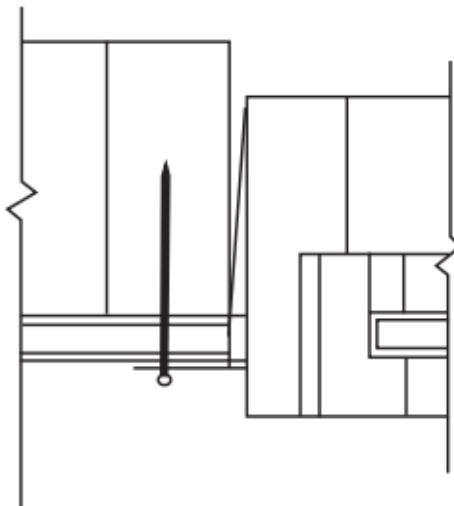


FIGURE R609.7.2(8)

THROUGH THE FLANGE

R609.7.2.1 Masonry, Concrete or Other Structural Substrate

Where the wood shim or buck thickness is less than $1\frac{1}{2}$ inches (38 mm), window and glass door assemblies shall be anchored through the jamb, or by jamb clip and anchors shall be embedded directly into the masonry, concrete or other substantial substrate material. Unless otherwise tested, bucks shall extend beyond the interior face of the window or door frame such that full support of the frame is provided. Anchors shall adequately transfer load from the window or door frame into the rough opening substrate [see Figures R609.7.2(1) and R609.7.2(2)].

Where the wood shim or buck thickness is $1\frac{1}{2}$ inches (38 mm) or more, and the buck is securely fastened to the masonry, concrete or other substantial substrate, window and glass door assemblies shall be anchored through the jamb, or by jamb clip, or through the flange to the secured wood buck. Unless otherwise tested, bucks shall extend beyond the interior face of the window or door frame such that full support of the frame is provided. Anchors shall be embedded into the secured wood buck to adequately transfer load from the window or door frame assembly [see Figures R609.7.2(3), R609.7.2(4) and R609.7.2(5)].

R609.7.2.2 Wood or Other Approved Framing Material

Where the framing material is wood or other *approved* framing material, window and glass door assemblies shall be anchored through the frame, or by frame clip, or through the flange. Anchors shall be embedded into the frame construction to adequately

transfer load [see Figures R609.7.2(6), R609.7.2(7) and R609.7.2(8)].

R609.8 Mullions

Mullions shall be tested by an *approved* testing laboratory in accordance with AAMA 450, or be engineered in accordance with accepted engineering practice. Mullions tested as stand-alone units or qualified by engineering shall use performance criteria cited in Sections R609.8.1, R609.8.2 and R609.8.3. Mullions qualified by an actual test of an entire assembly shall comply with Sections R609.8.1 and R609.8.3.

R609.8.1 Load Transfer

Mullions shall be designed to transfer the design pressure loads applied by the window and door assemblies to the rough opening substrate.

R609.8.2 Deflection

Mullions shall be capable of resisting the design pressure loads applied by the window and door assemblies to be supported without deflecting more than $L/175$, for spans up to and including 13 foot 6 inches, and $L/240 + 1/4$ inch for spans over 13 foot 6 inches, where L is the span of the mullion in inches.

R609.8.3 Structural Safety Factor

Mullions shall be capable of resisting a load of 1.5 times the design pressure loads applied by the window and door assemblies to be supported without exceeding the appropriate material stress levels. If tested by an *approved* laboratory, the 1.5 times the design pressure load shall be sustained for 10 seconds, and the permanent deformation shall not exceed 0.4 percent of the mullion span after the 1.5 times design pressure load is removed.

R609.9 Masonry Rough Openings

Masonry rough opening dimensions shall be within the tolerances specified in Section R606.3.1.1 and, in addition, shall provide for a window perimeter sealant joint a maximum of $1/4$ inch (6.4 mm) in width.

R609.10 Door Components

Substitution of door components shall comply with either:

1. ANSI/WMA 100; or
2. Be evaluated by an approved product evaluation entity, certification agency, testing laboratory, or engineer and may be interchangeable in exterior door assemblies if the components provide equal or greater structural performance as demonstrated by accepted engineering practices.

R609.11 Impact-Resistant Coverings

R609.11.1

Impact-resistant coverings shall be tested at 1.5 times the design pressure (positive or negative) expressed in pounds per square feet as determined by the *Florida Building Code, Building* Section 1609 for which the specimen is to be tested. The design pressures, as determined from Section 1609 of the *Florida Building Code, Building* or ASCE 7, are permitted to be multiplied by 0.6.

R609.11.1.1

Impact-resistant coverings shall be labeled in accordance with the provisions of Section R609.11.

R609.11.2 Labels

A permanent label shall be provided by the product approval holder on all impact-resistant coverings.

R609.11.2.1

The following information shall be included on the labels on impact-resistant coverings:

1. Product approval holder name and address.
2. All applicable methods of approval. Methods of approval include, but, are not limited to Miami-Dade NOA; Florida Building Commission, TDI Product Evaluation; ICC-ES.

3. The test standard or standards specified in Section R301.2.1.2, including standards referenced within the test standards specified in Section R301.2.1.2 used to demonstrate code compliance.
4. For products with a Florida product approval number or a Miami-Dade County Building and Neighborhood Compliance Department Notice of Acceptance Number (NOA), such numbers shall be included on the label.

R609.11.3 Location of Label

The location of the label on the impact-resistant covering shall be as follows:

1. Accordions: Bottom of the locking bar or center mate facing the exterior or outside.
2. Rollup: On the bottom of the hood facing the exterior or outside or on the bottom slat facing the exterior or outside.
3. Bahama Awning or Colonial Hinged: On the bottom, placed on the back of the shutter.
4. Panels: For metal and plastic panels the label may be embossed or printed spaced not more than every 3 lineal feet (914 mm) on each panel. The label shall be applied by the holder of the product approval and shall face the exterior or outside.
5. Framed products: The label shall be on the side or bottom facing the exterior or outside.
6. Labels on all other products shall face the exterior or outside.

R609.11.4 Installation

All impact-resistant coverings shall be installed in accordance with the manufacturer's installation instructions and in accordance with the product approval. Installation instructions shall be provided and shall be available to inspection personnel on the job site. Opening protection components, fasteners, and other parts evaluated by an approved product evaluation entity, certification agency, testing laboratory, architect, or engineer and approved by the holder of the product approval may be interchangeable in opening protection assemblies provided that the opening protection component(s) provide equal or greater structural performance and durability as demonstrated by testing in accordance with approved test standards.

Section R610 Structural Insulated Panel Wall Construction

R610.1 General

Structural insulated panel (SIP) walls shall be designed in accordance with the provisions of this section. Where the provisions of this section are used to design structural insulated panel walls, project drawings, typical details and specifications are not required to bear the seal of the architect or engineer responsible for design, unless otherwise required by the state law of the *jurisdiction* having authority.

R610.2 Applicability Limits

The provisions of this section shall control the construction of exterior structural insulated panel walls and interior load-bearing structural insulated panel walls for buildings not greater than 60 feet (18 288 mm) in length perpendicular to the joist or truss span, not greater than 40 feet (12 192 mm) in width parallel to the joist or truss span and not greater than two stories in height with each wall not greater than 10 feet (3048 mm) high. Exterior walls installed in accordance with the provisions of this section shall be considered as load-bearing walls. Structural insulated panel walls constructed in accordance with the provisions of this section shall be limited to sites where the ultimate design wind speed (V_{ult}) is not greater than 155 miles per hour (69 m/s), Exposure B or 140 miles per hour (63 m/s) Exposure C.

R610.3 Materials

SIPs shall comply with the requirements of ANSI/APA PRS 610.1:

R610.3.1 Lumber

The minimum lumber framing material used for SIPs prescribed in this document is NLGA graded No. 2 Spruce-pine-fir. Substitution of other wood species/grades that meet or exceed the mechanical properties and specific gravity of No. 2 Spruce-pine-fir shall be permitted.

R610.3.2 SIP Screws

Screws used for the erection of SIPs as specified in Section R610.5 shall be fabricated from steel, shall be provided by the SIP manufacturer and shall be sized to penetrate the wood member to which the assembly is being attached by not less than 1 inch (25

mm). The screws shall be corrosion resistant and have a minimum shank diameter of 0.188 inch (4.7 mm) and a minimum head diameter of 0.620 inch (15.5 mm).

R610.3.3 Nails

Nails specified in Section R610 shall be common or galvanized box unless otherwise stated.

R610.4 SIP Wall Panels

SIPs shall comply with Figure R610.4 and shall have minimum panel thickness in accordance with Tables R610.5(1) and R610.5(2) for above-grade walls. SIPs shall be identified by grade mark or certificate of inspection issued by an *approved* agency in accordance with ANSI/APA PRS 610.1.

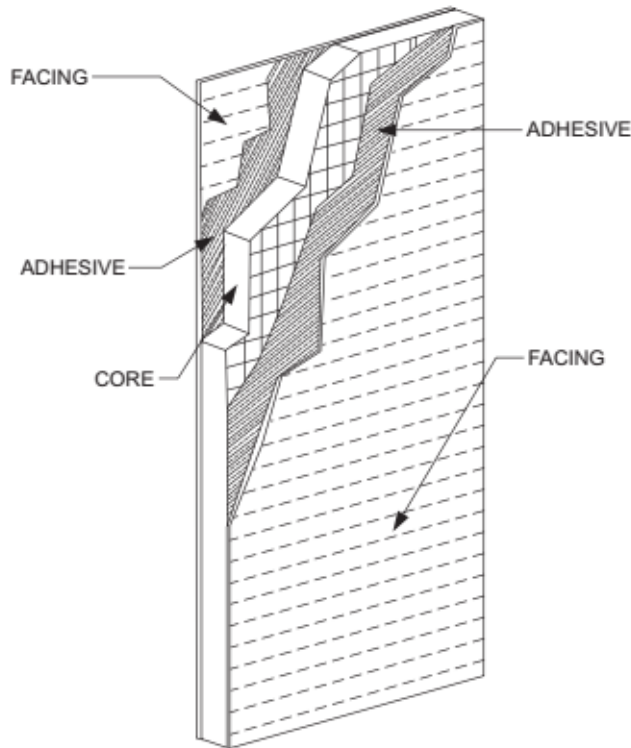


FIGURE R610.4
SIP WALL PANEL

R610.5 Wall Construction

Exterior walls of SIP construction shall be designed and constructed in accordance with the provisions of this section and Tables R610.5(1) and R610.5(2) and Figures R610.5(1) through R610.5(5). SIP walls shall be fastened to other wood building components in accordance with Section R602.3.

Framing shall be attached in accordance with Section R602.3 unless otherwise provided for in Section R610.

TABLE R610.5(1)

MINIMUM THICKNESS FOR SIP WALL SUPPORTING SIP OR LIGHT-FRAME ROOF ONLY (inches)^a

BUILDING WIDTH (ft)																	
ULTIMATE DESIGN WIND SPEED V_{ult} (mph)		24			28			32			36			40			
Exp. B	Exp. C	Wall Height (feet)			Wall Height (feet)			Wall Height (feet)			Wall Height (feet)			Wall Height (feet)			
		8	9	10	8	9	10	8	9	10	8	9	10	8	9	10	

110	—	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5
115	—	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5
130	110	4.5	4.5	6.5	4.5	4.5	6.5	4.5	4.5	6.5	4.5	4.5	DR	4.5	4.5	DR
140	120	4.5	6.5	DR	4.5	6.5	DR	4.5	DR	DR	4.5	DR	DR	4.5	DR	DR

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa, 1 mile per hour = 0.447 m/s.

DR = design required.

a. Design assumptions:

Maximum deflection criteria: $L/240$.

Maximum roof dead load: 10 psf.

Maximum roof live load: 70 psf.

Maximum ceiling dead load: 5 psf.

Maximum ceiling live load: 20 psf.

Wind loads based on Table R301.2 (2).

Strength axis of facing material applied vertically.

TABLE R610.5(2)

MINIMUM THICKNESS FOR SIP WALL SUPPORTING SIP OR LIGHT-FRAME ONE STORY AND ROOF ONLY (inches)^a

BUILDING WIDTH (ft)																
ULTIMATE DESIGN WIND SPEED V_{ult} (mph)		24			28			32			36			40		
Exp. B	Exp. C	Wall Height (feet)			Wall Height (feet)			Wall Height (feet)			Wall Height (feet)			Wall Height (feet)		
		8	9	10	8	9	10	8	9	10	8	9	10	8	9	10
110	—	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	6.5	4.5	4.5	DR	4.5	4.5	DR
115	—	4.5	4.5	4.5	4.5	4.5	6.5	4.5	4.5	DR	4.5	4.5	DR	4.5	DR	DR
130	110	4.5	4.5	6.5	4.5	4.5	DR	4.5	4.5	DR	4.5	DR	DR	4.5	DR	DR
140	120	4.5	6.5	DR	4.5	DR	DR	4.5	DR	DR	DR	DR	DR	DR	DR	DR

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa, 1 mile per hour = 0.447 m/s.

DR = Design required.

a. Design assumptions:

Maximum deflection criteria: $L/240$.

Maximum roof dead load: 10 psf.

Maximum roof live load: 70 psf.

Maximum ceiling dead load: 5 psf.

Maximum ceiling live load: 20 psf.

Maximum second-floor dead load: 10 psf.

Maximum second-floor live load: 30 psf.

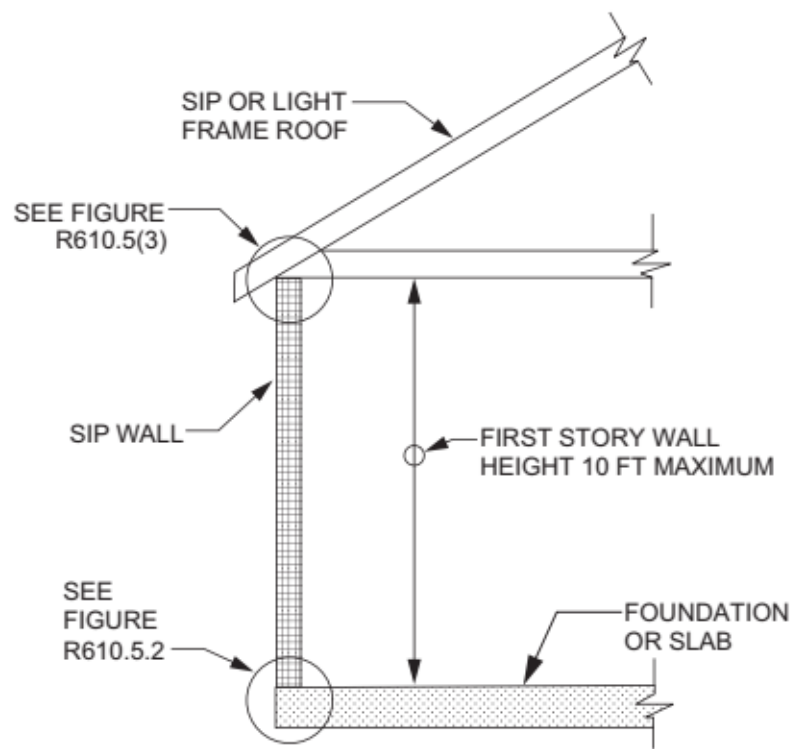
Maximum second-floor dead load from walls: 10 psf.

Maximum first-floor dead load: 10 psf.

Maximum first-floor live load: 40 psf.

Wind loads based on Table R301.2 (2).

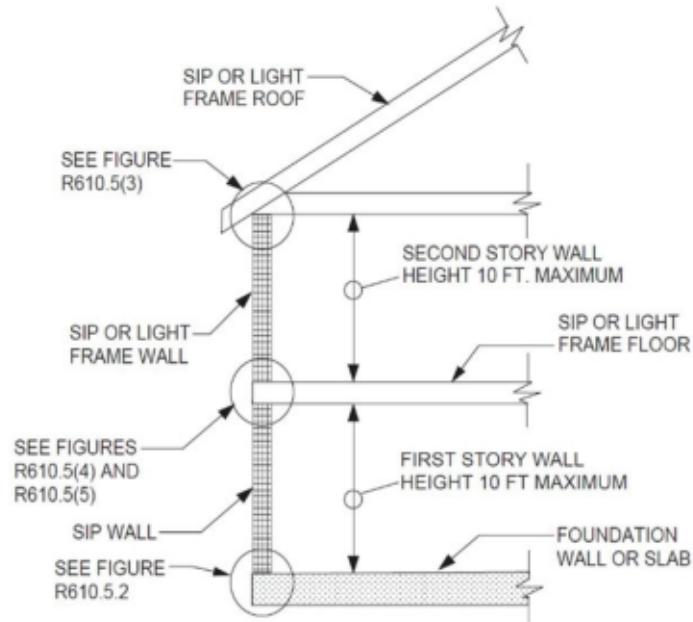
Strength axis of facing material applied vertically.



For SI: 1 foot = 304.8 mm.

Note: Figures illustrate SIP-specific attachment requirements. Other connections shall be made in accordance with Section R602.3, as appropriate.

FIGURE R610.5(1)
MAXIMUM ALLOWABLE HEIGHT OF SIP WALLS

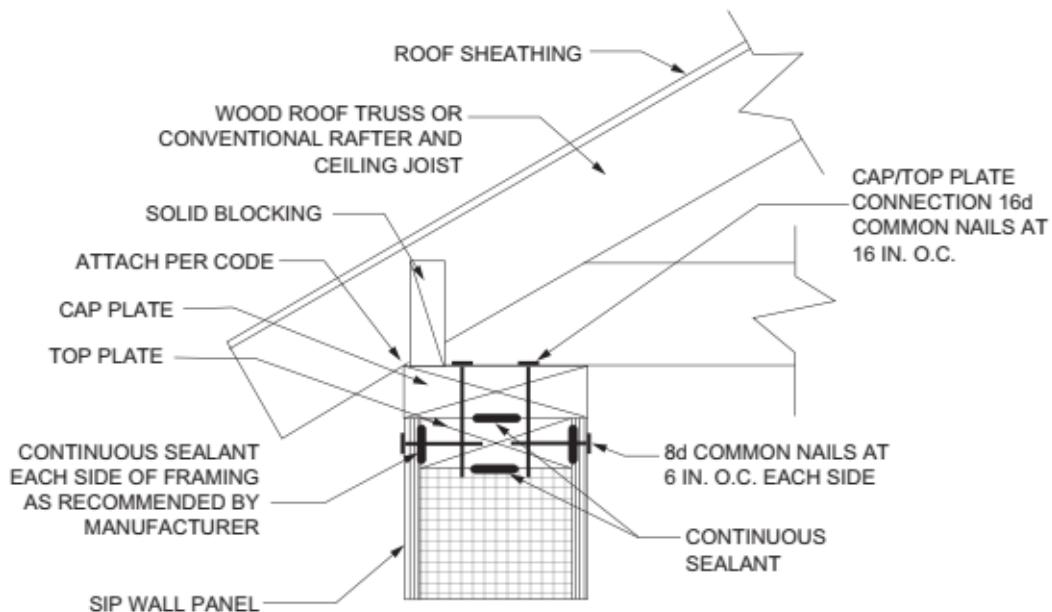


For SI: 1 foot = 304.8 mm.

Note: Figures illustrate SIP-specific attachment requirements. Other connections shall be made in accordance with Section R602.3, as appropriate.

FIGURE R610.5(2)

MAXIMUM ALLOWABLE HEIGHT OF SIP WALLS

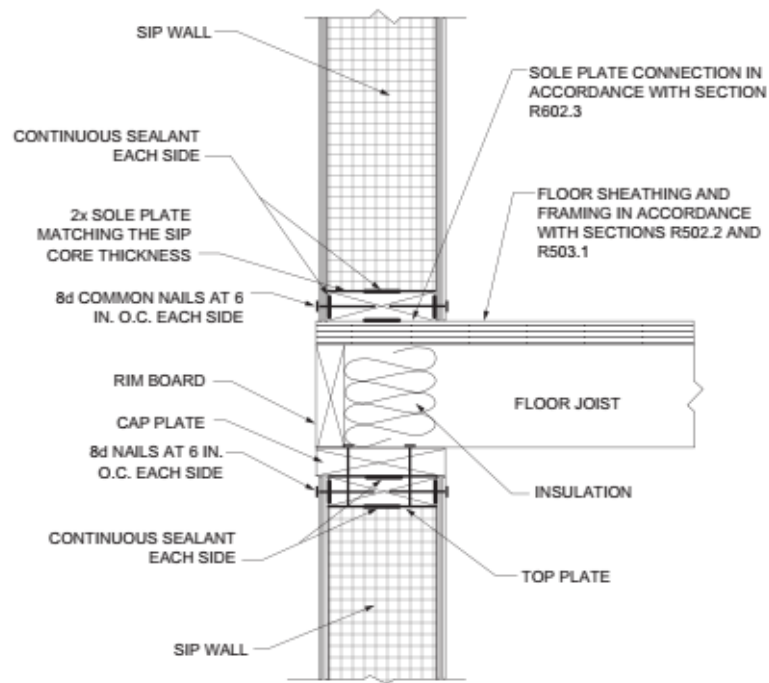


For SI: 1 inch = 25.4 mm.

Note: Figures illustrate SIP-specific attachment requirements. Other connections shall be made in accordance with Section R602.3, as appropriate.

FIGURE R610.5(3)

TRUSSED ROOF TO TOP PLATE CONNECTION

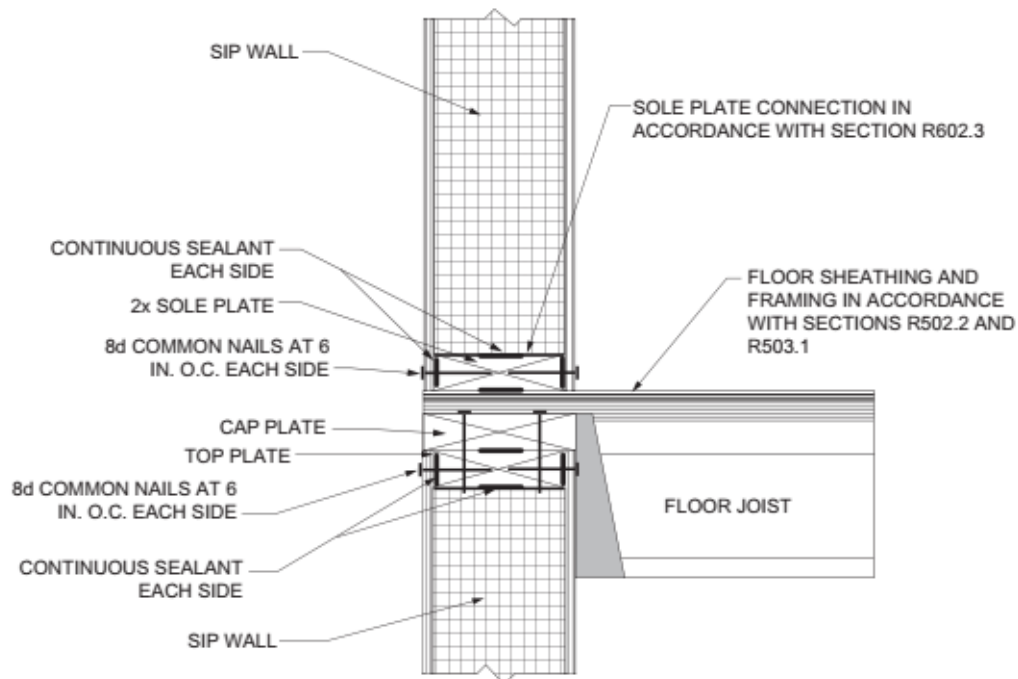


For SI: 1 inch = 25.4 mm.

Note: Figures illustrate SIP-specific attachment requirements. Other connections shall be made in accordance with Section R602.3, as appropriate.

FIGURE R610.5(4)

SIP WALL-TO-WALL PLATFORM FRAME CONNECTION



For SI: 1 inch = 25.4 mm.

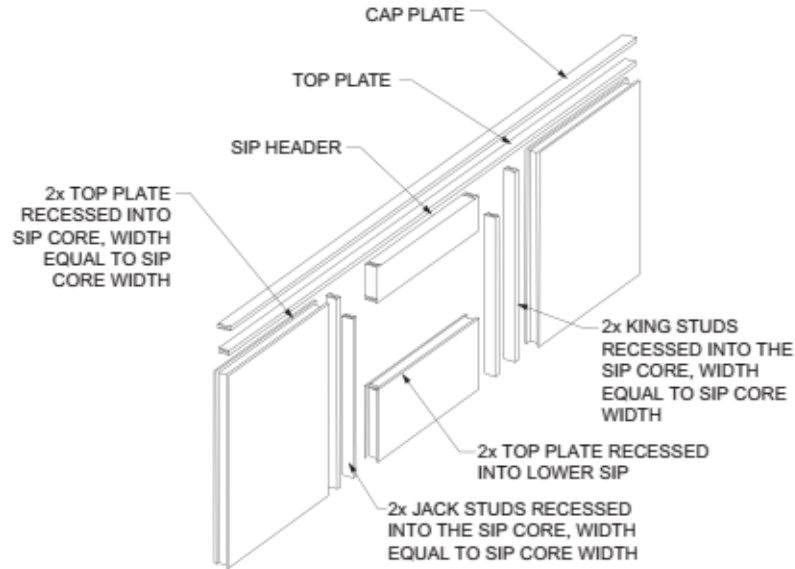
Note: Figures illustrate SIP-specific attachment requirements. Other connections shall be made in accordance with Section R602.3, as appropriate. I-joist floor shown for illustration only.

FIGURE R610.5(5)

SIP WALL-TO-WALL HANGING FLOOR CONNECTION

R610.5.1 Top Plate Connection

SIP walls shall be capped with a double top plate installed to provide overlapping at corner, intersections and splines in accordance with Figure R610.5.1. The double top plates shall be made up of a single 2 by top plate having a width equal to the width of the panel core, and shall be recessed into the SIP below. Over this top plate a cap plate shall be placed. The cap plate width shall match the SIP thickness and overlap the facers on both sides of the panel. End joints in top plates shall be offset not less than 24 inches (610 mm).



For SI: 1 inch = 25.4 mm.

Notes:

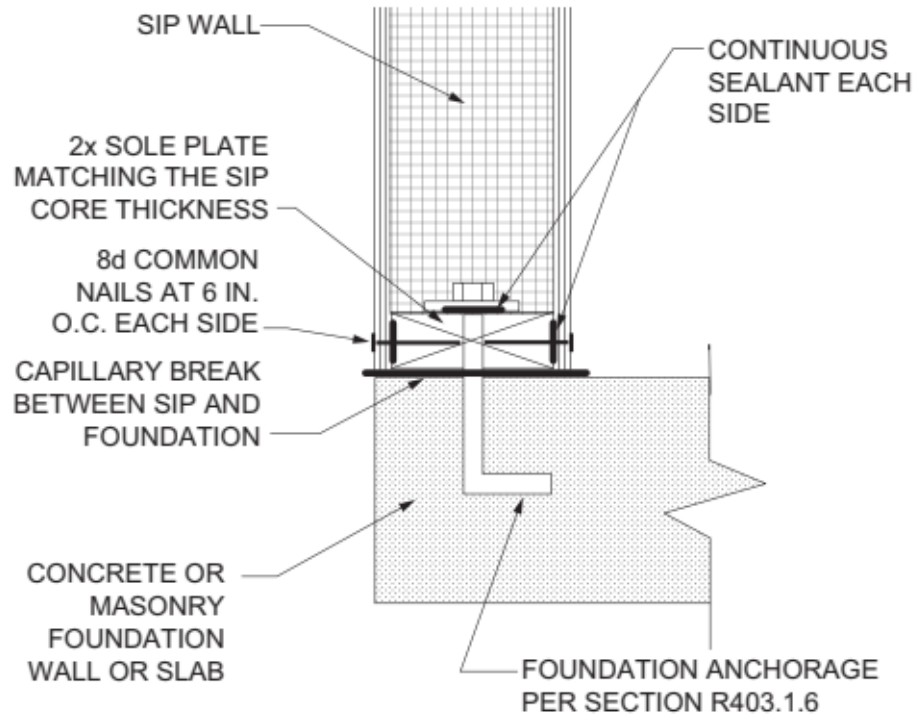
1. Top plates shall be continuous over header.
2. Lower 2x top plate shall have a width equal to the SIP core width and shall be recessed into the top edge of the panel. Cap plate shall be placed over the recessed top plate and shall have a width equal to the SIPs width.
3. SIP facing surfaces shall be nailed to framing and cripples with 8d common or galvanized box nails spaced 6 inches on center.

FIGURE R610.5.1

SIP WALL FRAMING CONFIGURATION

R610.5.2 Bottom (Sole) Plate Connection

SIP walls shall have full bearing on a sole plate having a width equal to the nominal width of the foam core. Where SIP walls are supported directly on continuous foundations, the wall wood sill plate shall be anchored to the foundation in accordance with Figure R610.5.2 and Section R403.1.



For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

FIGURE R610.5.2

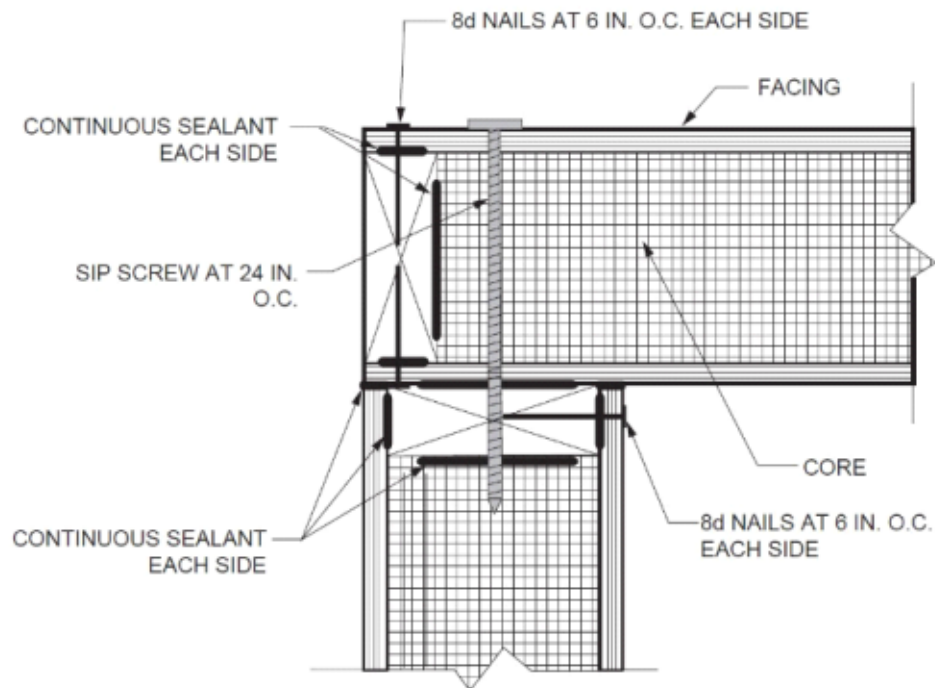
SIP WALL TO CONCRETE SLAB FOR FOUNDATION WALL ATTACHMENT

R610.5.3 Panel-to-Panel Connection

SIPs shall be connected at vertical in-plane joints in accordance with Figure R610.8 or by other approved methods.

R610.5.4 Corner Framing

Corner framing of SIP walls shall be constructed in accordance with Figure R610.5.4.



For SI: 1 inch = 25.4 mm.

FIGURE R610.5.4

SIP CORNER FRAMING DETAIL

R610.5.5 Wall Bracing

SIP walls shall be braced in accordance with Section R602.3. SIP walls shall be considered continuous wood structural panel sheathing (Bracing Method CS-WSP) for purposes of computing required bracing. SIP walls shall meet the requirements of Section R602.3 except that SIP corners shall be fabricated as shown in Figure R610.8. Where SIP walls are used for wall bracing, the SIP bottom plate shall be attached to wood framing below in accordance with Section R602.3.

R610.6 Interior Load-Bearing Walls

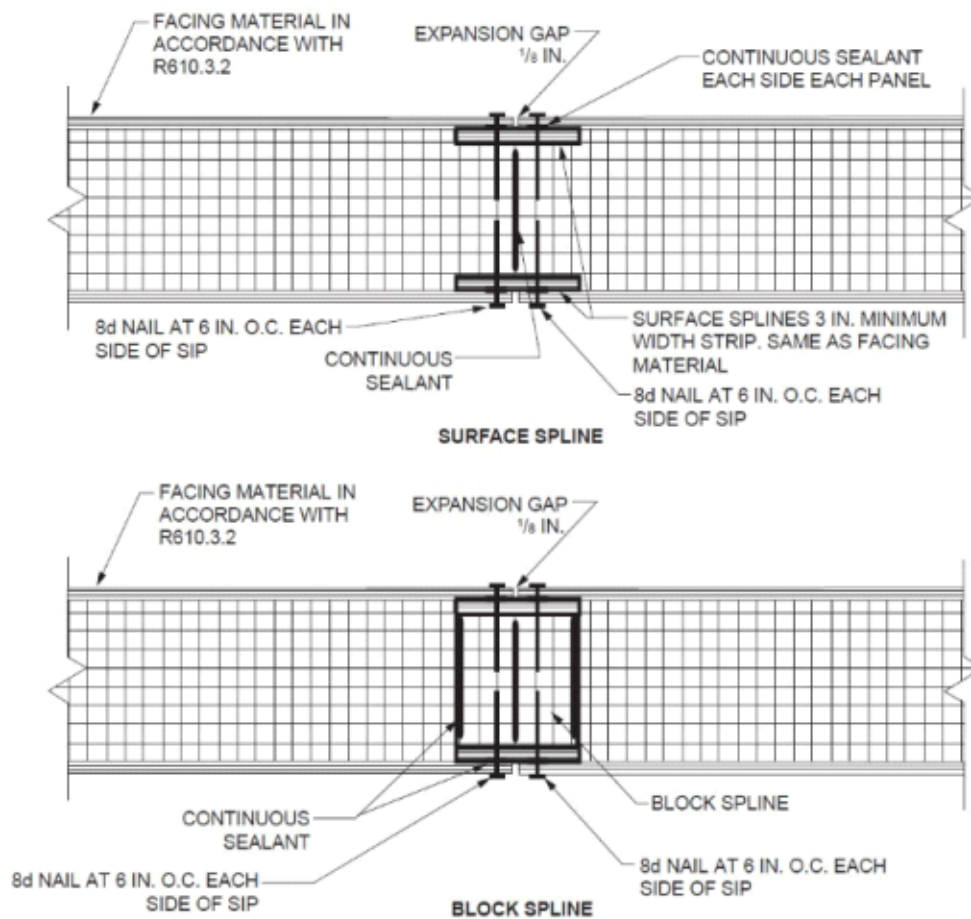
Interior load-bearing walls shall be constructed as specified for exterior walls.

R610.7 Drilling and Notching

The maximum vertical chase penetration in SIPs shall have a maximum side dimension of 2 inches (51 mm) centered in the panel. Vertical chases shall have a minimum spacing of 24 inches (610 mm) on center. A maximum of two horizontal chases shall be permitted in each wall panel—one at 14 inches (360 mm) plus or minus 2 inches (51 mm) from the bottom of the panel and one at 48 inches (1220 mm) plus or minus 2 inches (51 mm) from the bottom edge of the SIPs panel. Additional penetrations are permitted where justified by analysis.

R610.8 Headers

SIP headers shall be designed and constructed in accordance with Table R610.8 and Figure R610.5.1. SIP headers shall be continuous sections without splines. Headers shall be not less than $1\frac{7}{8}$ inches (302 mm) deep. Headers longer than 4 feet (1219 mm) shall be constructed in accordance with Section R602.3. The strength axis of the factors on the header shall be oriented horizontally.



For SI: 1 inch = 25.4 mm.

FIGURE R610.8

TYPICAL SIP WALL PANEL-TO-PANEL CONNECTION DETAILS

TABLE R610.8

MAXIMUM SPANS FOR 11⁷/₈-INCH OR DEEPER SIP HEADERS (feet)^{a, c, d}

LOAD CONDITION	BUILDING ^b width (feet)				
	24	28	32	36	40
Supporting roof only	4	4	4	4	2
Supporting roof and one-story	2	2	DR	DR	DR

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa.

DR = Design Required.

a. Design assumptions:

Maximum deflection criterion: $L/240$.

Maximum roof dead load: 10 psf.

Maximum ceiling load: 5 psf.

Maximum ceiling live load: 20 psf.

Maximum second-floor live load: 30 psf.

Maximum second-floor dead load: 10 psf.

Maximum second-floor dead load from walls: 10 psf.

Maximum first-floor dead load: 10 psf.

Wind load based on Table R301.2(2).

Strength axis of facing material applied horizontally.

b. Building width is in the direction of horizontal framing members supported by the header.

c. The table provides for roof slopes between 3:12 and 12:12.

d. The maximum roof overhang is 24 inches (610 mm).

R610.8.1 Wood Structural Panel Box Headers

Wood structural panel box headers shall be allowed where SIP headers are not applicable. Wood structural panel box headers shall be constructed in accordance with Section R602.3.